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Early View

Task force report

Clinical practice guideline for long COVID prevention and treatment

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Clinical Practice Guideline for Long COVID Prevention and Treatment

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Abstract

Background

This guideline aims to address key clinical questions of long COVID, and to provide evidence-base recommendations. The target population is adults with long COVID. The primary users of the guideline are clinical physicians, clinical pharmacists, nurses, and general practitioners in community healthcare institutions worldwide.

Methods

The guideline was registered at the Practice guideline REgistration for transPAREncy platform (PREPARE-2024CN123) and followed a pre-specified protocol. A multidisciplinary working group was established and comprised 60 members from 10 countries and 10 area of expertise, with a strong background in long COVID research and clinical practice, and methodology of guideline development. Through a two-step process, we determined the 8 PICO questions focusing on prevention and treatment of long COVID. After comprehensively searching literature, conducting systematic reviews, and investigating patients' values and preferences, three rounds of Delphi survey were conducted among 24 international experts to reach consensus. The Grading of Recommendations Assessment, Development,

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and Evaluation (GRADE) approach was applied to rate the certainty of evidence and determine the strength of recommendations.

Results

The guideline presents 10 specific recommendations, each supported by existing, updated, or newly conducted systematic reviews. The key recommendations are pertinent to the following issues: 1) suggestion of vaccination or use of antiviral agents during the acute phase of COVID-19 to prevent long COVID; 2) suggestions against the use of nirmatrelvir-ritonavir and glucocorticoids (patients with persistent respiratory symptoms and olfactory disorders) for long COVID treatment; 3) suggestions supporting the use of multi-species probiotics, cognitive behavioral therapy (patients with fatigue), and personalized rehabilitation after excluding post-exertional malaise for long COVID treatment.

Conclusions

This guideline provides evidence-based recommendations for the prevention and treatment of long COVID. Given the limited and often low-methodological quality evidence, all recommendations are supported by very low to moderate certainty. Further high-quality studies are needed to strengthen the evidence base.

Introduction

SARS-CoV-2 has circulated globally for more than five years, resulting in approximately 779 million infections and more than 7 million deaths globally[1]. Although most individuals recover completely from COVID-19, at least 10% of recovered patients continue to experience symptoms such as fatigue, palpitations, and dyspnea three months after disease onset[2,3]. The World Health Organization (WHO) defines post COVID-19 condition, commonly referred to as long COVID, as the presence of persistent or new symptoms three months after initial infection, lasting for at least two months and not explained by alternative diagnoses.

The estimated global cumulative incidence of long COVID rose steadily from 2020 to 2023[4]. Long COVID presents with a wide range of symptoms affecting multiple organ systems. Growing evidence also links SARS-CoV-2 infection with an elevated risk of long-term complications[5-8]. These long-term effects significantly impair patients' quality of life and functional capacity, contributing to increased sick leave and posing substantial economic burdens on individuals, families, and society [9].

With advances in the understanding of SARS-CoV-2 and its long-term consequences, increasing attention has been paid to long COVID. In response, multiple organizations and institutions have issued guidelines or established ongoing task forces on this condition[10-16]. However, these guidelines vary in quality, often lack comprehensiveness, and show significant inconsistency in content. Notably, there is still an lack of recent long COVID guidelines developed using contemporary evidence-based methodologies and shaped by multidisciplinary collaboration.

Given the growing body of evidence, we established an international, multidisciplinary working group to develop the clinical practice guideline for long COVID. This group aims to address key clinical

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questions of long COVID, mainly including prevention and treatment in adults. During the development of the guideline, the certainty of evidence and strength of recommendations was clearly defined, and patients' values and preferences were incorporated.

Methodology

This international guideline was developed following the WHO guidebook, utilizing the Appraisal of Guidelines for Research and Evaluation (AGREE) II tool, and adhering to the Reporting Items for Practice Guideline in Healthcare (RIGHT) statement[17,18]. Methodological support was provided by the Lanzhou University GRADE (Grading of Recommendations Assessment, Development, and Evaluation) Center and WHO Collaborating Centre for Guideline Implementation and Knowledge Translation. The guideline has been registered at the Practice guideline REgistration for transPAREncy platform (http://guidelines-registry.org/, registration No. PREPARE-2024CN123).

A multidisciplinary working group comprising 60 members from 10 areas of expertise and 10 countries formulated eight clinical questions (Table 1), conducted systematic evidence searches, assessed risk of bias with validated tools, and applied the GRADE approach to rate certainty and develop recommendations. Three rounds of Delphi surveys were used to reach consensus, and patient partners contributed to values and preferences. A total of 10 recommendations were formulated after three rounds of Delphi survey, considering the certainty of the evidence, balance of benefits and harms, patient values and preferences, acceptability, and cost considerations (Table 2)[19]. Plans for dissemination, evaluation, and future updates were also defined. Detailed methodological procedures are provided in Appendix 1, Methods Supplement.

Respiratory Manifestations of Long COVID

Evidence summary

Explanation

Pulmonary involvement is a common and clinically significant component of long COVID, closely related to the severity and extent of pulmonary involvement during the acute phase of SARS-CoV-2 infection[35]. Long-term abnormalities in lung imaging and pulmonary function, particularly impaired diffusing capacity, have been reported to persist for years after acute infection[36,37]. These findings suggest sustained lung parenchymal injury and microvascular dysfunction, which are considered key pathophysiological substrates underlying pulmonary long COVID[16].

Prevention of pulmonary long COVID

Current preventive strategies for long COVID, including vaccination and acute-phase pharmacological treatments, may reduce the overall risk of long COVID [38-41]; however, evidence specifically addressing the prevention of pulmonary long COVID remains limited. Available data suggest that the potential protective effects of vaccination and antiviral therapy on respiratory sequelae are largely

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indirect and mediated through attenuation of acute disease severity, rather than targeted prevention of chronic lung injury[42]. In our current guideline, an evidence synthesis evaluating the association between acute-phase antiviral treatment and long COVID-related dyspnea did not demonstrate a significant protective effect, and evidence for other pulmonary symptoms is also lacking. Greater attention is still needed to investigate preventive strategies for other clinical phenotypes of pulmonary long COVID.

Pharmacological treatment of pulmonary long COVID

Pharmacological treatment options for established pulmonary long COVID are also limited. The findings based on limited evidence suggest that antiviral therapy has not demonstrated clear benefit for respiratory symptoms in patients with long-standing disease and negative markers of viral persistence[20], although they do not exclude potential effect in other long COVID populations. Multiple ongoing clinical studies are evaluating antiviral drugs (NMV/r, amantadine, ensitrelvir, etc.) [43-49] for long COVID. Future studies should focus on identifying patients most likely to benefit through biomarker-based risk stratification for viral persistence, earlier treatment initiation, and longer treatment duration [50].Similarly, evidence supporting the use of monoclonal antibodies, which are highly specific proteins produced through B cell cloning that bind to distinct epitopes on pathogens, is extremely sparse and confined to isolated reports in immunocompromised patients[16].

Anti-inflammatory and immunomodulatory therapies have mainly been evaluated in small, uncontrolled studies, often in patients with interstitial lung disease-like phenotypes. For long COVID adult patients with persistent respiratory impairment, meta-analysis was not performed due to substantial heterogeneity across studies. Although available evidence suggests that glucocorticoids may improve certain lung imaging and function outcome (e.g., 6MWD, FEV_1, FVC, and DLCO), results are inconsistent regarding the primary outcome, and the magnitude of improvement is generally limited. Thus, the clinical benefit of glucocorticoids in this population remains uncertain. Additionally, while no serious adverse events were reported during follow-up periods, other non-serious adverse events were not systematically assessed, and the potential long-term risks of glucocorticoids therapy, including infections, osteoporosis, gastrointestinal bleeding, and mental health disorders, remain a concern. The recent identification of neutrophil-driven small airway inflammation and epithelial barrier dysfunction through single cell sequencing also raises the possibility that anti-inflammatory strategies, including small-molecule inhaled corticosteroids, may have therapeutic potential for dyspnea and chronic cough in pulmonary long COVID. However, this hypothesis requires validation in well-designed clinical trials [51].Therefore, based on current evidence, we suggest against the use of glucocorticoids for improving cardiopulmonary function in adult long COVID patients with persistent respiratory symptoms.

Among the agents studied, MMF has shown potential effectiveness in improving mMRC score, saturation (SpO_2), and pulmonary function test performance in long COVID patients with interstitial lung disease, but MMF was used in combination with corticosteroids and the study lacked proper control without immunosuppressive therapy [29]. JAK inhibitors have demonstrated benefits in reducing

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inflammatory biomarker levels and reducing time to liberation from oxygen support in long COVID patients in addition to steroids[30]. Tacrolimus has also been reported to improve pulmonary symptoms, although supporting evidence is limited to two case reports [31,32], and steroids were administered as basic treatment. After removing disease definition restrictions for long COVID, a living SR published in 2024 in *The BMJ* [34] described similar symptom improvement among patients receiving 8-week subcutaneous leronlimab compared with patients in placebo group.

Non-pharmacological treatment of pulmonary long COVID

Non-pharmacological interventions currently play a central role in the management of pulmonary long COVID. Among these, rehabilitation exercise programs (including breathing exercises, resistance training, strength training, aerobic exercise, and stretching), demonstrate the most consistent evidence for improving dyspnea, exercise capacity, lung function, and quality of life among long COVID patients without PEM, although the magnitude of benefit is generally modest and the certainty of evidence is low[33,34,52]. Specifically, the effects of inspiratory muscle training in improving dyspnea and lung function make it an optional intervention for patients with pulmonary long COVID, although high-quality evidence is still needed. Rehabilitation programs should be supervised, individualized, and adapted to specific symptom severity. The type and intensity of rehabilitation should be adapted to the patient's condition. For example, old patients or individuals who have been in intensive care for severe COVID-19 should receive more cautious and personalized rehabilitation plan developed in consultation with rehabilitation specialists, to avoid unnecessary or redundant interventions [12]. In contrast, evidence supporting cognitive behavioral therapy for pulmonary outcomes is lacking, as available studies primarily address fatigue and psychological symptoms rather than respiratory endpoints[53-55].

Studies have indicated that probiotics can inhibit various respiratory viruses and reduce viral load in vivo[56], while modulating both the innate and adaptive immune response [57] and regulating dynamic balance between pro-inflammatory and immunoregulatory cytokines[58], thereby attenuating respiratory infection severity. An in vivo study reported that fecal microbiota transplantation (FMT) from individuals with long COVID into germ-free mice induced lung inflammation in the absence of SARS-CoV-2[59]. In long COVID patients, probiotics have shown potential benefits for respiratory symptoms in small studies, including shortness of breath and cough-two manifestations that remain of particular concern to pulmonologists given their high prevalence and clinical relevance. However, substantial heterogeneity in formulations and limited availability of standardized preparations preclude clinical recommendation [22,60].

Despite increasing recognition of pulmonary long COVID, substantial evidence gaps remain. Respiratory endpoints such as dyspnea, pulmonary function, and imaging abnormalities are infrequently prioritized as primary outcomes in clinical trials. Given the marked heterogeneity in pulmonary long COVID phenotypes, further refinement and validation of distinct subtypes are needed. Notably, the identification of 129Xe MRI-defined clusters reflecting abnormalities in ventilation, membrane diffusion, and red blood cell gas transfer highlights the potential value of advanced imaging-based phenotyping in

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refining the diagnostic approach to pulmonary long COVID and enabling more mechanism-oriented clinical evaluation[61]. Data on high-risk populations, including older adults, critically ill survivors, and patients with fibrotic or vascular lung sequelae, are particularly limited. Future research should prioritize standardized pulmonary outcome measures, phenotype-driven trial designs, and biomarker-informed patient stratification to advance targeted prevention and treatment strategies for pulmonary long COVID.

Extra-Respiratory Manifestations of Long COVID

PICO question 1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID?

Evidence summary

Explanation

Preventing long COVID is of considerable clinical importance, and current research primarily focuses on vaccination and pharmacologic interventions administered during the acute phase of SARS-CoV-2 infection (within 4 weeks of infection). Several SRs have suggested that vaccination, whether administered before or after SARS-CoV-2 infection, may reduce the risk or odds of long COVID [38-41]. However, high heterogeneity across studies and the lack of large-scale RCTs result in very low certainty of evidence. Vaccine effectiveness is influenced by age, comorbidities, individual immune responses, and viral variants [66], and the optimal timing and dosing remain unclear. Therefore, vaccination decisions should follow local public health recommendations and individual clinical assessments.

Pharmacologic treatment during the acute phase may also reduce the risk of long COVID [67,68]. Our updated SR supports a potential benefit of selected antiviral agents, although the certainty of evidence remains low to very low, and varies by drug type, target symptoms, and patient risk profiles. Evidence does not support routine use of non-antiviral agents for long COVID prevention. Clinical decision-making should integrate individual risk profiles, drug-drug interactions, and practical contextual factors. Several clinical trials are currently underway and are expected to provide further insights into preventive strategies for long COVID [69-71].

PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?

Evidence summary

Explanation

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Persistent viral presence has been proposed as one potential mechanism for long COVID [2]. Nirmatrelvir-ritonavir (NMV/r), an inhibitor of SARS-CoV-2 main protease, is approved for the treatment of COVID-19 in adult patients at risk of progression to severe COVID-19 [72]. Case series and reports [73-76] have suggested that antiviral agents, including NMV/r may alleviate symptoms in some patients with long COVID; however, high-quality evidence remains limited. The included RCT demonstrated that NMV/r was safe but did not result in significant symptomatic improvement in long COVID patients with symptoms nearly 1.5 years after their acute infection and negative stool viral RNA at baseline.

PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?

Evidence summary

Explanation

In the context of long COVID, most therapeutic monoclonal antibodies are neutralizing agents that target specific viral epitopes to inhibit viral entry and replication. Evidence supporting their use in long COVID is extremely limited. Our SR found no robust evidence demonstrating clinical benefit in the general long COVID population. One case report suggested possible symptomatic improvement including pulmonary symptom in immunocompromised patients. The status of immunocompromise can potentially indicate viral reactivation or re-infection[16], providing a biological rationale for why exogenous neutralizing antibodies may facilitate symptom improvement in this population. Given the lack of efficacy and safety data, high cost, and uncertainty regarding optimal patient selection, current evidence does not support a recommendation for their routine use in long COVID. Several clinical trials investigating monoclonal antibodies, including AER002 and casirivimab-imdevimab, are currently ongoing [77-80].

PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?

Evidence summary

Explanation

Probiotics may influence host immunity and inflammation through modulation of the gut microbiota[82], and have shown potential protective effects in respiratory diseases, mainly through direct antiviral activity and immune modulation. For long COVID patients, some studies [22,60,81] suggest that probiotics might help alleviate persistent symptoms [83]. However, these studies differ significantly in terms of probiotic strains, dosages, and treatment duration, and individual responses may vary based on baseline microbiome composition and other host factors. Moreover, most probiotic formulations are

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not readily available as pharmaceutical preparations. As a result, evidence remains insufficient to support routine probiotic use in long COVID.

PICO question 5: In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?

Evidence summary

Explanation

Glucocorticoids, as a class of steroid hormones, regulate multiple physiological processes and are widely used during acute COVID-19, but recommendations for their use in long COVID are limited. Regarding olfactory disorders, both local and systemic glucocorticoids therapy have been shown to improve olfactory scores in long COVID adult patients. However, no significant differences were observed in patients' self-reported olfactory recovery. Furthermore, the magnitude of score improvement does not meet the threshold for clinical significance (defined as an increase of more than 2 points on a 10-point scale). A recent small-scale trial suggested that endoscopic topical therapy is feasible and safe for treating olfactory dysfunction post SARS-CoV-2 infection, supporting future larger trials[88]. Two SRs [89,90] on inhaled glucocorticoids for persistent post-COVID olfactory dysfunction were identified; although their definitions of long COVID differed from the WHO adopted in this guideline, their findings are largely consistent with ours.

PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?

Evidence summary

Explanation

Immunomodulators are agents that modulate the immune system by either enhancing or suppressing immune responses. Currently, clinical guidelines do not provide established recommendations for the use of immunomodulator in long COVID management. This guideline identified six studies suggesting potential benefits of immunomodulators in adult long COVID patients, but the overall evidence remains limited and restricted to combination with glucocorticoid steroids. Given the heterogeneity of available studies, small sample sizes, and a lack of consensus regarding key aspects such as dosing regimens, duration, combination therapy, and safety precautions, no recommendation can be made for or against immunomodulators in long COVID. Clinical trial registries indicate that multiple studies are underway investigating immunomodulators such as anakinra, intravenous immunoglobulin, and baricitinib for adult long COVID patients [93-98].

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PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?

Evidence summary

Explanation

CBT, a therapeutic approach centered on modifying cognitive patterns and behaviors, has shown potential in alleviating certain symptoms associated with long COVID by reshaping cognitive frameworks, improving emotional regulation, and promoting behavioral activation[53]. However, current evidence was limited, with studies featured small sample sizes, inconsistent outcome measures, and a lack of large-scale RCTs. Our meta-analysis showed high heterogeneity across studies, with most outcomes showing non-significant improvements supported by low-certainty evidence. Fatigue was the only symptom for which CBT demonstrated statistically significant benefits. Notably, current assessment tools largely focus on somatic symptoms and functional limitations, potentially neglecting cognitive and social dimensions of mental fatigue. These tools are subject to bias and often fail to clearly distinguish between physical and mental fatigue. In addition, while CBT shows potential for anxiety, depression, and social functioning in patients with long COVID, its overall effectiveness remains inconclusive and requires further validation through robust clinical trials [53-55].

Given the frequent co-occurrence of psychological symptoms with persistent physical discomfort, reduced quality of life, and other complex presentations in long COVID, pharmacological interventions may offer faster and more effective relief during acute or severe episodes. As such, this guideline suggests CBT as an adjunctive therapy, particularly for patients who do not respond adequately to pharmacological treatments. From a safety perspective, CBT is a non-pharmacological intervention with a favorable risk profile. No serious adverse events have been reported. In addition, the implementation of CBT requires trained psychotherapists, and availability of such resources may be limited in certain regions, potentially restricting its practical application. To date, existing guidelines have not provided clear recommendations regarding the use of CBT in long COVID management. Although a 2023 expert consensus endorsed CBT as first-line treatment for insomnia in long COVID patients [100], due to the lack of evidence, this guideline does not offer a recommendation regarding the use of CBT in the context of insomnia.

Despite these limitations, CBT remains a promising therapeutic option for long COVID patients, particularly those experiencing prominent fatigue and psychological symptoms, due to its potential benefits and low risk of harm.

PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?

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Evidence summary

Explanation

Rehabilitation exercise is a key component of long COVID management for patients without PEM. SRs included in this guideline indicate that rehabilitation interventions may improve physical capacity and quality of life in long COVID patients [33,34,52]. However, the magnitude of benefit is generally modest, and the certainty of most evidence is low to very low [33,34]. Patients can receive these interventions from healthcare providers online or face-to-face, with telerehabilitation and in-person rehabilitation demonstrating differential effects across specific outcomes [52].

A study demonstrated the cost-effectiveness of rehabilitation programs for long COVID patients previously hospitalized for COVID-19, with data on non-hospitalized patients remain limited[101]. Based on current evidence, we suggest that patients with long COVID should receive personalized rehabilitation training under the guidance and supervision of healthcare professionals, particularly those with a history of hospitalization for COVID-19 or symptoms that significantly affect activities of daily living and functional status. These recommendations are consistent with existing long COVID guidelines [10-12,102]. Patients with PEM or post-exertional symptom exacerbation (PESE) often demonstrate limited tolerance to rehabilitation training. In such cases, the main focus should be on activity management (rest/pacing) rather than physical exertion [103]. More researches on this topic are needed.

Suggestions for future research

To advance the understanding and management of long COVID, a coordinated research agenda is urgently needed. Uptake of standardized long COVID definition and core outcome is crucial to ensure comparability across studies and strengthen evidence base. Despite increasing attention, data remain sparse, particularly among children and vulnerable populations, hindering the development of inclusive and equitable clinical guidance. High-priority research areas include elucidating disease burden and subtypes, identifying risk factors and preventive strategies, improving diagnostic and evaluation tools, and generating robust evidence on treatment effectiveness through well-designed clinical trials (Appendix 1 Table S2).

Limitations

This guideline has several limitations. First, among the 26 steering and consensus group members, nine (35%) are affiliated with institutions in Europe or North America; Patient involvement was limited to three representatives, potentially missing perspectives from low- and middle-income countries. While the majority of panel members were based in Asia, we do not believe this compromises the methodological rigor of the guideline development process. Second, existing systematic reviews and one monoclonal antibody studies using non-WHO definitions of long COVID were included, broadening the evidence base but adding heterogeneity. Third, most evidence came from observational studies of

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variable quality, leading to low to very low certainty of evidence for many recommendations due to bias, inconsistency (e.g., high I^2 values in meta-analyses), and imprecision. Fourth, language restrictions were not imposed during the evidence search. However, inclusion criteria was limited to English publications, which may introduce language bias. Finally, the guideline mainly addresses adults, and does not cover emerging variants or outcomes beyond the search cutoff.

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Table 1 Eight PCIO questions for prevention and treatment of long COVID

PICO questions	Population	Intervention	Comparison	Main outcomes
PICO question 1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID?	Adult patient with long COVID	Vaccines or pharmacological interventions (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.)	No vaccines or no pharmacological interventions	Risk of developing long COVID and related symptoms
PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?		Antiviral drug	No antiviral drug	Long COVID symptoms
PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?		Monoclonal antibodies	No monoclonal antibodies	Long COVID symptoms, time to symptom resolution, safety
PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?		Probiotics (multi-species)	No probiotics / placebo	Long COVID symptoms, quality of life, safety
PICO question 5: In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?		Glucocorticoids (systemic or local administration)	No glucocorticoids	Cardiopulmonary function, olfactory recovery, safety
PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?		Immunomodulators including mycophenolate mofetil (MMF), Janus kinase (JAK) inhibitors, tacrolimus, others)	No immunomodulators	Pulmonary function, systemic symptoms, safety
PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?		Cognitive behavioral therapy (CBT)	No CBT / usual care	Fatigue, anxiety, depression, social functioning, stress, self-efficacy

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PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?		Rehabilitation exercise (e.g., breathing, resistance, aerobic, strengthening, or combined interventions; delivered face-to-face or via telerehabilitation)	No rehabilitation exercise / usual care	Exercise capacity, dyspnea, quality of life, safety

Table 2 Working group recommendations for PCIO questions on the prevention and treatment of long COVID

Recommendations	Strength of recommendation	Certainty of evidence	Remarks
Prevention			
PICO question 1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID?			
Irrespective of prior SARS-CoV-2 infection, vaccination against COVID-19 is suggested to prevent the development of long COVID. The specific vaccination dosage and timing should be determined by taking into account the availability, requirements of the local department, and patient's condition, while closely monitoring any potential adverse events.	Conditional	Very low	Currently, COVID-19 antivirals are primarily indicated for treating acute SARS-CoV-2 infection, with indications varying by drug. The recommendation should not be interpreted as superseding or contradicting current clinical guidelines for antiviral use during acute phase. Use of antivirals to prevent long COVID should involve shared decision-making, considering the patient's clinical condition, risk factors, and the specific indications and availability of each agent.
We suggest using antiviral drugs during the acute phase of COVID-19 to prevent the development of long COVID, especially in patients who are at higher risk.	Conditional	Low	
Treatment			
PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?			
We suggest against the use of nirmatrelvir-ritonavir for the treatment of long COVID in adult patients.	Conditional	Low	Given the lack of evidence for other antiviral drugs in long COVID treatment, this recommendation primarily pertains to nirmatrelvir-ritonavir.
PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?			

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We have no recommendation for or against the use of neutralizing monoclonal antibodies in adult patients with long COVID.	/	Very low	
PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?			
We suggest using the multi-species probiotics to alleviate symptoms for adult patients with long COVID.	Conditional	Moderate	The availability of standardized preparations varies across settings. The recommendation should be interpreted in the context of local availability.
PICO question 5: In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?			
We suggest against the use of glucocorticoids for improving cardiopulmonary function in adult long COVID patients with persistent respiratory symptoms.	Conditional	Very low	
We suggest against the use of glucocorticoids for the treatment of olfactory disorders in adult patients with long COVID.	Conditional	Low	
PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?			
We have no recommendation for or against the use of immunomodulators (mycophenolate mofetil, JAK inhibitors, tacrolimus) for adult patients with long COVID who have interstitial lung disease.	/	Very low	
PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?			
We suggest using cognitive behavioral therapy in adult patients with long COVID to alleviate fatigue symptoms.	Conditional	Low	The recommendation of cognitive behavioral therapy does not imply psychological causation of fatigue among long COVID patients.
PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?			
We suggest personalized rehabilitation training for adult patients with long COVID, following appropriate screening to exclude those with post-exertional malaise (PEM), and under the guidance and supervision of healthcare professionals	Conditional	Low	

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Table 3 Summary of evidence for the prevention and treatment of respiratory manifestations of long COVID

Study included	Study participants	Summary
PICO question 1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID?		
Acute-phase drugs (within 4 weeks post-infection)		
Updated meta-analysis including fifty studies (six RCTs, two non-randomized trials, 40 cohort studies, and two case-control studies) (Included studies listed in Appendix 2)		
Updated meta-analysis	7,060,422 patients	Antivirals had no significant effect on dyspnea (OR: 0.94, 95% CI: 0.72-1.23).
PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?		
One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of shortness of breath of long COVID (Change of severity score from baseline: -20.60 vs -24.50).
PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?		
One case report	One immunocompromised	This case report showed the patient who received casirivimab-imdevimab (dosage not reported) experienced

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	adult patient with long COVID (thymoma with pleural metastasis) who had cough	complete resolution of cough, suggesting potential benefit in immunocompromised individuals [21].
PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?		
One RCT	463 adult patients with long COVID (285 patients had shortness of breath and 249 patients had cough)	Compared with placebo, probiotics (e.g., SIM01, an oral synbiotic containing three bacterial strains) alleviated [22]: • Shortness of breath (OR: 1.87, 95% CI: 1.16-3.03) Cough (OR: 1.85, 95% CI: 1.04-3.29)
PICO question 5: In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?		
Six studies (1 RCT and 5 observational study)	3,189 long COVID adults with respiratory symptoms	• Results were inconsistent from observational studies: some reported modest improvements in 6-minute walking distance (6MWD)[23,24], forced expiratory volume in 1 second (FEV_1)[25], forced vital capacity (FVC)[23-25], and diffusion capacity of carbon monoxide (DLCO)[25,26], while others found no significant effect[26,27]. However, caution is warranted in application considering the inconsistency.

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		• One RCT [28] comparing high-dose vs. low-dose prednisone found no significant difference in lung function or radiological outcomes. • Overall, findings were heterogeneous, with limited and inconsistent benefits. Note: Oral glucocorticoids dosages for the treatment of persistent respiratory symptoms in long COVID are shown in Appendix 1 Table S1.
PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?		
One cohort study, one case-control study, and two case reports	95 adult patients with long COVID	• MMF: In a cohort study[29], patients with interstitial lung disease receiving immunosuppressive therapy (glucocorticoid alone, or MMF combined with corticosteroids) showed improvements in lung function and oxygenation over 6 months, with good tolerability. However, the study did not include patients who did not receive immunosuppressive therapy as control.
One case-control study	41 adult patients with long COVID	• JAK inhibitors: A case-control study[30] of patients with hypoxia, active inflammation and lung involvement found that adding JAK inhibitors (baricitinib or tofacitinib) to glucocorticoid steroids allowed more rapid corticosteroid tapering and fewer

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		adverse events, though no difference in oxygen support requirements was observed.
Two case reports	2 adult patients with long COVID	• Tacrolimus: Two case reports [31,32] reported improved pulmonary function and successful withdrawal from oxygen therapy with combination of corticosteroid and tacrolimus.
PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?		
None.		
PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?		
Two SRs	14 RCTs, 1,244 adult patients with long COVID	Compared with usual care, rehabilitation interventions (breathing exercises [six trials, 815 participants], breathing exercises in combination with resistance and/or aerobic training [five trials, 298 participants], strengthening and aerobic exercises [two trials, 71 participants], or aerobic exercises alone [one trial, 60 participants]) may improve[33]: • Dyspnea (SMD: -1.00, 95% CI: -1.94 to -0.10) No significant differences were observed in: • FEV_1 (SMD: -0.16; 95% CI: -0.42 to 0.11) • FVC (SMD: -0.13, 95% CI: -0.38 to 0.12)

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	Eight RCTs, 985 adult patients with long COVID	Compared with usual care, inspiratory muscle training may improve[34]: • Dyspnea (MD: 1.10, 95% CI: 0.44 to 1.76) • Breathlessness and activities (MD: 2.40, 95% CI: -2.66 to 7.46) Compared with standard physiotherapy, multicomponent exercise of progressively increasing intensity may improve [34]: • Dyspnea (mMRC dyspnea scale: MD: -0.76, 95% CI: -1.20 to -0.32, low certainty of evidence; multidimensional dyspnea profile: MD: -18.61, 95% CI: -27.40 to -9.82) • Breathing discomfort (MD: -1.74, 95% CI: -2.79 to -0.69) • Emotional response to dyspnea (MD: -6.95, 95% CI: -12.44 to -1.46) Sensory dimension of dyspnea (MD: -9.93, 95% CI: -14.56 to -5.30)

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Table 4 Summary of evidence on vaccination and acute-phase pharmacological interventions for the prevention of long COVID

Study included	Study participants	Summary
Vaccination		
Three high-quality SRs including 44 observational studies [38-40]		
Risk/Odds of long COVID		
Three SRs	14,842,052 participants	● Before infection: Two doses significantly reduced odds of long COVID (OR=0.76, 95% CI: 0.65-0.89, I^2=93.4%); one or three doses showed no significant benefit.[40] Another SR suggested that the prevalence of post-COVID-19 syndrome symptoms was lower in vaccinated individuals (9.5%) compared to unvaccinated (14.6%), with a notable decrease in activity-limiting symptoms (adjusted OR 0.59, 95% CI 0.48-0.73).[38] ● After infection: One dose post-infection reduced odds of long COVID (OR=0.85, 95% CI: 0.73-0.98, I^2=77.9%); two doses showed no significant effect.[40] Another SR showed that the odds ratio of developing long covid with one dose of vaccine ranged from 0.22 to 1.03; with two doses, odds ratios were 0.25-1; with three doses, 0.16; and with any dose, 0.48-1.01. Five studies reported on vaccination after infection, with odds ratios of 0.38-0.91.[39]

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		● Vaccine types: Both mRNA (BNT162b2) and adenoviral vector vaccines (ChAdOx1 nCoV-19) reduced risk of long COVID, regardless of the timing of administration.[40]
Acute-phase drugs (within 4 weeks post-infection)		
Updated meta-analysis including fifty studies (six RCTs, two non-randomized trials, 40 cohort studies, and two case-control studies) (Included studies listed in Appendix 2)		
Risk/Odds of long COVID		
Updated meta-analysis	7,060,422 patients	● Antivirals: Overall, antiviral use was associated with reduced odds of long COVID (19 studies, OR=0.69, 95% CI: 0.64-0.74, I^2=98.0%) (Appendix 2). Nirmatrelvir/ritonavir, molnupiravir, and favipiravir showed significant benefit; whereas remdesivir, ensitrelvir, and azvudine did not. ● Monoclonal antibodies: No significant effect on long COVID or individual symptoms. ● Corticosteroids: No significant preventive effect on long COVID. ● Metformin: One RCT showed reduced risk (HR: 0.59, 95% CI: 0.39-0.89) of long COVID; two large retrospective studies also supported the benefit.[62-64]
Respiratory system		
Updated meta-analysis	7,060,422 patients	● Antivirals had no significant effect on dyspnea (OR: 0.94, 95% CI: 0.72-1.23).

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Neuropsychological system		
Updated meta-analysis	7,060,422 patients	● Antivirals reduced odds of cognitive impairment (OR: 0.83, 95% CI: 0.74-0.94) and sleep disorders (OR: 0.94, 95% CI: 0.89-0.98).
Musculoskeletal system		
Updated meta-analysis	7,060,422 patients	● Antivirals had no significant effect on myalgia (OR: 0.88, 95% CI: 0.70-1.10).
Systemic symptoms		
Updated meta-analysis	7,060,422 patients	● Antivirals reduced odds of fatigue (OR : 0.90, 95% CI : 0.84-0.97).
Safety: No significant adverse events were reported in any of the included RCTs, despite the use of different vaccinations. For monoclonal antibodies, one study suggested potential pulmonary fibrosis risk with rituximab.[65]		

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Table 5 Summary of evidence on antiviral drugs for the treatment of long COVID

Study included in SR	Study participants	Summary
One RCT[20]		
Gastrointestinal system		
One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of gastrointestinal symptom of long COVID (Change of severity score from baseline: -19.60 vs -11.30).
Neuropsvchological system		
One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of brain fog of long COVID (Change of severity score from baseline: -28.40 vs -47.20).
Musculoskeletal system		

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One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of body aches of long COVID (Change of severity score from baseline: -22.50 vs -20.80).
Cardiovascular system		
One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of cardiovascular of long COVID (Change of severity score from baseline: -23.50 vs -20.80).
Systemic symptoms		
One RCT[20]	155 adult patients with history of COVID-19 and long COVID (No SARS-CoV-2 RNA detected in baseline stool samples)	At 10 weeks, there was no statistically significant difference between the nirmatrelvir-ritonavir group and the placebo-ritonavir group in the severity of fatigue of long COVID (Change of severity score from baseline: -23.50 vs -43.40).
Safety: The incidence of adverse events was similar between the nirmatrelvir-ritonavir and placebo-ritonavir groups.		

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Table 6 Summary of evidence on monoclonal antibody for the treatment of long COVID

Study included	Study participants	Summary
One case report [21]		
Systemic symptoms		
One case report	One immunocompromised adult patient with long COVID (thymoma with pleural metastasis) who had fever and malaise	This case report showed the patient who received casirivimab-imdevimab (dosage not reported) experienced complete resolution of fever and malaise, suggesting potential benefit in immunocompromised individuals [21].
Safety: No adverse events were reported in the study.		

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Table 7 Summary of evidence on probiotics for the treatment of long COVID

Study included	Study participants	Summary
Three RCTs (two published in peer-reviewed journals and one available as a preprint) [22,60,81]		
Musculoskeletal system		
One RCT	463 adult patients with long COVID (291 patients had muscle pain, 262 patients had joint pain and 189 patients had inability to exercise)	Compared with placebo, probiotics (e.g., SIM01, an oral synbiotic containing three bacterial strains) alleviated [22]: • Joint pain (OR: 2.10, 95% CI: 1.27-3.47) • Inability to exercise (OR: 2.32, 95% CI: 1.23-4.38) • Muscle pain (OR: 1.71, 95% CI: 1.07-2.72)
Neuropsychological system		
One RCT	463 adult patients with long COVID (323 patients had difficulty in concentration, 305 patients had insomnia, and 369 patients had memory loss)	Compared with placebo, probiotics (e.g., SIM01, an oral synbiotic containing three bacterial strains) alleviated [22]: • Difficulty in concentration (OR: 2.64, 95% CI: 1.69-4.14) • Memory loss (OR: 1.97, 95% CI: 1.27-3.04) • Insomnia (OR: 1.76, 95% CI: 1.12-2.77)
Gastrointestinal system		

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One RCT	463 adult patients with long COVID (374 patients had gastrointestinal upset)	Compared with placebo, probiotics (e.g., SIM01, an oral synbiotic containing three bacterial strains) alleviated [22]: • Gastrointestinal upset (OR: 1.99, 95% CI: 1.30-3.05)
Systemic symptoms		
Three RCTs	463 adult patients with long COVID (298 patients had general unwellness and 398 patients had fatigue)	Compared with placebo, probiotics (e.g., SIM01, an oral synbiotic containing three bacterial strains) alleviated [22]: • Fatigue (OR: 2.27, 95% CI: 1.52-3.40) • General unwellness (OR: 2.36, 95% CI: 1.43-3.90)
	38 patients with long COVID	This RCT [88] assessed VSL#3® (consisting Lactobacilli, Bifidobacteria, and Streptococcus thermophilus), and reported a greater reduction in long COVID-related fatigue compared with placebo, measured by Chalder Fatigue Scale Likert score at 4 weeks.
	26 patients with long COVID	This RCT [81] evaluated a synbiotic mixture (including L. rhamnosus DSM 32550, Humiome® L. plantarum DSM 34532, B. lactis DSM 32269, and B. longum DSM 32946). After 3 months of intake, participants showed significant reductions in:

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		• General fatigue (P = 0.03) • Physical fatigue (P = 0.03) • Mental fatigue (P = 0.03)
Safety: No significant adverse events were reported in any of the included RCTs, despite the use of different probiotics.		

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Table 8 Summary of evidence on glucocorticoids for the treatment of long COVID

Study included	Study participants	Summary
10 studies (4 RCTs, 3 cohort studies, 3 before-and-after studies); Updated meta-analysis conducted for olfactory disorders		
Gustatory		
Three RCTs and one before-and-after study.	225 long COVID adults with olfactory dysfunction	Olfactory disorders: • Meta-analysis from three RCTs[84-86] showed modest improvement in olfactory scores (mean difference: 1.66, 95% CI: 1.10-2.23), but no effect on self-reported recovery (Appendix 2). One study[87] combining glucocorticoids with other therapies suggested some benefit, especially when treatment started early, but results were not robust.
Safety: • No serious adverse events were reported across studies. • Local injections occasionally caused mild bleeding and pain. • Long-term safety (e.g., risk of infections, osteoporosis, gastrointestinal bleeding) was not adequately studied.		

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Table 9 Summary of evidence on immunomodulators for the treatment of long COVID

Study included	Study participants	Summary
One cohort study, one case-control study, one case series, and three case reports [29-32,91,92]		
Musculoskeletal system		
One case series	5 adult patients with long COVID	• Methotrexate and leflunomide were tried in three patients with joint pain [92], but two discontinued due to adverse events.
Systemic symptoms		
One case report	1 adult patients with long COVID	• Anakinra was reported effective and well tolerated in a case of steroid-dependent multisystem inflammatory syndrome [91].
Safety: No serious adverse events were reported for MMF, JAK inhibitors, or tacrolimus in the available studies.		

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Table 10 Summary of evidence on cognitive behavioral therapy for the treatment of long COVID

Study included in SR	Study participants	Summary
Mental Health		
Updated meta-analysis including two before-after study	93 adult patients with long COVID	Compared with baseline (e.g., pre-post comparison), CBT had no significant improvement in anxiety symptoms and depression [53,54]: • Anxiety (SMD: 0.32, 95% CI: -0.19 to 0.83) • Depression (SMD: 0.25, 95% CI: -0.10 to 0.61)
One Quasi-experimental study	40 adult patients with long COVID	Compared with usual care: • Stress :Due to the limited number of studies for each outcome, meta-analysis was not feasible. But the results showed that CBT significantly reduced stress level. [55].
Systemic symptoms		
Updated meta-analysis including one RCT and one before-after study	87 adult patients with long COVID	Compared with baseline (e.g., pre-post comparison), CBT had no significant improvement in social functioning [54,99]: • Social Functioning (SMD: 5.63, 95% CI: -3.91 to 15.17; $I^2 = 99.4\%$)

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Updated meta-analysis including one RCT and one before-after study	183 adult patients with long COVID	Compared with no CBT or baseline (pre-post comparison), CBT significantly alleviated general fatigue symptoms [53,99]: • Fatigue (SMD: 4.01, 95% CI: 1.17 - 6.85)
One Quasi-experimental study	40 adult patients with long COVID	Compared with no CBT: • Self-efficacy: Due to the limited number of studies for each outcome, meta-analysis was not feasible. But the result showed that CBT significantly increased self-efficacy [55].

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Table 11 Summary of evidence on rehabilitation exercise for the treatment of long COVID

Study included	Study participants	Summary
Three SRs (one SR, one living SR, one network meta-analysis) [33] [34] [52]		
Musculoskeletal system		
One SR	14 RCTs, 1,244 adult patients with long COVID	Compared with usual care, rehabilitation interventions (breathing exercises [six trials, 815 participants], breathing exercises in combination with resistance and/or aerobic training [five trials, 298 participants], strengthening and aerobic exercises [two trials, 71 participants], or aerobic exercises alone [one trial, 60 participants]), no significant differences were observed in [33]: • Lower limb muscle strength and endurance (SMD: -0.77, 95% CI: -1.94 to 0.44)
Neuropsychological system		
One SR	Eight RCTs, 985 adult patients with long COVID	Compared with standard physiotherapy, multicomponent exercise of progressively increasing intensity may have little or no important effect on [34]: • Mental health (MD: 2.06, 95% CI: -3.52 to 7.64) Compared with standard physiotherapy, the active breathing techniques may have little or no important effect on [34]: • Mental health (MD: 0, 95% CI: -3.69 to 3.69)
Systemic symptoms		

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Three SRs	14 RCTs, 1,244 adult patients with long COVID	Compared with usual care, rehabilitation interventions (breathing exercises [six trials, 815 participants], breathing exercises in combination with resistance and/or aerobic training [five trials, 298 participants], strengthening and aerobic exercises [two trials, 71 participants], or aerobic exercises alone [one trial, 60 participants]) may improve [33]: • Exercise capacity (SMD: -0.56, 95% CI: -0.87 to -0.22) • Quality of life (SMD: -0.41, 95% CI: -0.73 to -0.06)
	Eight RCTs, 985 adult patients with long COVID	Compared with usual care, inspiratory muscle training may improve [34]: • Quality of life (MD: -1.30, 95% CI: -5.90 to 3.30) Compared with standard physiotherapy, multicomponent exercise of progressively increasing intensity may improve [34]: • Physical function (MD: 6.96, 95% CI: 2.7 to 11.22) Compared with standard physiotherapy, the active breathing techniques may improve [34]: • Fatigue (MD: -9.97, 95% CI: -11.17 to -8.77) Compared with continuous aerobic exercise, intermittent aerobic exercise probably improves [34]: • Physical function (MD: 3.80, 95% CI: 1.12 to 6.48)

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		Compared with high-intensity aerobic strength training, low-intensity aerobic strength training may improve [34]: • Quality of life (8 weeks, MD: 8.50, 95% CI: 8.08 to 8.92; 24 weeks, MD: 10.40, 95% CI: 9.97 to 10.83)
	10 RCTs, 765 adult patients with long COVID	Compared with usual care, face-to-face rehabilitation may both improve[52]: • Physical function (MD: 29.06, 95% CI: 2.76 to 55.37) • Quality of life (physical domain: MD: 6.58, 95%CI: -0.14 to 13.30; Mental domain: MD: 2.14, 95%CI: -4.36 to 8.64) Compared with usual care, telerehabilitation may both improve [52]: • Physical function (MD: 3.13, 95% CI: -26.57 to 32.82) may both improve physical function • Quality of life (physical domain: MD: 5.17, 95%CI: -1.52 to 11.86; Mental domain: MD: 7.12, 95%CI: 0.61 to 13.64) Indirect comparison results showed that face-to-face rehabilitation was more effective than telerehabilitation in [52]: • Physical function evaluated by 6MWT or 30 seconds sit-to-stand test. (MD: -25.94, 95% CI: -65.61 to 13.73)

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		• The physical domain of quality of life (MD: 1.41, 95%CI: -8.07 to 10.90) Whereas face-to-face rehabilitation was less effective than telerehabilitation in: • The mental domain of quality of life (MD: -4.98, 95%CI -14.19 to 4.22)
Safety: No significant adverse events were reported in any of the included SR.		

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Appendix 1

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Methods Supplement

Establishment of the guideline working group

The guideline working group was established in December 2023 and comprised 60 members from 10 area of expertise and 10 countries (China, the United States, Switzerland, France, Spain, Japan, Korea, Singapore, Canada, and the United Kingdom). The group included experts in infectious disease, respiratory disease, public health, epidemiology, mental health, neurology, nephrology, pediatrics, cardiovascular disease, and guideline methodology. Members were allocated to four specific groups: (1) a steering group consisting of two members, including the chair (Bin Cao), and a chief methodologist (Yaolong Chen); (2) a consensus group of 24 members; (3) an evidence synthesis and evaluation group of 34 members; and (4) a secretariat of three members. Appendix 1 Table S3 presents detailed information about the guideline working group. A patient partner group comprised of two adult patients with long COVID was also introduced.

Declaration and management of conflict of interests

All guideline members have been required to disclose any potential conflict of interests prior to participation in the working group and engagement in the detailed process and meeting. This included completing a standard declaration of relevant financial and non-financial interest form. All declarations were reviewed and assessed based on predefined criteria to determine the severity and relevance of any potential conflict of interests. When panel members had declared interests related to a particular recommendation, they were asked to abstain from voting on that specific recommendation. Appendix 1 Table S4 presents a summary of the disclosures and information on how conflict of interests was managed.

Scope of the guideline

The target population of the guideline are adults with long COVID, as defined by the WHO. The primary users include clinical physicians, clinical pharmacists, nurses, and general practitioners in community healthcare institutions worldwide. The guideline mainly covers the prevention and treatment of long COVID.

Development of clinical questions

The secretariat proposed an initial draft list of clinical questions following a two-step process: (1) clinical physicians generated preliminary questions in the PICO (population, intervention, comparison, and outcome) format, leveraging their clinical expertise in long COVID, (2) guideline methodologists complemented and refined

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the list by reviewing existing long COVID guidelines and high-quality systematic reviews. Finally, the steering and consensus group reviewed, prioritized, and finalized eight clinical questions for the guideline.

Evidence retrieval, evaluation, and synthesis

We developed recommendations for each clinical question based on evidence derived from existing, updated, or newly conducted systematic reviews (SRs). We initially performed systematic searches for published SRs relevant to each question. Eligible SRs were assessed according to the following criteria: relevance to clinical question, timeliness, methodological quality (using Assessing the Methodological Quality of Systematic Reviews tool), and adequacy of reported results[1]. Where necessary, updated searches were conducted to ensure the inclusion of the most current evidence. For questions with no available SR(s), the evidence synthesis and evaluation group conducted new SRs to synthesize the latest evidence. We systematically searched the following databases: MEDLINE (via PubMed), Web of Science Core Collection, Embase, the Cochrane Library, China Biology Medicine disc (CBM), WanFang Data, and Epistemonikos COVID-19 L·OVE platform, covering the period from January 1, 2020, to November 30, 2024. We also searched preprint platforms (Medrxiv, SSRN, Researchsquare), clinical trial registers, and Google to identify studies not indexed in the aforementioned databases. For PICO question 8 (cognitive behavioral therapy), we also searched the PsycINFO. Language restrictions were not imposed during the evidence search. However, when applying the inclusion and exclusion criteria, we limited eligible studies to those published in English. We prioritized studies that defined long COVID according to the WHO criteria (Appendix 1 Table S5).[2] In the absence of such studies, we considered evidence using alternative definitions.

Risk of bias for included studies was assessed using appropriate tools according to study design: the Cochrane Risk of Bias tool (ROB 1.0) for randomized controlled trials (RCTs), Risk Of Bias In Non-randomised Studies-of Interventions (ROBINSI) for non-randomised trials, Newcastle-Ottawa Scale (NOS) for cohort studies and case-control studies, Agency for Healthcare Research and Quality (AHRQ) tool for cross-sectional studies, and checklist proposed by Murad et al for case series and case reports. Meta-analyses were performed using Stata software (Stata Corp LLC).[3] The GRADE approach was applied to rate the certainty of evidence and determine the strength of recommendations (Appendix 1 Table S6).[4] Appendix 2 provided detailed information for each question, including study selection flow diagrams, list of included studies, risk of bias assessments, and meta-analysis results (where applicable).

To ensure methodological rigor and minimize bias, all key steps of the systematic reviews and meta-analyses were conducted in duplicate by independent reviewers. Specifically, this included title and abstract screening, full-text screening, data extraction, and risk of bias assessments.

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Formulation of the recommendations

A total of 10 recommendations were formulated after three rounds of Delphi survey, considering the certainty of the evidence, balance of benefits and harms, patient values and preferences, acceptability, and cost considerations (Table 2) [5]. We used the GRADE Evidence-to-Decision (EtD) framework to structure the decision-making process (Appendix 3). From April 1, 2025, to April 21, 2025, the first round of Delphi survey was conducted to gather opinions from guideline consensus group. The questionnaire was sent to 24 international experts, with a response rate of 75%. Responses from this first round were independently evaluated by secretariat and subsequently reported to the evidence synthesis and evaluation group, which used the findings to develop revised recommendations for the second round (from May 1, 2025 to May 11, 2025). Consensus group members were asked to rate their level of agreement with each recommendation on either agree, disagree and unsure. The percentage of agreement was recorded, and consensus was predefined as at least 70% agreement among the panel members. A total of 104 suggestions were collected by panelists throughout the two rounds of Delphi. The final version of recommendations was approved after the third round of Delphi survey (from September 5, 2025 to September 10, 2025). The Delphi survey was conducted anonymously, as experts were unaware of each other's responses across three rounds. The details of guideline consensus group members participation and the voting results across three rounds of Delphi surveys are shown in Appendix 1, Table S7 and Table S8, respectively.

Investigation of patients' value and preference

We invited two long COVID patient representatives to provide feedback and external input on the recommendations [6], especially for those supported with low and very low certainty of evidence. The secretariat was responsible to communicate with patients, send the questionnaire, and collect their feedback. All results were presented to the guideline working group for a full consideration while formulating recommendations.

Dissemination and implementation of the guideline

After the publication, we plan to disseminate and promote the guideline through the following strategies: (I) introduce the guideline in specialized journals, websites, and academic platforms, (II) strategically organize special session to promote the guideline, ensuring the clinical physicians, clinical pharmacists, and nurses gain a comprehensive understanding and apply the guideline accurately, (III) establish dedicated outpatient clinics in various levels of healthcare institution to meet the healthcare needs of patients, (IV) monitor the dissemination of the guideline and conduct an evaluation within 1 to 2 years to release the impact on the diagnosis, treatment, and management of long COVID. This guideline is planned to be updated within the next three years to incorporate updated evidence.

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To evaluate the guideline's dissemination and impact on long COVID diagnosis, treatment, and management within 1-2 years, we will: (1) conduct annual surveys among healthcare professionals and stakeholders to assess adoption, utility, and barriers; (2) track usage metrics (e.g., downloads, citations) via journal and social media platforms; (3) perform observational studies in clinical settings (Asia, Europe, North America) to evaluate patient outcomes (e.g., symptom relief, quality of life). Results will be published annually, with updates every 1-2 years based on feedback and new evidence.

To enhance applicability across diverse healthcare settings, we summarized practical considerations for implementing the recommendations in both low- and high-resource environments (Appendix 1 Table S9).

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Table S1 Oral glucocorticoids dosages for the treatment of persistent respiratory symptoms in long COVID

No.	Detailed drug and dosages	Doses category
[1]	Start with prednisone at 20-80 mg/day and gradually reduce the dosage.	Low to High
[2]	Prednisolone or equivalent methylprednisolone, with an average initial dose of 27.6 mg/day, reduced to 20 mg/day after 14 days, and decreased by 5 mg every week.	Low
[3]	Start with 36 mg/day of deflazacort tablets and gradually reduce the dosage.	Medium
[4]	Prednisone 20-40/day, combined with mycophenolate mofetil.	Low to Medium
[5]	The initial dose of prednisolone was 26.6 mg/day, and it was rapidly reduced within three weeks.	Low
[6]	- High-dose group: 40 mg/day (for 1 week) → 30 mg/day (for 1 week) → 20 mg/day (for 2 weeks) → 10 mg/day (for 2 weeks) - Low-dose group: 10 mg/day	Low to Medium Low

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Table S2 Research priorities and future directions for long COVID

Research area	Key priorities
Understanding disease burden and subtypes	Longitudinal studies with large sample size are still urgently needed to elucidate the clinical course of long COVID and its long-term consequences. These studies should comprehensively track symptoms, functional impairments, imaging abnormalities, and incident comorbidities using standardized assessment tools and appropriate control groups. In parallel, the development and validation of robust subtyping models will facilitate patient stratification and guide individualized management. Subtyping efforts should account for symptom heterogeneity, differences in data sources (e.g., electronic health records, cohort data), and variations across age groups and vulnerable populations.
Prevention and risk factor identification	The guideline is largely informed by observational studies, as randomized controlled trial (RCT) evidence is still lacking. Future research within the framework of RCTs is especially needed to evaluate whether antiviral use during the acute phase can effectively prevent the development of long COVID. Clarifying how to prevent long COVID remains a research priority. Key questions include determining the optimal vaccination strategies—such as type, dosage, and timing—with rigorous adjustment for baseline immunity and viral variants. Similarly, individualized antiviral strategies during the acute phase warrant further investigation, including dosage and duration, particularly given differential benefits across different risk groups. Moreover, the relationship between age and long COVID risk across diverse population subgroups, considering its inconsistent evidence for now, should be more precisely characterized to inform targeted prevention.
Diagnosis and Evaluation	To improve diagnostic accuracy and streamline clinical care, comparative research is needed to assess the sensitivity, specificity, and cost-effectiveness of various diagnostic tools—including questionnaires, functional assessments, imaging techniques, and laboratory markers—across the broad symptom spectrum of long COVID. In addition, the global accessibility of these tools should be examined to support the development of resource-stratified diagnostic strategies, especially for low-resource settings. Further validation of assessment scales for symptoms such as fatigue and cognitive impairment is needed, including the establishment of culturally and regionally appropriate normative values. Research should also evaluate how well current tools perform in specific subgroups (e.g., those with severe dyspnea or cognitive dysfunction), and whether they sufficiently capture the multidimensional nature of symptoms (e.g., distinguishing physical vs. mental fatigue, or characterizing sleep disturbances beyond insomnia).

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	Objective biomarkers from blood, imaging, or physiological monitoring may hold promise for improving diagnosis, risk stratification, and monitoring of long COVID. Their clinical utility must be evaluated alongside subjective symptom reporting to enhance diagnostic precision and inform clinical decisions.
Treatment strategies and therapeutic trials	There is a pressing need for well-designed clinical trials to assess the safety and effectiveness of candidate therapies for long COVID, including antiviral agents, monoclonal antibodies, immunomodulators, corticosteroids, probiotics, CBT, and rehabilitation interventions. The expert panel acknowledges that the current evidence for the treatment of long COVID remains limited. During discussions on specific therapeutic interventions, it was frequently noted that the lack of sufficient high-quality studies—particularly those focused on defined subgroups of long COVID patients—posed challenges to issuing more specific or definitive recommendations. The panel encourages further research to evaluate the effectiveness and safety of targeted treatments in well-characterized patient populations. Research should identify population-specific needs and determine safe and effective strategies tailored to the severity and nature of symptoms. Future studies should emphasize optimal patient selection, sufficient sample sizes, multicenter collaboration, and extended follow-up to capture sustained effects and late outcomes.
Pediatric Long COVID	Future research on pediatric Long COVID should focus on generating pediatric-specific evidence for prevention strategies, such as conducting multicenter RCTs to evaluate the preventive effects of COVID-19 vaccines (including booster doses) in children, comparing different doses/types, and monitoring long-term safety (e.g., impact on growth). Development and validation of pediatric-specific clinical assessment tools, such as a core outcome set for pediatric Long COVID, including symptom screening scales (e.g., pediatric PHQ-9) and functional tests (e.g., pediatric 6-minute walk test). Evaluation of safety and efficacy of therapeutic interventions in children, such as initiating pediatric RCTs to assess the effectiveness of antiviral drugs (e.g., nirmatrelvir/ritonavir) for Long COVID symptoms in high-risk children, with close monitoring of adverse events. Optimization of pediatric management and rehabilitation strategies, such as assessing the role of cognitive behavioral therapy (CBT) in addressing psychological symptoms among children.

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Table S3 Details of the guideline working group and group of patient representatives

Name	Country	Affiliation	Specialty
Steering Group			
Bin Cao (Chair)	China	China-Japan Friendship Hospital	Respiratory disease
Yaolong Chen (Chief Methodologist)	China	WHO Collaborating Centre for Guideline Implementation and Knowledge Translation, GIN Asia, Lanzhou University GRADE Centre	Guideline methodology
Guideline Coordinator			
Xiaoying Gu	China	China-Japan Friendship Hospital	Respiratory epidemiology
Consensus group			
Roger Paredes	Spain	Hospital Germans Trias I Pujol & irsiCaixa, Universitat Politècnica de Catalunya-BarcelonaTech	Infectious disease
Guiqiang Wang	China	Department of Infectious Diseases, Peking University First Hospital	Infectious disease
Quanyi Wang	China	Beijing Center for Disease Prevention and Control	Epidemiology
Joan B Soriano	Spain	Instituto de Investigación Hospital Universitario de la Princesa (IISP) Universidad Autónoma de Madrid	Respiratory disease/Epidemiology
Yang Jin	China	Union Hospital, Tongji Medical College of Huazhong University of Science and Technology	Respiratory disease
Li Li	China	Army Medical University	Respiratory disease
Norio Ohmagari	Japan	Managing Director at National Center for Global Health and Medicine	Infectious disease
Eu Suk Kim	Korea	Infectious disease consultant at Bundang Seoul National University Hospital	Infectious disease
Yee-Sin Leo	Singapore	National Centre for Infectious Diseases; Lee Kong Chien School of Medicine, Nanyang Technological University; Yong Loo Lin School of Medicine, National University Singapore	Infectious disease
Dingyu Zhang	China	Union Hospital, Tongji Medical College of Huazhong University of Science and Technology	Infectious disease
Hongzhou Lu	China	The Third People's Hospital of Shenzhen	Infectious disease
Maosheng Ran	China	West China Hospital, Sichuan University	Mental health
Ziyad Al-Aly	US	Clinical Epidemiology Center, Veterans Affairs Saint Louis Health Care System	Nephrology/Clinical epidemiology

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Janne Estill	Switzerland	University of Geneva	Public health/Epidemiology
Weizhong Yang	China	School of Population Medicine, Chinese Academy of Medical Sciences and Peking Union Medical College	Infectious disease/Epidemiology
Karen Spruyt	France	Université Paris Cité – INSERM NeuroDiderot	Pediatrics/Statistics/Sleep
Enmei Liu	China	Children's Hospital of Chongqing Medical University	Pediatric respiratory medicine
Yanjiang Wang	China	Daping Hospital	Neurology
Rachael Evans	UK	University of Leicester	Respiratory disease
Lee E. Errett	Canada	University of Toronto	Cardiovascular disease
Wenhong Zhang	China	Huashan Hospital, Fudan University	Infectious disease
James D. Chalmers	UK	Ninewells Hospital and Medical School	Respiratory disease
Michael Peluso	US	Zuckerberg San Francisco General Hospital; University of California	Infectious disease
Jaehyun Jeon	Korea	National Medical Center, Republic of Korea	Infectious disease
Evidence synthesis and evaluation group			
Qianling Shi	China	The First School of Clinical Medicine, Lanzhou University	Evidence-based medicine
Xiaoying Gu	China	China-Japan Friendship Hospital	Respiratory epidemiology
Jiuyang Xu	China	China-Japan Friendship Hospital	Respiratory disease
Hui Zhang	China	China-Japan Friendship Hospital	Respiratory disease
Wenhao Cao	China	China-Japan Friendship Hospital	Respiratory disease
Rongling Zhang	China	China-Japan Friendship Hospital	Respiratory disease
Yang Jin	China	China-Japan Friendship Hospital	Respiratory disease
Qing Zhang	China	China-Japan Friendship Hospital	Respiratory disease
Jingya Li	China	China-Japan Friendship Hospital	Respiratory disease
Bingyi Wang	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine

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Zijun Wang	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Jungang Zhao	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Jie Zhang	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Junxian Zhao	China	Clinical Medicine Research Center, The First Hospital of Lanzhou University	Evidence-based medicine
Meihua Wu	China	School of Public Health, Lanzhou University	Public health
Yishan Qin	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Yueyan Li	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Xu Wang	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Dongrui Peng	China	School of Public Health, Lanzhou University	Public health
Xufei Luo	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Fan Wang	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Qi Zhou	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Hui Liu	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Zijin Wang	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Hanxiang Liu	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Xueping Li	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Zhenyu Tang	China	Children's Hospital of Chongqing Medical University	Evidence-based medicine
Huayu Zhang	China	School of Public Health, Lanzhou University	Public health
Haodong Li	China	School of Public Health, Lanzhou University	Public health
Tianhu Liang	China	Clinical Medicine Research Center, The First Hospital of Lanzhou University	Evidence-based medicine
Luyuan Sun	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Ling Wang	China	Peking Union Medical College, Chinese Academy of Medical Sciences	Evidence-based medicine
Shouyuan Wu	China	Evidence-Based Medicine Center, School of Basic Medical Sciences, Lanzhou University	Evidence-based medicine
Di Zhu	China	School of Public Health, Lanzhou University	Public health
Secretariat			
Xiaoying Gu	China	China-Japan Friendship Hospital	Respiratory epidemiology

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Qianling Shi	China	The First School of Clinical Medicine, Lanzhou University	Evidence-based medicine
Rongling Zhang	China	China-Japan Friendship Hospital	Respiratory disease
Group of Patient Representatives			
Qianqi Zhou	41-year-old female		
Wenchang Zhang	67-year-old male		

/: the information is not provided in consideration of patient representatives' privacy.

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Table S4 Summary of declarations of interest from the guideline working group members

Name	Disclosure of interest	Conflict of interest and management
Bin Cao	None declared	Not applicable
Yaolong Chen	The Co-Founder and Co-Chair of RIGHT (Reporting Items for Practice Guidelines in Healthcare) working group	The conflict was not considered serious enough to affect guideline working group and recommendations.
Roger Paredes	Advisory Board in Pfizer, Gilead, Astra Zeneca	Pfizer: Participated in the advisory meeting regarding collection of long COVID data on ibuzatrelvir trial and on development of a patient reported outcome tool which has been published (https://pubmed.ncbi.nlm.nih.gov/40833451/) Gilead: Contributed to the interpretation and drafting of a manuscript on remdesivir use during the acute phase of COVID-19 and its association with long COVID risk (https://pubmed.ncbi.nlm.nih.gov/40721590/) The collaboration with Gilead specifically addressed remdesivir and its potential role in preventing long COVID. Roger's vote for Recommendation 2 was not counted.
Guiqiang Wang	None declared	Not applicable
Quanyi Wang	None declared	Not applicable
Joan B Soriano	Scientific Director and Board Director of REiCOP, the Spanish Network of Research in Long COVID https://reicop.org/. Members of the ERJ editorial board.	The conflict was not considered serious enough to affect guideline working group and recommendations
Yang Jin	None declared	Not applicable

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(Consensus group)		
Li Li	None declared	Not applicable
Norio Ohmagari	None declared	Not applicable
Eu Suk Kim	None declared	Not applicable
Yee-Sin Leo	None declared	Not applicable
Dingyu Zhang	None declared	Not applicable
Hongzhou Lu	None declared	Not applicable
Maosheng Ran	None declared	Not applicable
Ziyad Al-Aly	None declared	Not applicable
Janne Estill	None declared	Not applicable
Weizhong Yang	None declared	Not applicable
Karen Spruyt	None declared	Not applicable
Enmei Liu	None declared	Not applicable
Yanjiang Wang	None declared	Not applicable
Rachael Evans	Attended ECCMID conference for non-promotional talk at a Moderna symposium. An expert witness for the UK national Covid Inquiry.	The conflict was not considered serious enough to affect guideline working group and recommendations.
Lee E. Errett	None declared	Not applicable
Wenhong Zhang	None declared	Not applicable
James D. Chalmers	Members of the ERJ editorial board.	The conflict was not considered to affect guideline working group and recommendations.
Michael Peluso	Participated in one-time consulting sessions for Gilead Sciences and	The conflict was not considered serious enough to affect guideline working group and recommendations.

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	AstraZeneca related to the importance of Long COVID.	
Jaehyun Jeon	None declared	Not applicable
Xiaoying Gu	None declared	Not applicable
Qianling Shi	None declared	Not applicable
Jiuyang Xu	None declared	Not applicable
Hui Zhang	None declared	Not applicable
Bingyi Wang	None declared	Not applicable
Zijun Wang	None declared	Not applicable
Jungang Zhao	None declared	Not applicable
Jie Zhang	None declared	Not applicable
Junxian Zhao	None declared	Not applicable
Meihua Wu	None declared	Not applicable
Yishan Qin	None declared	Not applicable
Yueyan Li	None declared	Not applicable
Xu Wang	None declared	Not applicable
Dongrui Peng	None declared	Not applicable
Xufei Luo	None declared	Not applicable
Fan Wang	None declared	Not applicable
Qi Zhou	None declared	Not applicable
Hui Liu	None declared	Not applicable
Zijin Wang	None declared	Not applicable
Hanxiang Liu	None declared	Not applicable

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Xueping Li	None declared	Not applicable
Zhenyu Tang	None declared	Not applicable
Huayu Zhang	None declared	Not applicable
Haodong Li	None declared	Not applicable
Tianhu Liang	None declared	Not applicable
Luyuan Sun	None declared	Not applicable
Ling Wang	None declared	Not applicable
Shouyuan Wu	None declared	Not applicable
Di Zhu	None declared	Not applicable
Wenhao Cao	None declared	Not applicable
Yang Jin (Evidence group)	None declared	Not applicable
Rongling Zhang	None declared	Not applicable
Qing Zhang	None declared	Not applicable
Jingya Li	None declared	Not applicable

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Table S5 Definitions of long COVID provided by different institutions and organizations

Term	Institutions/ Organizations	Definition
Post COVID-19 condition Long COVID	WHO	Continuation or development of new symptoms 3 months after the initial SARS-CoV-2 infection, with these symptoms lasting for at least 2 months with no other explanation.
Long COVID	NASEM	An infection-associated chronic condition (IACC) that occurs after SARS-CoV-2 infection and is present for at least 3 months as a continuous, relapsing and remitting, or progressive disease state that affects one or more organ systems.
Post-COVID-19 syndrome	NICE	Signs and symptoms that develop during or after an infection consistent with COVID-19, continue for more than 12 weeks and are not explained by an alternative diagnosis.
Post-COVID conditions Long COVID	CDC	Version 1: Health issues that persist more than four weeks after a person is first infected with SARS-CoV-2. Version 2: A chronic condition that occurs after SARS-CoV-2 infection and is present for at least 3 months.

CDC, US Centers for Disease Control and Prevention; NASEM, National Academies of Sciences, Engineering, and Medicine. NICE, UK National Institute for Health and Care Excellence.

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Table S6 Grading of certainty of evidence and strength of recommendations

Category	Definition
Certainty of evidence	
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect
Very low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect
Strength of recommendation	
Strong	Guideline panel is confident that the desirable effects of an intervention outweigh its undesirable effects (strong recommendation for an intervention) or that the undesirable effects of an intervention outweigh its desirable effects (strong recommendation against an intervention)
Conditional	The desirable effects probably outweigh the undesirable effects (conditional recommendation for an intervention) or undesirable effects probably outweigh the desirable effects (conditional recommendation against an intervention) but appreciable uncertainty exists

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Table S7 Participation of guideline consensus group members across different stages of guideline development

Name	Country	Specialty	Formulation of clinical question or Guideline Discussion	1st round of Delphi	2nd round of Delphi	3rd round of Delphi
Roger Paredes	Spain	Infectious disease	√	√	√	√
Guiqiang Wang	China	Infectious disease	√	√	√	√
Quanyi Wang	China	Epidemiology	√	√	√	√
Joan B Soriano	Spain	Respiratory disease/Epidemiology	√	√	√	√
Yang Jin	China	Respiratory disease	√	√	√	√
Li Li	China	Respiratory disease	√	√	√	√
Norio Ohmagari	Japan	Infectious disease	√	√	√	√
Eu Suk Kim	Korea	Infectious disease	√	√	√	√
Yee-Sin Leo	Singapore	Infectious disease	√	√	√	
Dingyu Zhang	China	Infectious disease		√	√	√
Hongzhou Lu	China	Infectious disease		√	√	√
Maosheng Ran	China	Mental health		√	√	√
Ziyad Al-Aly	US	Nephrology/Clinical epidemiology	√	√	√	√
Janne Estill	Switzerland	Public health/Epidemiology	√	√	√	√
Weizhong Yang	China	Infectious disease/Epidemiology		√	√	√
Karen Spruyt	France	Pediatrics/Statistics/Sleep	√	√	√	
Enmei Liu	China	Pediatric respiratory medicine	√	√	√	√
Yanjiang Wang	China	Neurology		√	√	√
Rachael Evans	UK	Respiratory disease	√		√	√
Lee E. Errett	Canada	Cardiovascular disease	√			

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Wenhong Zhang	China	Infectious disease	√			
James D. Chalmers	UK	Respiratory disease	√			
Jaehyun Jeon	Korea	Infectious disease	√			√
Michael Peluso	US	Infectious disease				√

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Table S8 Voting results across three rounds of Delphi survey

	1st round of Delphi			2nd round of Delphi			3rd round of Delphi		
PICO question 1: What treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) or vaccines can prevent long COVID?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 1	88.89%	0.00%	11.11%	NA	NA	NA	NA	NA	NA
Recommendation 2	70.59%	11.76%	17.65%	NA	NA	NA	NA	NA	NA
PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 3	58.82%	11.76%	29.41%	94.44%	0.00%	5.56%	NA	NA	NA
PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 4	82.35%	5.88%	11.76%	88.89%	5.56%	5.56%	85.00%	10.00%	5.00%
PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 5	58.82%	5.88%	35.29%	94.44%	5.56%	0.00%	NA	NA	NA
PICO question 5: In patients with Long COVID, should glucocorticoids be used versus no glucocorticoids?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 6	58.82%	17.65%	23.53%	100.00%	0.00%	0.00%	NA	NA	NA
Recommendation 7	41.18%	23.53%	35.29%	94.44%	5.56%	0.00%	NA	NA	NA
PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 8	47.06%	11.76%	41.18%	83.33%	5.56%	11.11%	85.00%	10.00%	5.00%
PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?									
	Agree	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**	**Agree**	**Disagree**	**Unsure**
Recommendation 9	72.22%	11.11%	16.67%	NA	NA	NA	NA	NA	NA

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PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?									
	Agree	Disagree	Unsure	Agree	Disagree	Unsure	Agree	Disagree	Unsure
Recommendation 10	100.00%	0.00%	0.00%	NA	NA	NA	NA	NA	NA

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Table S9 Practical considerations for implementing the recommendations in both low- and high-resource environments

Intervention	High-Income Settings	Low- and Middle-Income / Resource-Limited Settings
Prevention (Vaccination, Antivirals, Metformin)	Broad access to vaccines and antiviral agents; infrastructure supports monitoring of safety and effectiveness	Limited vaccine and drug supply; high costs; prioritization of high-risk groups needed; metformin may be more feasible due to lower cost and wider availability
Antiviral drugs for long COVID	Availability of clinical trials and novel antivirals; adverse event monitoring feasible	Restricted or no access to antivirals; off-label or prolonged use often impractical; supportive care prioritized
Monoclonal antibodies	Available in specialized centers; potential targeted use in immunocompromised patients	Very limited or unavailable due to high cost and supply constraints; not feasible for routine use
Probiotics	Multiple strains and commercial preparations accessible; regulated quality control	Limited access to standardized formulations; reliance on affordable and locally available products; traditional fermented foods may serve as alternatives
Glucocorticoids	Can be prescribed with appropriate monitoring; imaging and lung function follow-up possible	Risk of overuse or misuse without monitoring; potential harms greater; generally not recommended outside specialist care
Immunomodulators	Access to mycophenolate, JAK inhibitors, and tacrolimus in tertiary centers with specialist oversight	Very limited availability; often unaffordable; risks outweigh potential benefits; supportive and symptomatic care emphasized
Cognitive Behavioral Therapy (CBT)	Access to trained therapists, structured programs, and digital/telehealth delivery	Shortage of mental health professionals; task-shifting to primary care, group sessions, or low-intensity/digital CBT approaches may be more practical
Rehabilitation exercise	Specialized rehabilitation centers and telerehabilitation available; individualized programs under supervision	Limited facilities; community- or home-based programs preferred; task-shifting to trained community health workers; pacing strategies emphasized for patients with PEM

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Appendix 2 Detailed information for evidence synthesis and summary

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PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators? 50

PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy? 56

PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise? 60

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PICO question 1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID?

Evidence summary

Current research on long COVID prevention focused on vaccination and acute-phase drugs following SARS-CoV-2 infection. The details are described as follows:

Vaccination

Several SRs have assessed the preventive effect of vaccination on the risk of long COVID. We conducted a systematic search (from January 1, 2020, to November 30, 2024) and identified three high-quality SRs that support the recommendation based on the methodological quality and content comprehensiveness [1-3]. These SRs included 44 observational studies with a total of 14,842,052 participants. However, there was notable heterogeneity in the long COVID definitions and demographic characteristics, with specific age distributions not clearly specified.

(1) Preventive effect of vaccination prior to SARS-CoV-2 infection: All three SRs showed that vaccination prior to SARS-CoV-2 infection could reduce the risk of long COVID[1-3]. Regarding specific dosage, one SR [3] found that two vaccine doses prior to infection significantly lowered the risk of long COVID (OR=0.76, 95% CI:0.65 to 0.89, I^2=93.4%, very low certainty of evidence). The reduction was particularly evident for fatigue, headache, olfactory dysfunction, and myalgia. However, one dose (OR: 1.01, 95% CI:0.88 to 1.15, I^2=63.0%, very low certainty of evidence) and three doses (OR: 0.31, 95% CI:0.05 to 1.84, I^2=97.3%, very low certainty of evidence) showed no statistically significant effect. No adverse events were reported.

(2) Preventive effect of vaccination after SARS-CoV-2 infection: Two SRs found that post-infection vaccination may decrease the risk of long COVID [1,2]. Regarding specific dosage, one SR [3] found that one dose after infection significantly reduced the risk of long COVID (OR: 0.85, 95% CI:0.73 to 0.98, I^2=77.9%, moderate certainty of evidence), while two doses did not have statistical significance (OR: 0.63, 95% CI:0.38 to 1.03, I^2=88.4%, very low certainty of evidence). No adverse events were reported.

(3) Vaccine types: Three reviews assessed mRNA and adenoviral vector vaccines (1-3 doses). A subgroup analysis [3] showed that BNT162b2 (OR: 0.79, 95% CI:0.67 to 0.92, I^2=93.0%) and ChAdOx1 nCoV-19 (OR: 0.59, 95% CI:0.50 to 0.70, I^2=0.0%) significantly reduced the risk of long COVID, regardless of vaccination timing relative to infection.

Acute-phase drugs (administered within the first 4 weeks post-infection)

Based on the SR by Sun G and colleagues [4], we updated the search and analysis (search period: January 1, 2020 to November 30, 2024). A total of 50 studies were included (including 6 RCTs, 2 non-randomized trials, 40 cohort studies, and 2 case-control studies) [5-54], involving 7,060,422 COVID-19 patients aged ⩾12 years. The study populations varied in disease severity (severe, moderate, and mild), and included both hospitalized and non-hospitalized individuals. Participants also represented diverse risk profiles, such as the presence of comorbidities (e.g., diabetes, hypertension) and other risk factors for development of long COVID.

The interventions assessed included antiviral drugs (molnupiravir, remdesivir, nirmatrelvir, nirmatrelvir/ritonavir, favipiravir, ensitrelvir, azvudine), corticosteroids (dexamethasone, prednisolone, methylprednisolone), monoclonal antibodies (bamlanivimab/etesevimab, casirivimab/imdevimab, sotrovimab), and metformin. Based on data availability, we conducted meta-analysis for 35 studies with long COVID as primary outcome. Given the variability in follow-up durations across studies, we synthesized results at the 6-month follow-up. The secondary outcomes included specific long COVID symptoms—fatigue, dyspnea, myalgia, sleep disturbances, and cognitive impairment—with no restriction on follow-up time points due to limited data availability.

Antiviral Drugs: (1) Primary outcome: acute-phase antiviral drugs significantly reduced the risk of long COVID (19 studies, OR: 0.69, 95% CI: 0.64 to 0.74, I^2=98.0%, low certainty of evidence) compared with the control group. Subgroup analyses showed that nirmatrelvir/ritonavir (13 studies, OR: 0.69, 95% CI: 0.60 to 0.78, I^2 = 98.5%, low certainty of evidence), nirmatrelvir (1 study, OR: 0.88, 95% CI: 0.87 to 0.89, I^2 = 0.0%, very low certainty of evidence), molnupiravir (3 studies, OR: 0.86, 95% CI: 0.77 to 0.96, I^2 = 91.3%, low certainty of evidence), and favipiravir (1 study, OR: 0.49, 95% CI: 0.34 to 0.70, I^2=0.0%, very low certainty of evidence) all significantly reduced the risk of long COVID; while remdesivir (2 studies, OR: 0.34, 95% CI: 0.09 to 1.23, I^2 = 94.6%, very low certainty of evidence), ensitrelvir (1 study, OR: 0.77, 95% CI: 0.53 to 1.13, I^2=0.0%, very low certainty of evidence), and azvudine (1 study, OR: 0.81, 95% CI: 0.57 to 1.16, I^2 = 0.0%, very low certainty of evidence) did not have a statistically significant effect on the risk of long COVID.

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(2) Secondary outcome: compared with the control group, acute-phase antiviral drugs significantly reduced the risk of fatigue (6 studies, OR: 0.90, 95% CI: 0.84 to 0.97, I^2 = 63.5%, very low certainty of evidence), cognitive impairment (5 studies, OR: 0.83, 95% CI: 0.74 to 0.94, I^2 = 82.1%, low certainty of evidence), and sleep disorders (5 studies, OR: 0.94, 95% CI: 0.89 to 0.98, I^2=0.0%, low certainty of evidence); however, for dyspnea (4 studies, OR: 0.94, 95% CI: 0.72 to 1.23, I^2 = 36.2%, low certainty of evidence) and myalgia (3 studies, OR: 0.88, 95% CI: 0.70 to 1.10, I^2 = 0.0%, low certainty of evidence), no statistically significant effects were observed. No subgroup analyses by drug type were performed for the secondary outcomes. (3) Safety: one study reported 14 serious adverse events with favipiravir (compared with 4 in the control group)[43], and no adverse events for other antivirals.

Monoclonal Antibodies: No significant effect on the risk of long COVID (7 studies, OR: 0.92, 95% CI: 0.69 to 1.22, I^2 = 65.20%, very low certainty of evidence), fatigue (1 study, OR: 0.79, 95% CI: 0.54 to 1.15, I^2=0.0%, very low certainty of evidence), dyspnea (1 study, OR: 0.74, 95% CI: 0.47 to 1.15, I^2 = 0.0%, very low certainty of evidence), cognitive impairment (1 study, OR: 0.81, 95% CI: 0.46 to 1.42, I^2 = 0.0%, very low certainty of evidence), and sleep disorders (1 study, OR: 1.10, 95% CI: 0.67 to 1.82, I^2 = 0.0%, very low certainty of evidence) was observed for monoclonal antibodies. No studies were identified regarding myalgia. On safety, one study suggested increased pulmonary fibrosis risk related to the use of rituximab [30]; another study reported mild to moderate adverse events unrelated to the use of tixagevimab/cilgavimab [45]. No severe adverse events were reported.

Corticosteroids: No significant effect on the risk of long COVID (5 studies, OR: 0.97, 95% CI: 0.74 to 1.25, I^2=63.5%, very low certainty of evidence) or dyspnea (1 study, OR: 0.74, 95% CI: 0.47 to 1.15, I^2=0.0%, very low certainty of evidence) was found for corticosteroid use during acute phase. No study addressed any secondary outcomes, and no severe adverse events were reported.

Metformin: We did not perform meta-analysis due to high heterogeneity of included studies. One RCT (involving 1,126 overweight or obese diabetic patients aged 30-85 years) [52] showed metformin significantly reduced the risk of long COVID at 10-month follow-up (hazard ratio [HR]: 0.59, 95% CI: 0.39 to 0.89). Two retrospective cohort studies (involving 90,000 adult patients with type 2 diabetes) [53,54]found metformin could reduce long COVID risk when compared to other non-insulin antidiabetic drugs (sulfonylureas, dipeptidyl peptidase-4 inhibitors or thiazolidinediones) [53,54]. Definitions and measurement methods for long COVID also varied across metformin studies. No adverse events were reported.

Evidence retrieval and synthesis

Section 1: antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.

Part 1: Search resources and result

Table S4-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	805
	Web of Science Core Collection	Nov 30, 2024	756
	Embase	Nov 30, 2024	352
	The Cochrane Library	Nov 30, 2024	84
	China Biology Medicine (CBM)	Nov 30, 2024	4
	Wanfang Database	Nov 30, 2024	80
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	67
Preprint platform	Medrxiv	Nov 30, 2024	-
	SSRN	Nov 30, 2024	-
	Researchsquare	Nov 30, 2024	-
Clinical trial registries	WHO ICTRP	Nov 30, 2024	-
	ClinicalTrials.gov	Nov 30, 2024	-
	COVID-19 Study Register	Nov 30, 2024	-
Google	-	Nov 30, 2024	2
Reference lists of relevant SRs	-	Nov 30, 2024	-

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Part 2: Flow diagram of the literature search

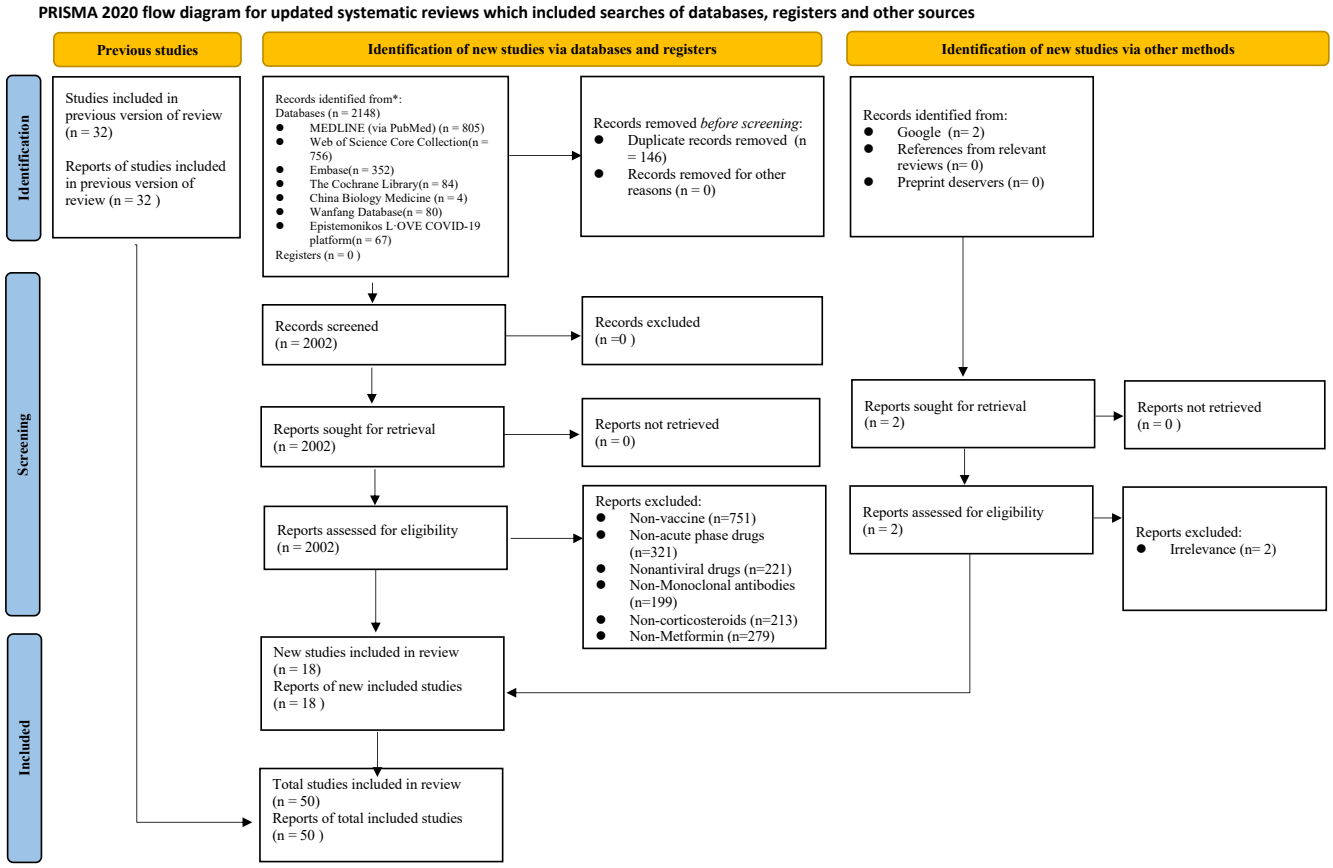


Figure S4-1 Flow diagram of the literature search for treatment*

Note: treatment includes antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin)

Part 3: list of included studies

Table S4-2 list of included studies

No.	Study title	Study design	DOI
1	Risk factors and incidence of long-COVID syndrome in hospitalized patients: does remdesivir have a protective effect?	Cohort study	10.1093/qjmed/hcab297
2	No Impact of Corticosteroid Use During the Acute Phase on Persistent Symptoms Post-COVID-19	Cohort study	10.2147/IJGM.S367273
3	Post-COVID depression and its multiple factors, does Favipiravir have a protective effect? A longitudinal study of indonesia COVID-19 patients	Cohort study	10.1371/journal.pone.0279184
4	Long-term respiratory outcomes after COVID-19: a Brazilian cohort study	Cohort study	10.26633/RPSP.2022.187
5	Monoclonal Antibodies and Antivirals against SARS-CoV-2 Reduce the Risk of Long COVID: A Retrospective Propensity Score-Matched Case–Control Study	Case control study	10.3390/biomedicines10123135
6	Corticosteroids for COVID-19 symptoms and quality of life at 1 year from admission	Cohort study	10.1002/jmv.27296
7	Lung-function trajectories in COVID-19 survivors after discharge: A two-year longitudinal cohort study	Cohort study	10.1016/j.eclinm.2022.101668
8	Effect of remdesivir post hospitalization for COVID-19 infection from the randomized SOLIDARITY Finland trial	Randomized controlled trial	10.1038/s41467-022-33825-5
9	The impact of early therapies for COVID-19 on death, hospitalization and persisting symptoms: a retrospective study	Cohort study	10.1007/s15010-023-02028-5
10	The Predictors of Long COVID in Southeastern Italy	Cohort study	10.3390/jcm12196303

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11	Efficacy of nirmatrelvir and ritonavir for post-acute COVID-19 sequelae beyond 3 months of SARS-CoV-2 infection	Cohort study	10.1002/jmv.28750
12	Association of Treatment with Nirmatrelvir and the Risk of Post-COVID-19 Condition	Cohort study	10.1001/jamainternmed.2023.0743
13	Nirmatrelvir/ritonavir and risk of long COVID symptoms: a retrospective cohort study	Cohort study	10.1038/s41598-023-46912-4
14	Paxlovid reduces the risk of Long COVID in patients six months after hospital discharge	Cohort study	10.1002/jmv.29014
15	The effect of nirmatrelvir-ritonavir on the long-term risk of neuropsychiatric sequelae following COVID-19	Cohort study	10.1002/jmv.28951
16	Molnupiravir and risk of post-acute sequelae of covid-19: cohort study	Cohort study	10.1136/bmj-2022-074572
17	Risk of hospitalization and sequelae in patients with COVID-19 treated with 3-day early remdesivir vs. controls in the vaccine and Omicron era: A real-life cohort study	Cohort study	10.1002/jmv.28660
18	Nirmatrelvir and Molnupiravir and Post-COVID-19 Condition in Older Patients	Cohort study	10.1001/jamainternmed.2023.5099
19	Effectiveness of oral antiviral agents on long-term cardiovascular risk in nonhospitalized patients with COVID-19: A multicenter matched cohort study	Cohort study	10.1002/jmv.28992
20	Effectiveness of Nirmatrelvir–Ritonavir Against the Development of Post–COVID-19 Conditions Among U.S. Veterans	Non-randomised study	10.7326/M23-1394
21	Distinct temporal trajectories and risk factors for Post-acute sequelae of SARS-CoV-2 infection	Cohort study	10.3389/fmed.2023.1227883
22	The effect of corticosteroids, antibiotics, and anticoagulants on the development of post-COVID-19 syndrome in COVID-19 hospitalized patients 6 months after discharge: a retrospective follow up study	Case control study	10.1007/s10238-023-01153-7
23	Long-term effect of tocilizumab on mortality, readmissions, persistent symptoms and lung function in SARS-CoV-2 patients 1 year after hospital discharge: A matched cohort study	Cohort study	10.1016/j.resmer.2023.101064
24	Real-world effectiveness of COVID-19 vaccines and anti-SARS-CoV-2 monoclonal antibodies against postacute sequelae of SARS-CoV-2: analysis of a COVID-19 observational registry for a diverse US metropolitan population	Cohort study	10.1136/bmjopen-2022-067611
25	Post-Acute SARS-CoV-2 Symptoms are Fewer, Less Intense Over Time in People Treated with Mono-Clonal Antibodies for Acute Infection	Cohort study	10.2147/IJGM.S406571
26	Pharmacotherapy and pulmonary fibrosis risk after SARS-CoV-2 infection: a prospective nationwide cohort study in the United States	Cohort study	10.1016/j.lana.2023.100566
27	Treatment of COVID-19 during the Acute Phase in Hospitalized Patients Decreases Post-Acute Sequelae of COVID-19	Cohort study	10.3390/jcm12124158
28	Modifiable in-hospital factors for 12-month global cognition, post-traumatic stress disorder symptoms, and depression symptoms in adults hospitalized with COVID-19	Cohort study	10.1111/irv.13197
29	Does monoclonal antibody treatment for COVID-19 impact short and long-term outcomes in a large generalisable population? A retrospective cohort study in the USA	Cohort study	10.1136/bmjopen-2022-069247
30	Impact of Paxlovid on in-hospital outcomes and post-COVID-19 condition in adult patients infected with SARS-CoV-2 Omicron variant: A non-randomized controlled clinical trial	Non-randomised study	10.1097/MD.0000000000036714
31	Association of nirmatrelvir for acute SARS-CoV-2 infection with subsequent Long COVID symptoms in an observational cohort study	Cohort study	10.1002/jmv.29333
32	Prognostic Factors for Predicting Post-COVID-19 Condition in Patients With COVID-19 in an Outpatient Setting	Cohort study	10.3346/jkms.2024.39.e23
33	Clinical effectiveness of nirmatrelvir plus ritonavir on the short- and long-term outcome in high-risk children with COVID-19	Cohort study	10.1002/jmv.29662
34	Real-World Effectiveness of Nirmatrelvir in Protecting Long COVID for Outpatient Adult Patients-A Large-Scale Observational Cohort Study from the RECOVER Initiative	Cohort study	10.21203/rs.3.rs-4536807/v1
35	Effect of paxlovid treatment during acute covid-19 on long covid onset: an EHR-based target trial emulation from the N3C and RECOVER consortia	Cohort study	10.1101/2024.01.20.24301525

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36	Nirmatrelvir/ritonavir treatment and the risk of post-COVID condition over 180 days in Malaysia	Cohort study	10.1186/s12879-024-09679-1
37	Nirmatrelvir-Ritonavir for Acute COVID-19 in Patients With Cardiovascular Disease and Postacute Sequelae of SARS-CoV-2 Infection	Cohort study	10.1016/j.jacadv.2024.100961
38	Effectiveness of azvudine against severe outcomes among hospitalized COVID-19 patients in Xinjiang, China: a single-center, retrospective, matched cohort study	Cohort study	10.1080/14787210.2024.2362900
39	Favipiravir for COVID-19 in adults in the community in PRINCIPLE, an open-label, randomised, controlled, adaptive platform trial of short- and longer-term outcomes	Randomized controlled trial	10.1016/j.jinf.2024.106248
40	Post-acute COVID-19 outcomes including participant-reported long COVID: amubarvimab/romlusevimab versus placebo in the ACTIV-2 trial	Randomized controlled trial	10.1016/j.eclinm.2024.102787
41	Safety, Efficacy and Pharmacokinetics of AZD7442 (Tixagevimab/Cilgavimab) for Treatment of Mild-to-Moderate COVID-19: 15-Month Final Analysis of the TACKLE Trial	Randomized controlled trial	10.1007/s40121-024-00931-4
42	Prevention of Post COVID-19 Condition by Early Treatment with Ensitrelvir in the Phase 3 SCORPIO-SR Trial	Randomized controlled trial	10.1016/j.antiviral.2024.105958
43	Nirmatrelvir plus ritonavir reduces COVID-19 hospitalization and prevents long COVID in adult outpatients	Cohort study	10.1038/s41598-024-76472-0
44	Acute COVID-19 treatment is not associated with health problems 2 years after hospitalization	Cohort study	10.1016/j.ijid.2024.02.009
45	Association between monoclonal antibody therapy, vaccination, and longer - term symptom resolution after acute COVID-19	Cohort study	10.1002/jmv.29541
46	Association of nirmatrelvir-ritonavir with post-acute sequelae and mortality among patients who are immunocompromised with COVID-19 in Hong Kong: a retrospective cohort study	Cohort study	10.1016/S2665-9913(24)00224-8
47	Association of nirmatrelvir–ritonavir with post-acute sequelae and mortality in patients admitted to hospital with COVID-19: a retrospective cohort study	Cohort study	10.1016/S1473-3099(24)00217-2
48	Prevalent Metformin Use in Adults With Diabetes and the Incidence of Long COVID: An EHR-Based Cohort Study From the RECOVER Program	Cohort study	10.2337/DCa24-0032
49	Risk of Post-Acute Sequelae of SARS-CoV-2 Infection (PASC) Among Patients with Type 2 Diabetes Mellitus on Anti-Hyperglycemic Medications	Cohort study	10.2147/CLEP.S458901
50	Outpatient treatment of COVID-19 and incidence of post-COVID-19 condition over 10 months (COVID-OUT): a multicentre, randomised, quadruple-blind, parallel-group, phase 3 trial	Randomized controlled trial	10.1016/S1473-3099(23)00299-2

Part 4: Risk of bias assessment of the included studies

Table S4-3 Risk of bias assessment of included studies

Risk of bias for the included cohort studies as assessed by the Newcastle-Ottawa Scale									
Study ID	Selection of exposure								
	Represent-ativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure to implants	Demonstration that outcome of interest was not present at start of study	Study controls the most important factor	Study controls for any additional factor	Assess-ment of outcome	Was follow up long enough for outcomes to occur	Adequacy of follow up of cohorts
Chuang 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Xie 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Durstenfeld 2024	Yes	Yes	No	Yes	Yes	Yes	No	Yes	No
Wang 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Congdon 2023	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	No
Liu 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Boglione 2022	Yes	Yes	No	Yes	No	No	No	Yes	Yes

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Risk of bias for the included cohort studies as assessed by the Newcastle-Ottawa Scale									
Study ID	Selection of exposure								
	Represent-ativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure to implants	Demonstration that outcome of interest was not present at start of study	Study controls the most important factor	Study controls for any additional factor	Assess-ment of outcome	Was follow up long enough for outcomes to occur	Adequacy of follow up of cohorts
Xie 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mazzitelli 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fung 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Chen 2023	Yes	Yes	No	Yes	No	No	Yes	Yes	No
Sui Ko A 2022	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No
Quaranta 2023	Yes	Yes	No	Yes	Yes	No	No	Yes	No
Bertuccio 2023	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No
Cardona-Pascual 2023	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Tannous 2023	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Price 2023	Yes	Yes	No	Yes	No	No	Yes	Yes	Yes
Adegunsoye 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Choi 2024	Yes	Yes	No	Yes	Yes	No	No	Yes	Yes
Herman 2022	Yes	Yes	No	Yes	Yes	No	No	Yes	Yes
Visconti 2022	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No
Bonet 2023	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Catalán 2022	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No
Wu 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Preiss 2024	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Low 2024	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Xie 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Kapar 2024	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Sharif-Askari 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Liu 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wang 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Berentschot 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Roberts 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Griffin 2023	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zhang 2022	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jin H. Han 2023	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No
Huwen Wang 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Guozhang Lin 2025	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Steven G. Johnson 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Oluwasolape Olawore 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No

Risk of bias for the included case-control studies as assessed by the Newcastle-Ottawa Scale	
Study ID	Selection

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	Is the Case Definition Adequate	Representative-ness of the Cases	Selection of Controls	Definition of Controls	Study controls the most important factor	Study controls for any additional factor	Ascertain-ment of Exposure	Same method of ascertain-ment for cases and controls	Non-Respons e Rate
Cimellaro 2022	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No
Davelaar 2023	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No

Risk of bias for the included Non-randomised studies as assessed by the Risk Of Bias In Non-randomised Studies-of Interventions								
Study ID	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviations from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	overall risk of bias
Xu 2023	Critical	Low	Low	No information	Low	Low	Low	Critical
Ioannou 2023	Low	Low	Low	Low	Low	Low	Low	Low

Risk of bias for the included Randomized controlled trials as assessed by the Cochrane risk-of-bias assessment tool for randomised trials								
Study ID	Random sequence generation	Allocation concealmen t	Blinding of participants	Blinding of personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other biases
Nevalainen 2022	Low	Low	No information	No information	No information	Low	Low	Low
Evering 2024	Low	Low	Low	Low	No information	Low	Low	Low
Hobbs 2024	Low	Low	Low	Low	Low	Low	Low	Low
Carolyn T Bramante 2023	Low	Low	Low	Low	Low	Low	Low	Low
Hobbs 2024	Low	Low	Low	Low	Low	Low	Low	Low
Yotsuyanagi 2024	No information	Low	Low	Low	Critical	Low	Low	Low

Risk of bias for the included systematic reviews as assessed by a Measurement Tool to Assess the Methodological Quality of Systematic Reviews/ Meta-analysis											
Study ID	item 1	item 2	item 3	item 4	item 5	item 6	item 7	item 8	item 9	item 10	item 11
Milena 2024	Yes	Yes	No	Yes	No	Yes	Yes	No	Yes	Yes	Yes
Oyungerel 2025	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Not applicable	No	No
Nick 2024	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes

Item 1: Was an “a priori” design provided?
Item 2: Was there duplicate study selection and data extraction?
Item 3: Was a comprehensive literature search performed?
Item 4: Was the status of publication (i.e. grey literature) used as an inclusion criterion?
Item 5: Was a list of studies (included and excluded) provided?
Item 6: Were the characteristics of the included studies provided?
Item 7: Was the scientific quality of the included studies assessed and documented?
Item 8: Was the scientific quality of the included studies used appropriately in formulating conclusions?
Item 9: Were the methods used to combine the findings of studies appropriate?
Item 10: Was the likely-hood of publication bias assessed?
Item 11: Was the conflict of interest stated

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Part 5: Evidence syntheses

5.1.1 Meta-analysis

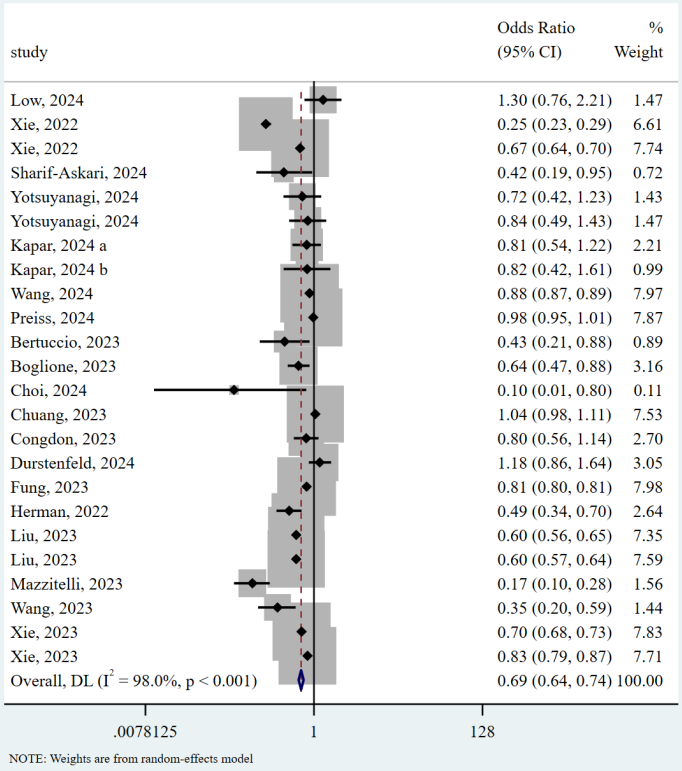


Figure S4-2 Forest plot presenting relationships between antiviral drugs treatment and long COVID incidence

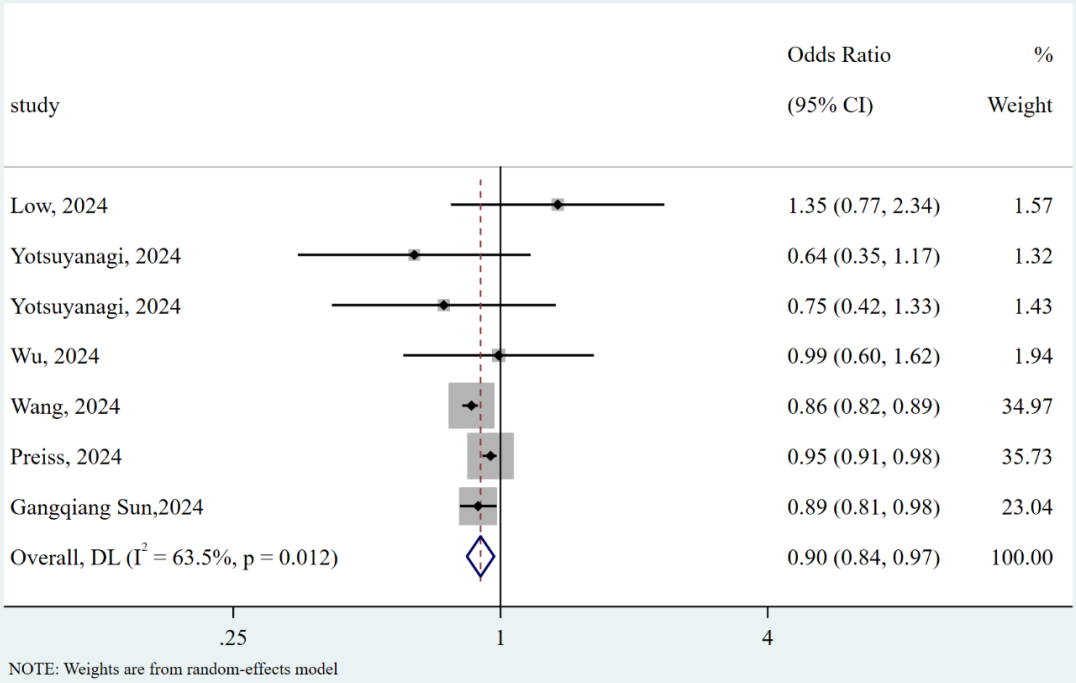


Figure S4-3 Forest plot presenting relationships between antiviral drugs treatment and fatigue

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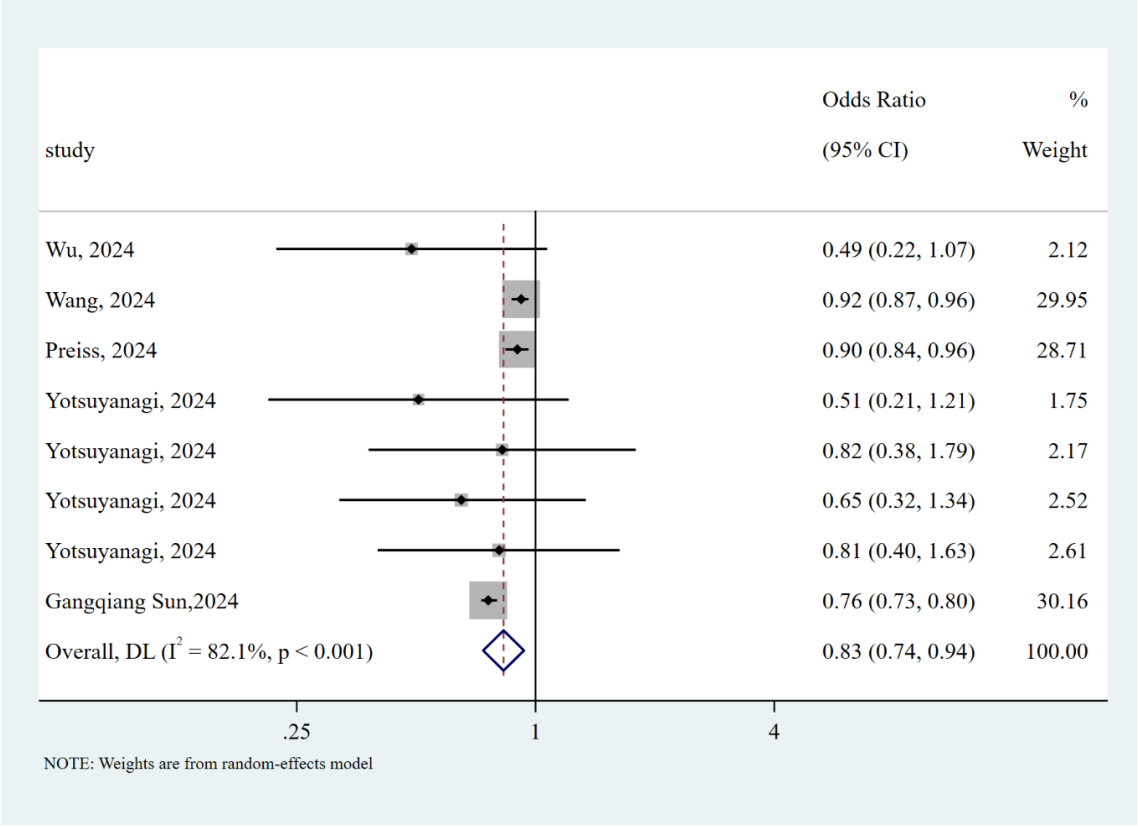


Figure S4-4 Forest plot presenting relationships between antiviral drugs treatment and cognitive impairment

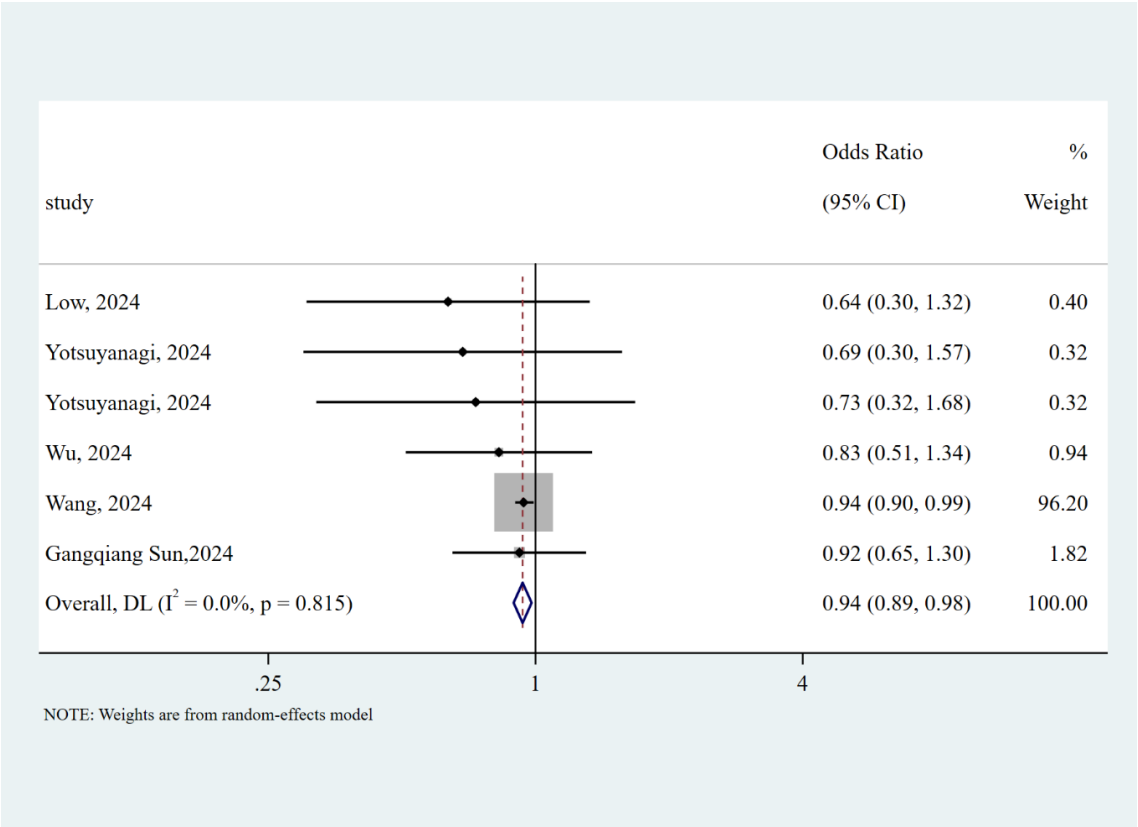


Figure S4-5 Forest plot presenting relationships between antiviral drugs treatment and sleep disorders

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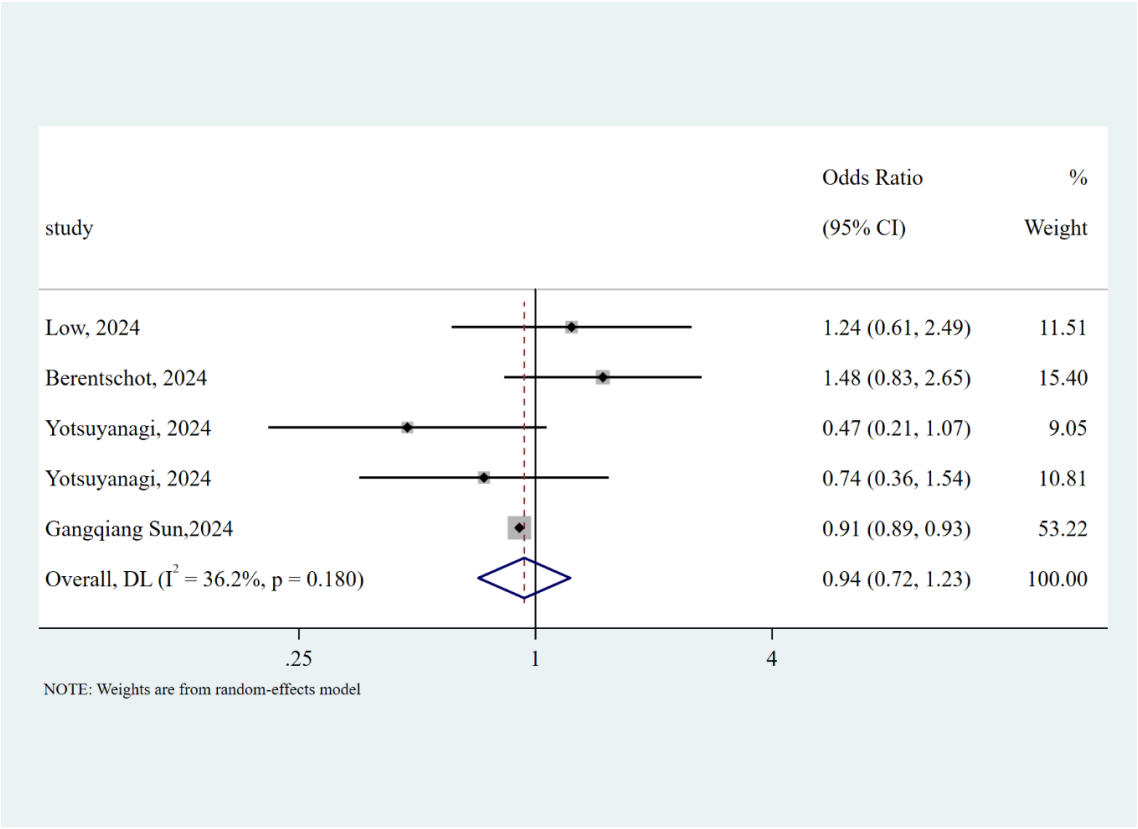


Figure S4-6 Forest plot presenting relationships between antiviral drugs treatment and dyspnea

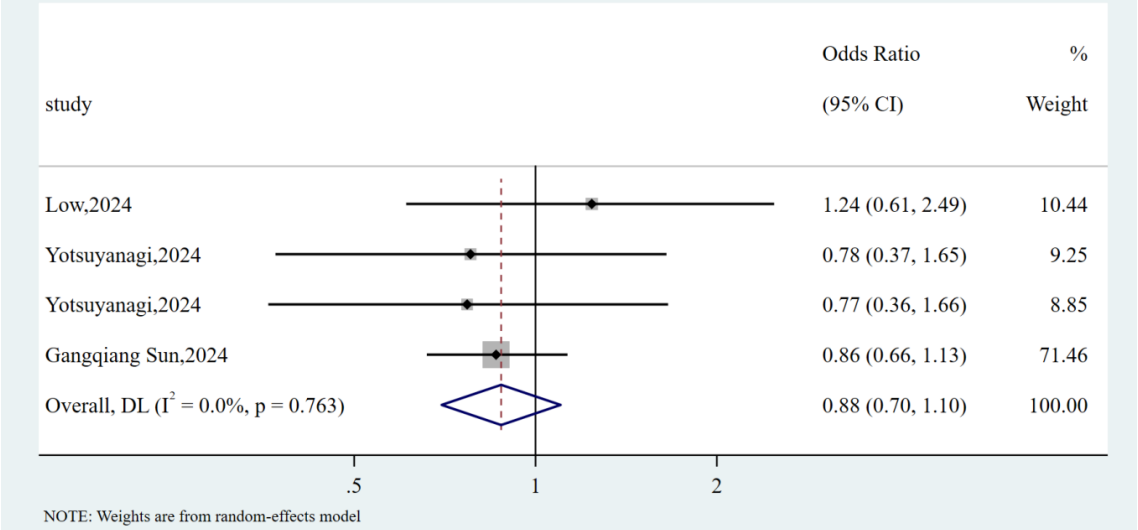


Figure S4-7 Forest plot presenting relationships between antiviral drugs treatment and myalgia

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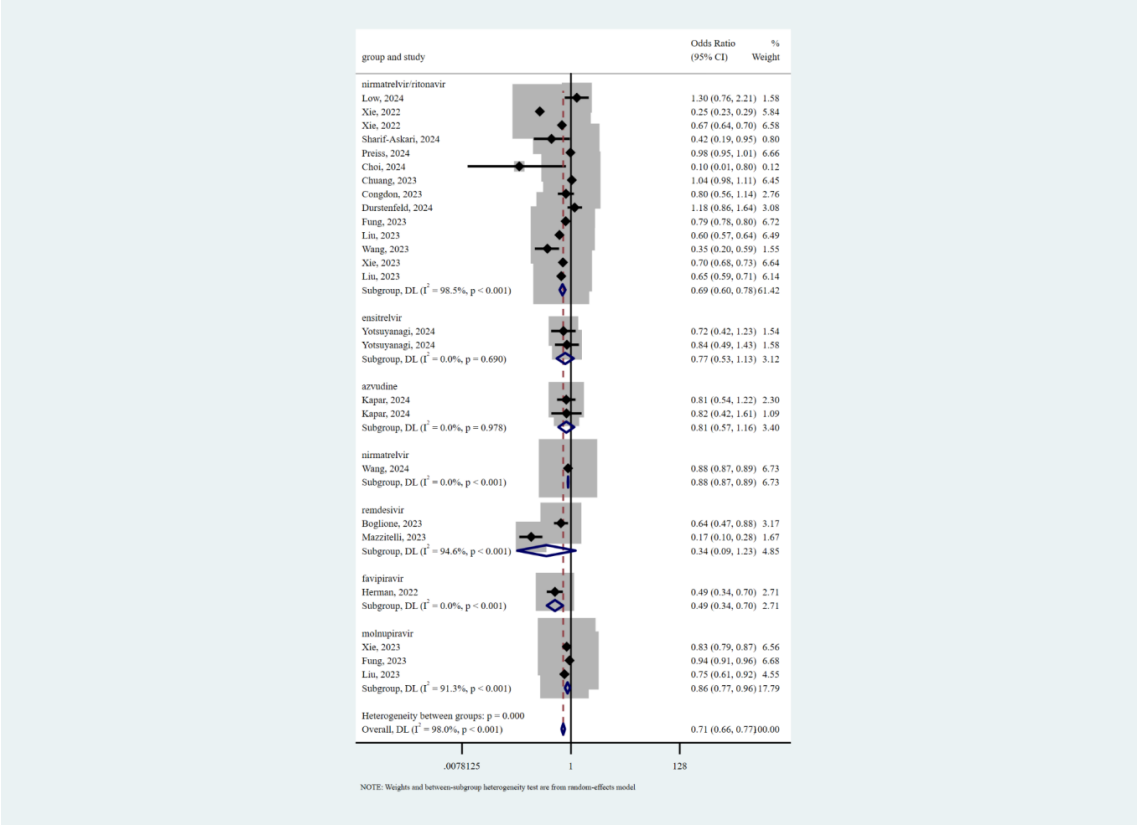


Figure S4-8 The subgroup analysis of the forest plot presenting the relationships between antiviral drugs treatment and long COVID incidence

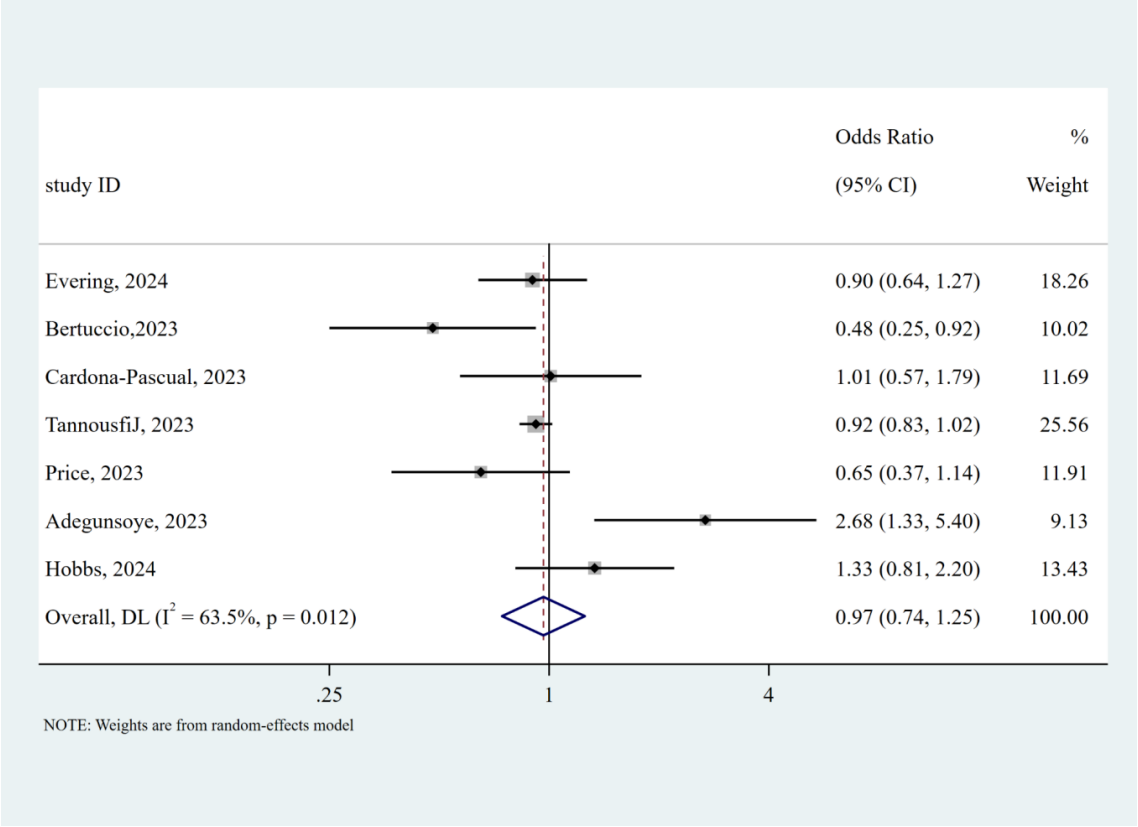


Figure S4-9 Forest plot presenting relationships between corticosteroids treatment and long COVID incidence

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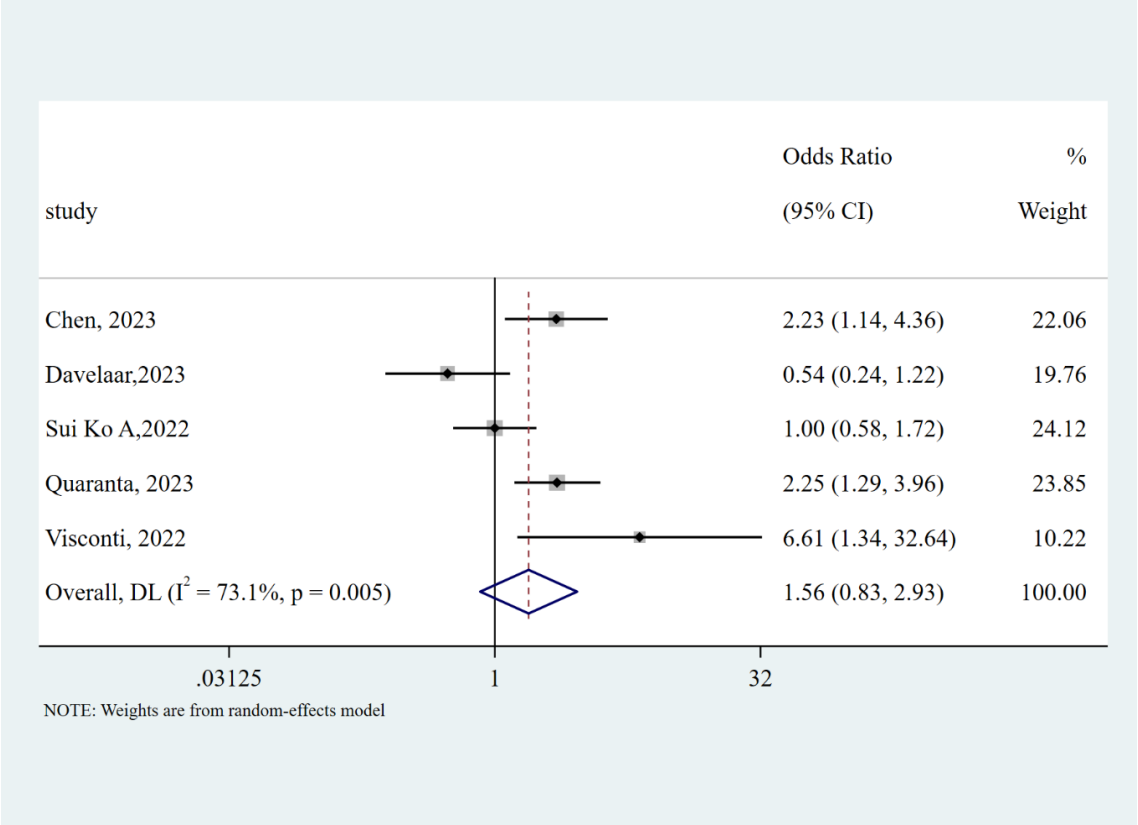


Figure S4-10 Forest plot presenting relationships between Monoclonal Antibodies treatment and long COVID incidence

5.1.2 Data qualitative summary

Table S4-4 Basic Characteristics of the Study on Metformin Treatment and long COVID

Title	Prevalent Metformin Use in Adults With Diabetes and the Incidence of Long COVID: An EHR-Based Cohort Study From the RECOVER Program	Risk of Post-Acute Sequelae of SARS-CoV-2 Infection (PASC) Among Patients with Type 2 Diabetes Mellitus on Anti-Hyperglycemic Medications	Outpatient treatment of COVID-19 and incidence of long COVID over 10 months (COVID-OUT): a multicentre, randomised, quadruple-blind, parallel-group, phase 3 trial
Study ID	Johnson 2024	Oluwasolape 2024	Bramante 2023
Publication Year	2024	2024	2023
Publication Journal	Diabetes Care	Clinical Epidemiology	The Lancet Infectious Diseases
Type of Research	Retrospective cohort study	Cohort study	Randomized controlled trials
Preventive Measures	Metformin	Metformin, sulfonylureas, or dipeptyl-peptidase-4-inhibitors	Metformin plus ivermectin, metformin plus fluvoxamine, metformin
LC Definition/Target Outcome	The postacute sequelae of SARS-CoV-2 infection (PASC), also called long COVID, occurs when there are ongoing, relapsing or new symptoms, or other health consequences occurring 4 weeks after the acute phase of SARS-CoV-2 infection	The outcome of interest was defined using both a clinical and a machine learning (ML) definition of PASC. The clinical definition required the presence of an ICD-10 diagnoses code for long-COVID (U09.9). If a patient had multiple diagnoses codes for the outcome, we used the earliest occurrence	The primary method for ascertaining long COVID was participant-reported receipt of a long COVID diagnosis from a medical provider
Outcomes	①Death ② Long COVID	Long COVID	①Severe COVID-19 by day 14 of study drug ② Long COVID
Outcome Indicator Measurement Time	6 months	3-and 6-months	days 180, 210, 240, 270, and 300
Research subject	Adults with type 2 diabetes mellitus (T2DM)	Adults 18 years or older with T2DM diagnosis and at least one healthcare encounter in the two 6-	Adults aged 30–85 years with overweight or obesity who had COVID-19 symptoms for fewer than 7 days and a documented

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		month periods before incident COVID-19 diagnosis	SARS-CoV-2 positive PCR or antigen test within 3 days before enrolment
Age	≥18 years	≥18 years	30–85 years
Conclusions	In the N3C, the hazard ratio (HR) for death or PASC with a U09.9 diagnosis code (PASC-U09.0) was 0.79 [0.71–0.88], and for death or N3C computable phenotype PASC (PASC-N3C) was 0.85 [0.78–0.92]. In PCORnet, the HR for death or PASC-U09.9 was 0.87 [0.66–1.14], and for death or PCORnet computable phenotype PASC (PASC-PCORnet) was 1.04 [0.97–1.11]. Incident PASC by diagnosis code was 1.6% metformin vs. 2.0% comparator in the N3C, and 2.1% metformin vs. 2.5% comparator in PCORnet. By computable phenotype, incidence was 4.8% metformin and 5.2% comparator in the N3C and 24.7%metformin vs. 26.1% comparator in PCORnet.	We did not find a significant difference in risk of PASC at 3 months (RR=0.86 [0.56, 1.32], RD =−3.06 per 1000 [−12.14, 6.01]), or at 6 months (RR =0.81 [0.55; 1.20], RD = −4.91 per 1000 [−14.75, 4.93]) comparing prevalent users of metformin to prevalent users of SU/DPP4i. Similar observations were made for the outcome definition using the ML algorithm.	The cumulative incidence of long COVID by day 300 was 6.3% [4.2–8.2] in participants who received metformin and 10.4% [7.8–12.9] in those who received identical metformin placebo (HR=0.59 [0.39–0.89]). The metformin beneficial effect was consistent across prespecified subgroups. When metformin was started within 3 days of symptom onset, the HR was 0.37 [0.15–0.95].

5.2. GRADE assessment (Summary of Findings table)

Table S4-5 Summary of Antiviral Drugs and the Incidence of Long COVID

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Impreci-sion	Public-ation bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Long COVID incidence (antiviral drugs)											
19	not serious	not serious	not serious	not serious	none	⊕⊕◯◯ Low	NA	NA	OR 0.69 (0.64 to 0.74)	NA	NA
Long COVID incidence (nirmatrelvir/ritonavir)											
13	not serious	not serious	not serious	not serious	none	⊕⊕◯◯ Low	NA	NA	OR 0.69 (0.60 to 0.78)	NA	NA
Long COVID incidence (ensitrelvir)											
1	not serious	not serious	not serious	serious[a]	none	⊕◯◯◯ Very low	NA	NA	OR 0.77 (0.53 to 1.13)	NA	NA
Long COVID incidence (azvudine)											
1	not serious	not serious	not serious	serious[a]	none	⊕◯◯◯ Very low	NA	NA	OR 0.81 (0.57 to 1.16)	NA	NA
Long COVID incidence (nirmatrelvir)											
1	not serious	not serious	not serious	serious[a]	none	⊕◯◯◯ Very low	NA	NA	OR 0.88 (0.87 to 0.89)	NA	NA

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Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Impreci-sion	Public-ation bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Long COVID incidence (remdesivir)											
2	not serious	not serious	not serious	serious[a]	none	⨁◯◯◯ Very low	NA	NA	OR 0.34 (0.09 to 1.23)	NA	NA
Long COVID incidence (favipiravir)											
1	not serious	not serious	not serious	serious[a]	none	⨁◯◯◯ Very low	NA	NA	OR 0.49 (0.34 to 0.70)	NA	NA
Long COVID incidence (molnupiravir)											
3	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	NA	NA	OR 0.86 (0.77 to 0.96)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to wide confidence interval of results.

Table S4-6 Summary of Antiviral Drugs and the Secondary Outcomes

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Fatigue (antiviral drugs)											
6	not serious	serious[a]	not serious	not serious	none	⨁◯◯◯ Very low	NA	NA	OR 0.90 (0.84 to 0.97)	NA	NA
Cognitive impairment (antiviral drugs)											
5	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	NA	NA	OR 0.83 (0.74 to 0.94)	NA	NA
Sleep disorders (antiviral drugs)											
5	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	NA	NA	OR 0.94 (0.89 to 0.98)	NA	NA
Dyspnea (antiviral drugs)											
4	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	NA	NA	OR 0.94 (0.72 to 1.23)	NA	NA
Myalgia (antiviral drugs)											

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Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
3	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	NA	NA	OR 0.88 (0.70 to 1.10)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to serious inconsistency because of statistical heterogeneity.

Table S4-7 Summary of Monoclonal Antibodies and the Incidence of Long COVID

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Long COVID incidence (Monoclonal Antibodies)											
7	not serious	serious[a]	not serious	not serious	none	⨁◯◯◯ Very low	NA	NA	OR 0.92 (0.69 to 1.22)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to serious inconsistency because of statistical heterogeneity.

Table S4-8 Summary of Monoclonal Antibodies and the Secondary Outcomes

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Fatigue (Monoclonal Antibodies)											
1	not serious	not serious	not serious	serious[a]	none	⨁◯◯◯ Very low	NA	NA	OR 0.79 (0.54 to 1.15)	NA	NA
Cognitive impairment (Monoclonal Antibodies)											
1	not serious	not serious	not serious	serious[a]	none	⨁◯◯◯ Very low	NA	NA	OR 0.81 (0.46 to 1.42)	NA	NA
Sleeping disorders (Monoclonal Antibodies)											
1	not serious	not serious	not serious	serious[a]	none	⨁⨁◯◯ Low	NA	NA	OR 1.10 (0.67 to 1.82)	NA	NA
Dyspnea (Monoclonal Antibodies)											

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Certainty assessment							Summary of findings				
1	not serious	not serious	not serious	serious[a]	none	⊕◯◯◯ Very low	NA	NA	OR 0.74 (0.47 to 1.15)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to wide confidence interval of results.

Table S4-9 Summary of Corticosteroids and the Incidence of Long COVID

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Long COVID incidence (Corticosteroids)											
5	not serious	serious[a]	not serious	not serious	none	⊕◯◯◯ Very low	NA	NA	OR 0.97 (0.74 to 1.25)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to serious inconsistency because of statistical heterogeneity.

Table S4-10 Summary of Corticosteroids and the Secondary Outcomes

Certainty assessment							Summary of findings				
Participants (studies) Follow-up	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Study event rates (%)		Relative effect (95% CI)	Anticipated absolute effects	
							With non-antiviral drugs	With antiviral drugs		Risk with non-antiviral drugs	Risk difference with antiviral drugs
Dyspnea (Corticosteroids)											
1	not serious	not serious	not serious	serious[a]	none	⊕◯◯◯ Very low	NA	NA	OR 0.74 (0.47 to 1.15)	NA	NA

CI: confidence interval; OR: odds ratio; OS: observational studies; NA: Not Applicable. Explanations: a. Downgraded one level due to wide confidence interval of results.

Section 2: vaccines

Part 1: Search resources and results

Table S4-11 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	805
	Web of Science Core Collection	Nov 30, 2024	756
	Embase	Nov 30, 2024	352
	The Cochrane Library	Nov 30, 2024	84
	China Biology Medicine (CBM)	Nov 30, 2024	4
	Wanfang Database	Nov 30, 2024	80
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	67
Preprint platform	Medrxiv	Nov 30, 2024	-
	SSRN	Nov 30, 2024	-
	Researchsquare	Nov 30, 2024	-
Clinical trial registries	WHO ICTRP	Nov 30, 2024	-
	ClinicalTrials.gov	Nov 30, 2024	-
	COVID-19 Study Register	Nov 30, 2024	-

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Google	-	Nov 30, 2024	2
Reference lists of relevant SRs	-	Nov 30, 2024	-

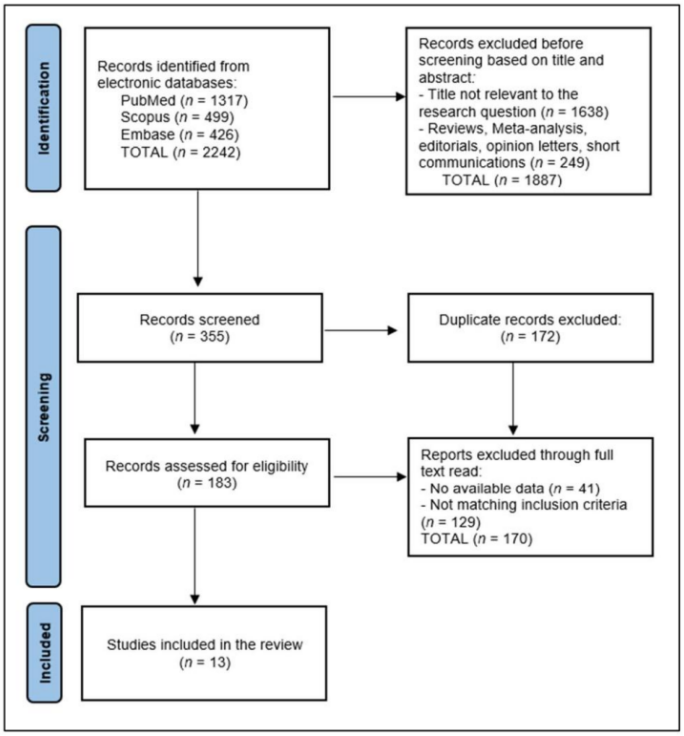
Part 2: SRs used to support the PICO question

Systematic Review 1: Impact of Pre-Infection COVID-19 Vaccination on the Incidence and Severity of Post-COVID Syndrome: A Systematic Review and Meta-Analysis

Published Journal: Vaccines (Basel). 2024;12(2):189.

Literature Search Period: The systematic review search was carried out in October 2023 across three electronic databases, including PubMed, Embase, and Scopus, focusing on literature published from 2020, since the inception of the COVID-19 pandemic, up to October 2023.

(1) Literature Screening Flowchart

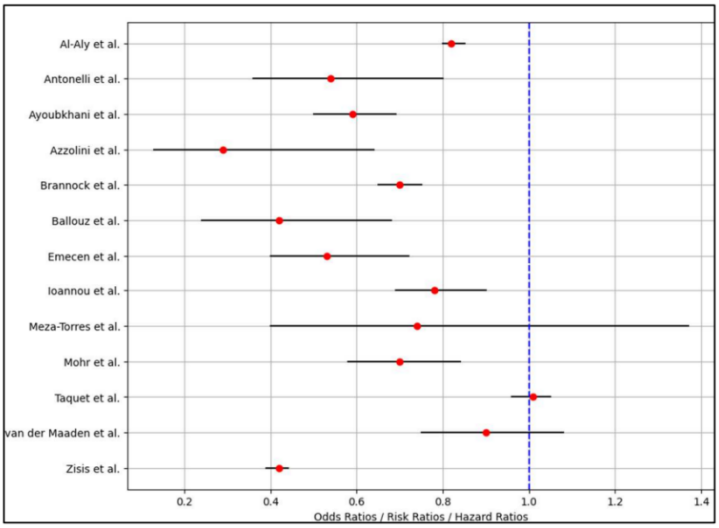


(2) Information List of Included Studies and Quality Assessment Results

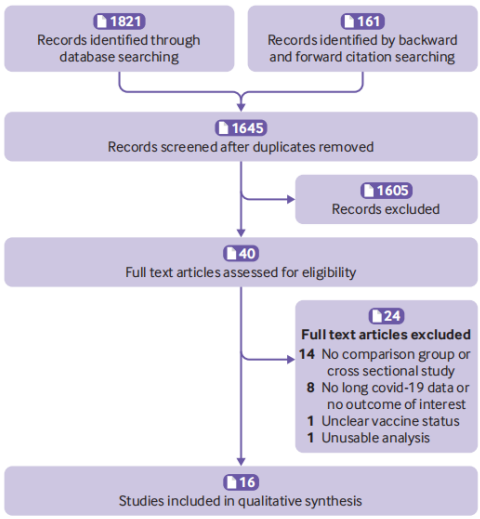
Reference Number and First Author	Country	Publication Year	Study Design	Study Quality
Al-Aly et al.	United States	2022	Retrospective cohort	Medium
Antonelli et al.	United Kingdom	2022	Prospective cohort	High
Ayoubkhani et al.	United Kingdom	2022	Retrospective cohort	Medium
Azzolini et al.	Italy	2022	Retrospective cohort	Medium
Brannock et al.	United States	2023	Retrospective cohort	High
Ballouz et al.	Switzerland	2023	Retrospective cohort	Medium
Emecen et al.	Turkey	2023	Prospective cohort	High
Ioannou et al.	United States	2022	Retrospective cohort	High
Meza-Torres et al.	United Kingdom	2022	Retrospective cohort	Medium

(3) Meta-analysis Results: Forest plot analysis of the risk for long COVID among the analyzed studies

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Systematic Review 2: Effect of covid-19 vaccination on long covid: systematic review
Published Journal: Vaccines (Basel). 2024;12(2):189.
Literature Search Period: PubMed, Embase, and Cochrane covid-19 trials for published studies, and Europe PubMed Central (Europe PMC) for preprints, from 1 January 2020 to 3 August 2022.
(1) Literature Screening Flowchart



(2) Information List of Included Studies of vaccines given before infection

Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
Al-Aly2022, USA	Retrospectively assembled US Veterans Health Ad-cohort 1 January - 1 December 2021	ministration electronic health databases	n = 33940 patients who were fully vaccinated 214 days before a positive coxid-19 test result, who were still alive 30 days after a positive test result	One dose of Janssen, or two doses of Mod erna or Pfizer	n - 113474 propensity score matched patients who were not vaccinated and alive 30 days after a positive cvoid-19 test result and no vaccination	Risk of 21 sequelae after the acute infection at 6 months

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Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
Antonelli 2022, UK	Prospective, community based, nested, case-control study, 8 December 2020 - 4 July 2021	Covid-19 symptom study app users	n - 3071 adult app users with a positive covid-19 test result ≥14 days after their first vaccine or at least seven days after their second vaccine; had no positive test result before vaccination and who had used the app for 214 consecutive days after the test	First or second dose of Pfizer, Moderna, or AstraZeneca	n - 3244 participants reporting a positive SARS-CoV-2 test result who were not vaccinated and who had used the app for 214 days after the test	Long duration (≥1 month) of symptoms after one dose. Most common symptoms
Ayoubkhani 2022, UK, preprint	Retrospectively assembled cohort, 26 April 2020 - 30 November 2021	Covid-19 infection survey participants	n = 3090 adult participants who tested positive for SARS-CoV-2 between 26 April2020 and 30 November 2021; who had received two vaccines at Moderna least two weeks before infection	Two or more doses of AstraZeneca, Pfizer, or	n - 3090 1:1 propensity score matched patients based on sociodemographic characteristics and time from infection to follow up for long covid who were not vaccinated	Rates of long covid >3 months after infection
Azzolini 2022, Italy	Retrospective cohort, March 2020 - April 2022	Healthcare workers from nine hospitals	n = 318 healthcare workers who received >1 dose of One, two, or three vaccine before SARS-CoV-2 infection	doses of Pfizer	n = 421 healthcare workers who were not vaccinated before infection	Rates of long covid >1 month after infection

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Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
loannou2022, USA	Retrospectively assembled cohort 1 Feb 2020 - 30 Apr 2021	US Veterans Health Ad ministration electronic health databases	n - 8357 people with 1 - 2 doses of vaccines with a positive SARS-CoV-2 test result between 1 February 2020 and 30 April 2021, who were still alive 3 months after infection, with no evidence of reinfection	One or two doses of Moderna or Pfizer	n - 58693 people who were not vaccinated at the time of infection, who were still alive three months after infection, with no evidence of reinfection	Rates and risk factors of long covid. Care at >3 months after infection
Mohr 2022, USA preprint	Prospective cohort, Decem ber 2020 - August 2021	Healthcare profes sionals in 12 states participating in vaccine effectiveness study (PREVENT trial)	n - 180 healthcare professionals who had two dos- Two doses of Moderna n-239healthcare professionals es of vaccines 14 days before could-19 disease	or Pfizer	who had covid-19 with no previ ous vaccination	Presence of symptoms 1.5 months after onset of covid-19 disease.
Pell 2022, UK preprint	Prospective cohort, April 2020 - May2021	Long Covid in Scotland Study (Long CISS)	n - 1154 adults with 1 - 4 doses of vaccine, who had Not specified a positive PCR test result		n - 32127 adults who were not vaccinated	Confusion and difficulty concentrating at 6 - 18 months
Simon 2021, USA, preprint	Retrospectively assembled cohort, February 2020 - May 2021	Arcadia data research dataset with >150 mil lion patient records	n - 2392 patients who had their first dose of vac cine before a diagnosis of covid-19	One dose of Pfizer Moderna, or Janssen	n - 220460patients who were not vaccinated before covid-19 and 12 weeks after	Odds of having long covid at 3 months
Tannous 2022 USA, preprint	Retrospectively assembled cohort, 3 March 2020 - 20 November 2021	Houston Methodist Covid-19 Surveillance and Outcomes Registry (CURATOR) electronic	n - 3781 adult patients vaccinated >14 days before Two doses of mRNA covid-19 disease	vaccines or one dose of Janssen	n - 49458 patients who were not vaccinated	Odds of having long covid >1 month. Most common symptoms

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Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
		health record database				
Taquet 2021, USA	Retrospectively assembled cohort, 21 September 2020 - 14 December 2021	TriNetX Analytics, a federated network of linked electronic health records, with 81 million patient records	n=25225 adult patients who were vaccinated with a confirmed diagnosis of covid-19 Not specified	>1 dose of Janssen or >2 doses of mRNA vaccine	n=25225 1:1 propensity score matched patients who were not vaccinated	Odds of having any long covid symptoms within 6 months
van der Maaden, Netherlands, pre- print	Prospective cohort, 19 May - 13 December 2021	Dutch prospective long covid study	n=3838 adult patients (aged <65), who were fully vaccinated three months after a positive SARS-CoV-2 test result	Pfizer, Moderna, or Janssen	n=528 patients (<65 years) who were not vaccinated, three months after a positive SARS-CoV-2 test result	Odds of at least one significantly raised long covid symptom at >2 months
Zisis 2022, USA	Retrospectively assembled cohort, 1 Janurary - 31 August 2021	TriNetX Analytics, a federated network of linked electronic health records, with 81 million patient records	n=9479 adults who received a covid-19 vaccine at least 2 weeks before SARS-CoV-2 infection	Pfizer, Moderna, or Janssen	n=9479 propensity matched patients who had received the influenza vaccine at any time	New onset of long covid symptoms such as fatigue at 3 month follow-up
Ayoubkhani 2022, UK	Interrupted time series, 3 February - 5 September 2021	Covid-19 infection survey participants	n=28356 patients with long covid, who had received at least one vaccine after diagnosis	AstraZeneca, Pfizer, or Moderna	Self-controlled (symptoms before vaccine)	Long covid of any severity after first and second dose ≥3 months. 10 most common symptoms

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Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
Simon 2021, USA, preprint	Retrospectively assembled cohort, February 2020 - May2021	Arcadia data research dataset with >150 million patient records	n=17796 patients with a diagnosis of covid-19, by PCR or ICD-10 code U07.1 at any time or B97.29 before May 2020, and who were vaccinated within 12 weeks after a diagnosis of covid-19	One dose of Pfizer, Moderna, or Janssen	n=220460 patients who were not vaccinated before covid-19 and 12 weeks after	Odds of having long covid at 3 months
Tran 2021, France, preprint	Prospective cohort, December 2020 - September 2021	ComPaRe (cohort of patients with chronic diseases)	n=455 adults with a confirmed or suspected SARS CoV-2 infection and at least one symptom attributable to long covid reported at baseline and persisting for >3 weeks	AstraZeneca, Pfizer,Moderna, or Janssen	n=455 1:1 propensity matched controls from the same cohort who were not vaccinated	Remission of all long covid symptoms by four months
Wisnivesky 2022, USA	Prospective cohort, 20 July 2020 - August 2021	Patients with covid-19 enrolled in prospective registry established at Mount Sinai Health System	n=324 adult patients with a history of laboratory confirmed covid-19, with one or more symptoms of long covid, treated at Mount Sinai and who were fully vaccinated	Pfizer,Moderna, or Janssen	n=129 patients from the same cohort who were not vaccinated	Fatigue at six months
Wynberg 2022, Netherlands	Prospective cohort, 11 May 2020 - 1 November 2021	RECOVERED study participants	n=36 patients with long covid, aged 16 - 85 years, with laboratory confirmed covid-19, Amsterdam residents,	AstraZeneca, Pfizer,Moderna, or Janssen	n=32 participants who were not vaccinated matched 1:1 on age, sex, obesity status, and time since illness onset to participants	Recovery from long covid symptoms at ≥3 months since onset of illness

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Study, year, country	Study type and timeframe	Patient or data source	Intervention population	Intervention	Comparator	Outcomes of interest and length of follow-up
			with >3 months of follow-up		who were vaccinated	

PCR=polymerase chain reaction; PREVENT=Project PREVENT (PReventing Emerging Infections through Vaccine EffectiveNess Testing Project) trial

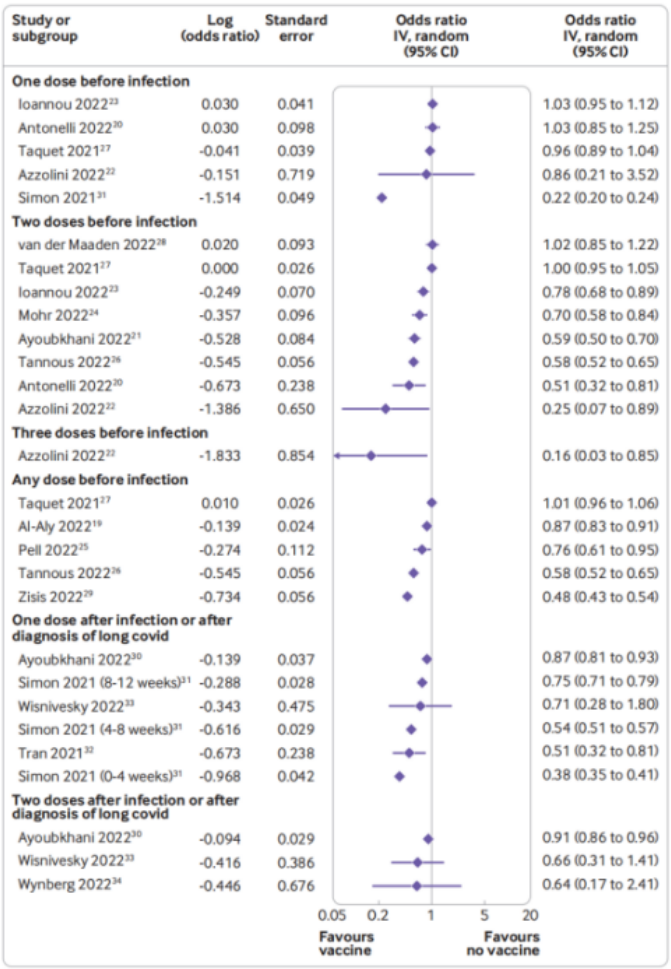
(3) Quality Assessment Results

Study, year	Confounding	Selection of participants	Classification of interventions	Deviations from intended intervention	Missing data	Measurement of outcomes	Overall risk of bias	Selection of reported result
Al-Aly2022	Low	Low	Low	Low	Low	Moderate	Low	Moderate
Antonelli2022	Moderate	Low	Low	Low	Serious	Serious	Low	Serious
Ayoubkhani2022	Critical	Low	Low	Low	Moderate	Low	Low	Critical
Azzolini2022	Moderate	Low	Moderate	Low	NI	Critical	Low	Critical
loannou2022	Moderate	Low	Low	Low	NI	Low	Moderate	Moderate
Mohr 2022	Serious	Serious	Low	Low	Serious	Moderate	Moderate	Serious
Pell2022	Serious	Low	Serious	Low	Serious	Moderate	Moderate	Serious
Tannous2022	Serious	Low	Low	Low	Low	Serious	NI	Serious
Taquet2021	Moderate	Low	Low	Low	NI	Low	Low	Moderate
van der Maaden 2022	Serious	Low	Moderate	Low	Low	Low	Low	Serious
Zisis 2022	Serious	Critical	Serious	Low	NI	Serious	NI	Critical
Ayoubkhani2022	Low	Low	Low	Low	Moderate	Low	Low	Moderate
Simon 2021	Serious	Low	Low	Low	Moderate	Low	Low	Serious
Tran 2021	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Wisnivesky 2022	Serious	Serious	Serious	Low	Low	Low	Low	Serious
Wynberg 2022	Serious	Moderate	Serious	Low	Low	Serious	Low	Serious

NI=no information

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(4) Meta-analysis Results: Forest plot of the effect of covid-19 vaccine doses on long covid

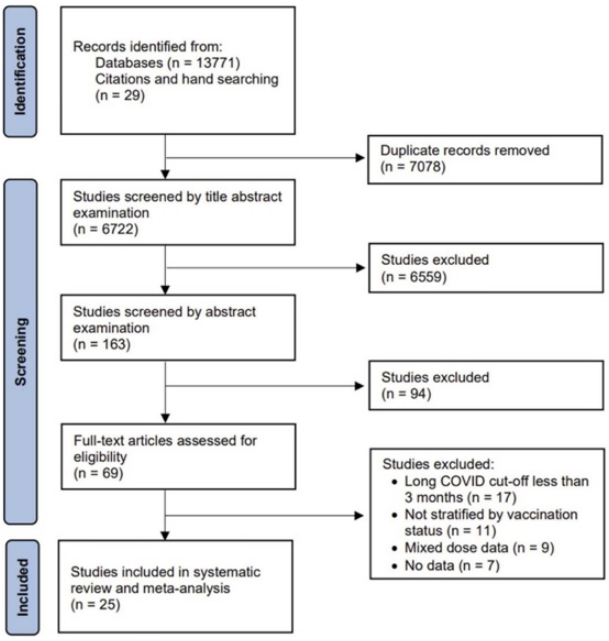


Systematic Review 3: The effect of pre-COVID and post-COVID vaccination on long COVID: A systematic review and meta-analysis

Published Journal: Journal of Infection. 2024;89(6):106358.

Literature Search Period: Included both peer-reviewed and pre-print studies published from May 25, 2021, to February 18, 2024, including case-control, prospective cohort, retrospective cohort, and cross-sectional studies that investigate the relationship between vaccination and long COVID symptoms.

(1) Literature Screening Flowchart



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(2) Information List of Included Studies

Study characteristics					Participant characteristics		
Author, year	Country	Study design	Study period	Study population	Sample size	Age	Male (%)
Al-Aly 2022	USA	Retrospective cohort	1 January - 1 December 2021	the Department of Veterans Affairs (VHA) population	13482547	Mean: 57.81 SD: 15.94	vac, inf: 91.03 unvac, inf: 86.92
Alghamdi 2022	Saudi Arabia	Retrospective cohort	Not specified	general population (online survey)	2218	12–17 (6.9%) 18–20 (17.7%) 21-30 (37.7%) 31-40 (19.5%) 41-50 (13.0%) 51-60 (4.0%) 61-70 (0.9%)	38.95
Ayoubkhani 2022	UK	Prospective cohort	26 April 2020 - 30 November 2021	general population (UK COVID-19 infection survey)	6180	Mean, SD - vaccinated, infected: 49.0, 12.0 - unvaccinated, infected: 46.7, 11.2	46.04
Azzolini 2022	Italy	Prospective cohort	March 2020 - April 2022	healthcare workers in 9 Italian health care facilities	739	Mean with long COVID: 44.3 without long COVID: 41.2	25.44
Brannock 2022	USA	Retrospective cohort	1 August 2021 - 31 January 2022	electronic health records from 11 healthcare institutions taking part in the NIH Researchng COVID to Enhance Recovery (RECOVER) Initiative, a project of the National Covid Cohort Collaborative (N3C)	47752	18-24 (9.5%) 25-34 (17.9%) 35-49 (26.1%) 50-64 (27.2%) 65+ (25.4%)	35
Budhiraja 2022	India	Prospective cohort	March 2020 - February 2022	Tertiary care hospitals in the National Capital Regon of Delhi	5529	< 18 (1.2%) 18-44 (26.9%) 45-59 (29.3%) 60-74 (31.3%) 75 or above (11.3%)	64.6
Carazo 2022	Canada	Retrospective cohort	July 2020 - May 2021	healthcare workers in Quebec	PCC4w analysis: 6,061 PCC12w analysis: 1,783 Control: 4,390 Total: 12234	Mean, SD Symptomatic hospitalized: 46.7, 11.9 Symptomatic non-hospitalized: 40.0, 12.1 Controls: 39.0, 10.4	Symptomatic hospitalized: 29.7 Symptomatic non-hospitalized: 20.8 controls: 13.0
Emecen 2022	Turkey	Prospective cohort	1 November 2020 - 31 May 2021	general population (from the DEU-COVIMER, a telephone interview-based COVID-19 follow-up center	5610	18–49 years (67.7%) 50–64 years (22.4%) ≥65 years (9.9%)	48.2

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				established in a tertiary care hospital)			
Fusco 2023	USA	Prospective cohort	21 January 2022 - 30 October 2022	adults testing for SARS-CoV-2 at one of 88 CVS Health test sites across the US	328	Mean, SD: 42, 14.5	26.2
Hajjaji 2022	France	Prospective cohort	April 2020 - August 2021	cancer patients treated at Oscar Lamret Cancer Center (Lille, France)	168	Long Covid <50 (24%) 50-59 (30%) 60-69 (35%) ≥70 (11%) Non-Long Covid <50 (25%) 50-59 (31%) 60-69 (24%) ≥70 (20%)	Long Covid: 19 No Long Covid: 31
Ioannou 2022	USA	Retrospective cohort	February 2020 - December 2021	data from Veterans Affairs Health Care System, which provides medical care at 171 medical centres throughout the US	198601	18-49 (26.3%) 50-59 (18.4%) 60-64 (10.5%) 65-69 (10.4%) 70-74 (17.6%) 75-79 (8.3%) 80-84 (3.9%) 85-89 (2.9%) ≥90 (1.7%)	89.1
Kim 2023	USA	Cross-sectional study	14 September 2022 - 14 November 2022	general population, using data from the US Census Bureau's Household Pulse Survey	153 543	not specified	not specified
Kuodi 2022	Isreal	Cross-sectional study	March 2020 - November 2021	≥18yo participants tested positive by RTPCR between 15th March 2020 and 15th November 2021 in three major government hospitals in Northern Israel	951	19-35 (30.3%) 36-60 (49.2%) >60 (20.5%)	29.76
Messiah 2022	USA	Prospective cohort	October 2020 - May 2022	general population (the Texas Coronavirus Antibody REsponse Survey, a statewide prospective population-based survey among 5-90 years old)	1801	not specified	not specified
Richard 2023	USA	Prospective cohort	28 February 2020 - 31 December 2021	data from Epidemiology, Immunology, and Clinical Characteristics of Emerging Infectious Diseases With Pandemic Potential (EPICC) study implemented in the US military health system (MHS)	1832	18-44 (66.9%) 45-64 (26.6%) 65 or above (6.5%)	61

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Taquet 2022	USA	Retrospective cohort	January 1 2021 - August 31 2021	US patients in the TrinNetX database: Primary cohort: confirmed SARS-CoV-2 infection occurred at least 14 days after a recorded administration of a COVID-19 vaccine Control cohort: patients were included only if no vaccine against COVID-19 was recorded before their SARS- CoV-2 infection and if they had received a vaccine against influenza at any time	18958	mean, SD vaccinated, infected: 56.5, 18.0 unvaccinated, infected: 57.6, 20.6	vac, inf: 40.1 unvac, inf: 39.2 total: 39.67
Van der Maaden 2022	Netherlands	Prospective cohort	19 May 2021 - 13 December 2021	general population (data were collected in the context of the Dutch prospective Long COVID study)	cases: 9199 symptomatic but test-negative controls: 1698 population controls: 3708 total: 14605	median (IQR) cases: 49 (37-61) test-negative controls: 55.3 (41-65) population controls: 52.2 (40-60)	cases: 36 test-negative controls: 33.9 population controls: 32.8
Wong 2023	China (Hong Kong)	Cross-sectional study	2 June 2022 - 28 June 2022	community-dwelling patients with COVID-19 included from Qualtrics platform in Beijing, Shanghai, Guangzhou and Hong Kong	2712	18-24 (5.1%) 25-34 (42.2%) 35-44 (38.1%) 45-54 (11.1%) 55-64 (2.6%) 65 or above (0.7%)	40.2
Zisis 2022	USA	Retrospective cohort	1 January 2021 - 31 August 2021	Patients' data from the TriNetX Research Network platform, a network of elec-tronic medical records. Patients were infected with SARS-CoV-2 (confirmed by PCR) and sought care in the United States from 21 September 2020 to 14 December 2021	50450	mean (SD) - vaccinated, infected: 54.82 (17.77) - unvaccinated, infected: 55.06 (17.96)	40.09
Ayoubkhani 2022	UK	Prospective cohort	3 February 2021 - 5 September 2021	Participants in the Office for National Statistics COVID-19 Infection Survey, a longitudinal survey of people aged 2 years or older in randomly sampled UK households, excluding hospitals, care homes, halls of residence, and prisions	28356	mean: 45 SD: 13.6	44
El Otmani 2022	Morocco	Case control	February 2021 - April 2021	Healthcare staff working in Ibn	236	median: 29 IQR: 21-54	29

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				University Hospital (doctors, nurses, medical assistants, technicians, physiotherapists, psychologists)			
Nehme 2022	Switzerland	Retrospective cohort	23 April 2021 - 27 July 2021	Individuals who previously tested positive for SARS-CoV-2 infection at the outpatient testing center of the Geneva University Hospitals, Switzerland	2094	mean: 43.5 SD: 13.7	44.7
Pereira 2023	UK	Prospective cohort	September 2020 - March 2021	Participants from the CLoCk Study, a national cohort study of SARS-CoV-2 PCR-positive children and young people aged 11-17 years	3395	11-14 (43%) 15-17 (57%)	37.7
Simon 2022	USA	Retrospective cohort	February 2020 - May 2021	Data for this analysis were collected from Arcadia Data Research, a electronic health record systems. Data were captured directly from electronic health record (EHR) systems and practice management systems.	240648	0-24 (15.8%) 25-44: (20.1%) 45-64 (28.6%) 65+ (28.2%)	40.1
Wynberg 2022	UK	Prospective cohort	11 May 2020 - 21 June 2021	Participants enrolled from RECoVERED, a cohort study of individuals with SARS-CoV-2 infection in the municipal region of Amsterdam, the Netherlands	316	median: 51	57

(3) Quality Assessment Results

Author, year	Confounding	Selection	Intervention classification	Deviation from intended intervention	Outcome classification	Loss of data	Selective reporting	Average score	Risk of bias
Al-aly 2022	2	3	1	1	1	1	1	1.43	Moderate
Alghamdi 2022	2	2	1	1	2	2	1	1.57	Moderate
Ayoubkhani 2022	2	1	1	1	2	1	1	1.29	Low
Azzolini 2022	1	2	1	1	2	1	1	1.29	Low
Brannock 2022	2	2	1	1	1	1	1	1.29	Low
Budhiraja 2022	2	3	1	1	2	1	1	1.57	Moderate
Carazo 2022	3	3	1	1	3	1	1	1.86	Serious
Emecen 2022	2	1	1	1	2	2	1	1.43	Moderate
Fusco 2023	2	1	1	1	2	2	1	1.43	Moderate
Hajjaji 2022	4	4	1	1	2	1	1	2.00	Serious
Ioannou 2022	2	3	1	1	1	1	1	1.43	Moderate

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Author, year	Confounding	Selection	Intervention classification	Deviation from intended intervention	Outcome classification	Loss of data	Selective reporting	Average score	Risk of bias
Kim 2023	2	2	1	1	2	1	1	1.43	Moderate
Kuodi 2022	2	1	1	1	2	1	1	1.29	Low
Messiah 2022	2	2	1	1	2	1	1	1.43	Moderate
Richard 2023	2	3	1	1	2	2	1	1.71	Moderate
Taquet 2022	2	1	1	1	1	1	1	1.14	Low
Van der Maaden 2022	1	1	1	1	2	1	1	1.14	Low
Wong 2023	1	2	1	1	2	1	1	1.29	Low
Zisis 2022	1	2	1	1	2	1	1	1.29	Low
Ayoubkhani 2022	1	1	2	1	2	1	1	1.29	Low
El Otmani 2022	2	1	2	1	2	1	1	1.43	Moderate
Nehme 2022	1	1	1	1	1	1	1	1.00	Low
Pereira 2023	1	3	1	1	2	1	1	1.43	Moderate
Simon 2022	1	1	2	1	2	1	1	1.29	Low
Wynberg 2022	2	2	1	1	1	1	1	1.29	Low

(4) Meta-analysis Results

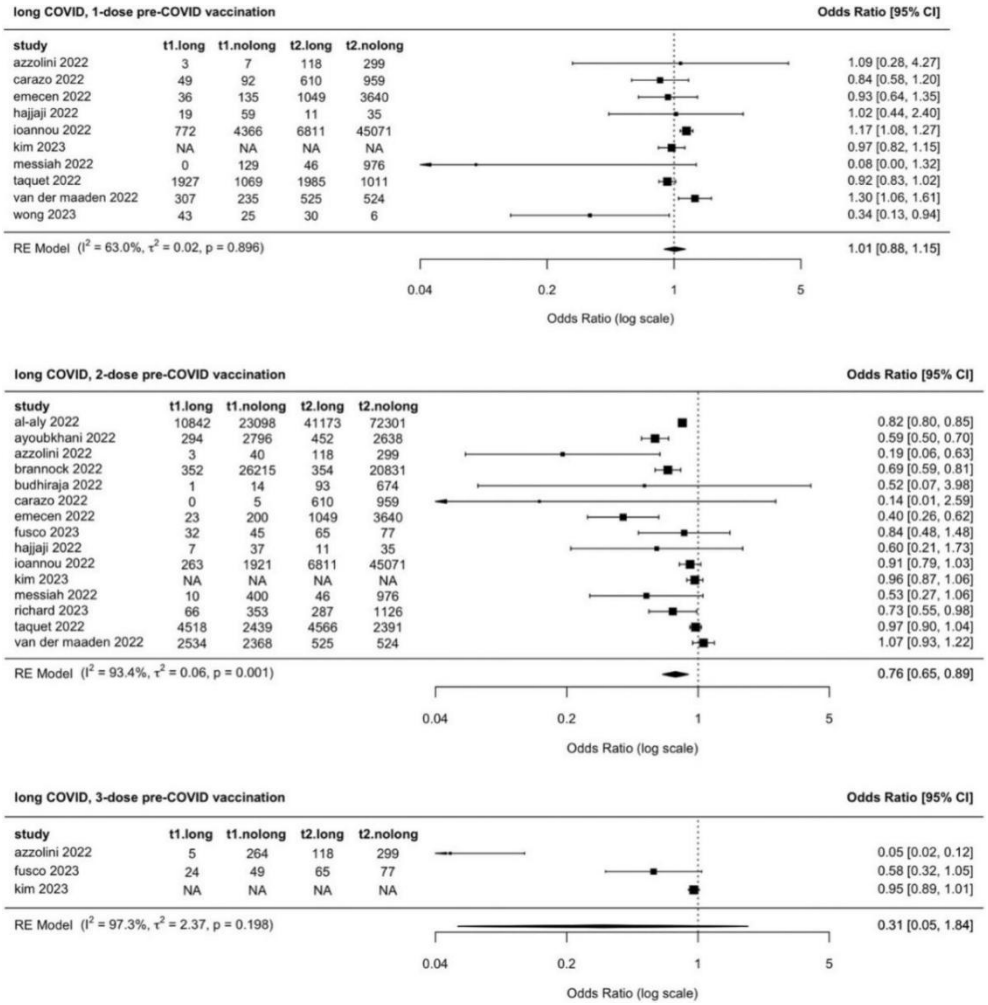


Fig. a-c: Forest plot of 1-dose, 2-dose, 3-dose pre-COVID vaccination against long COVID. The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long COVID; nolong = number of patients without long COVID; CI = confidence interval; NA = not available; RE model = random effects model; I^2 and τ^2 are heterogeneity measures; $P = P$ value of vaccine effectiveness.

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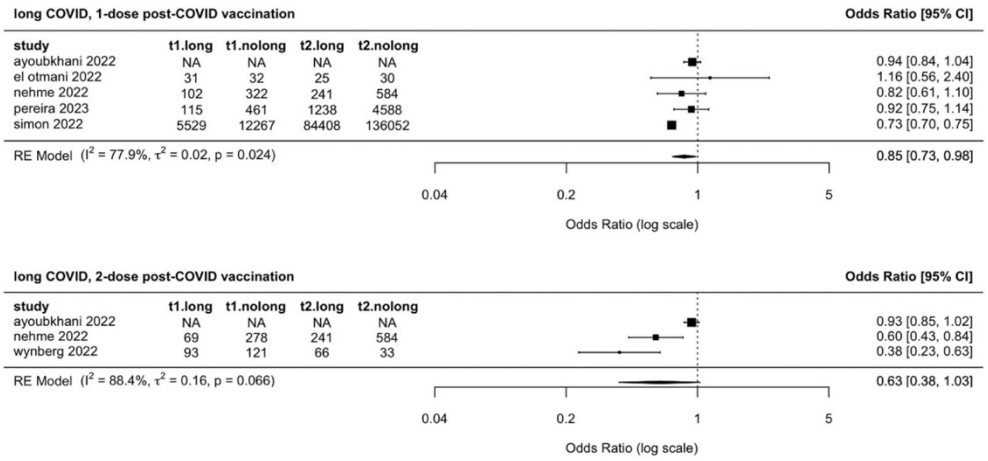


Fig. a-b: Forest plot of 1-dose and 2-dose post-COVID vaccination against long COVID. The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long COVID; nolong = number of patients without long COVID; CI = confidence interval; NA = not available; RE model = random effects model; I^2 and τ^2 are heterogeneity measures; P = P value of vaccine effectiveness.

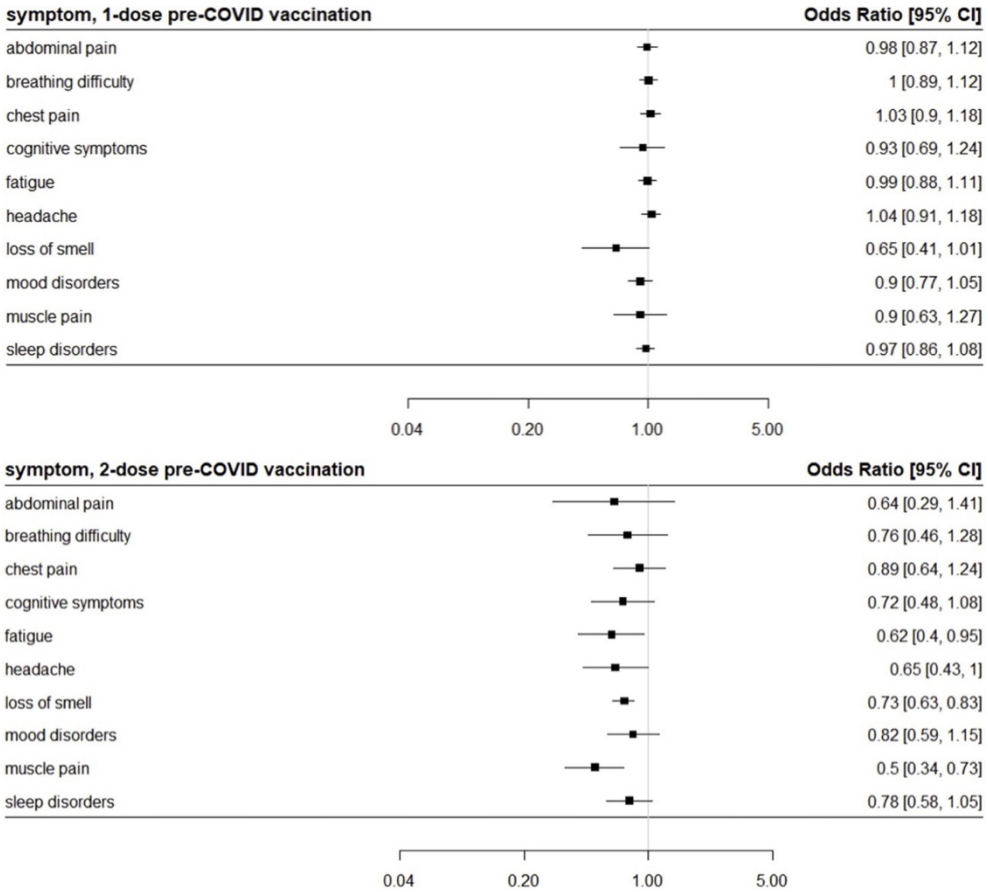
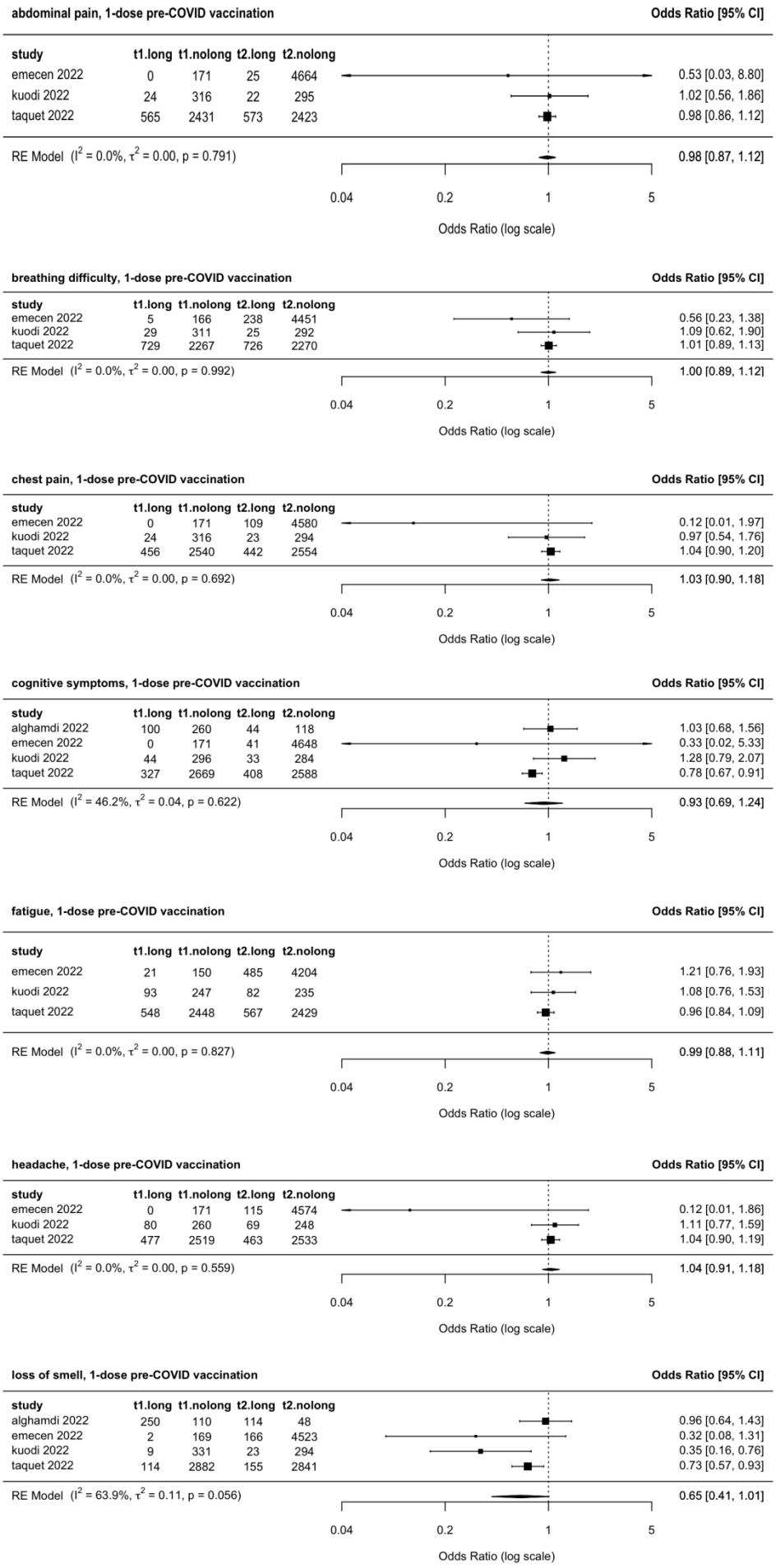


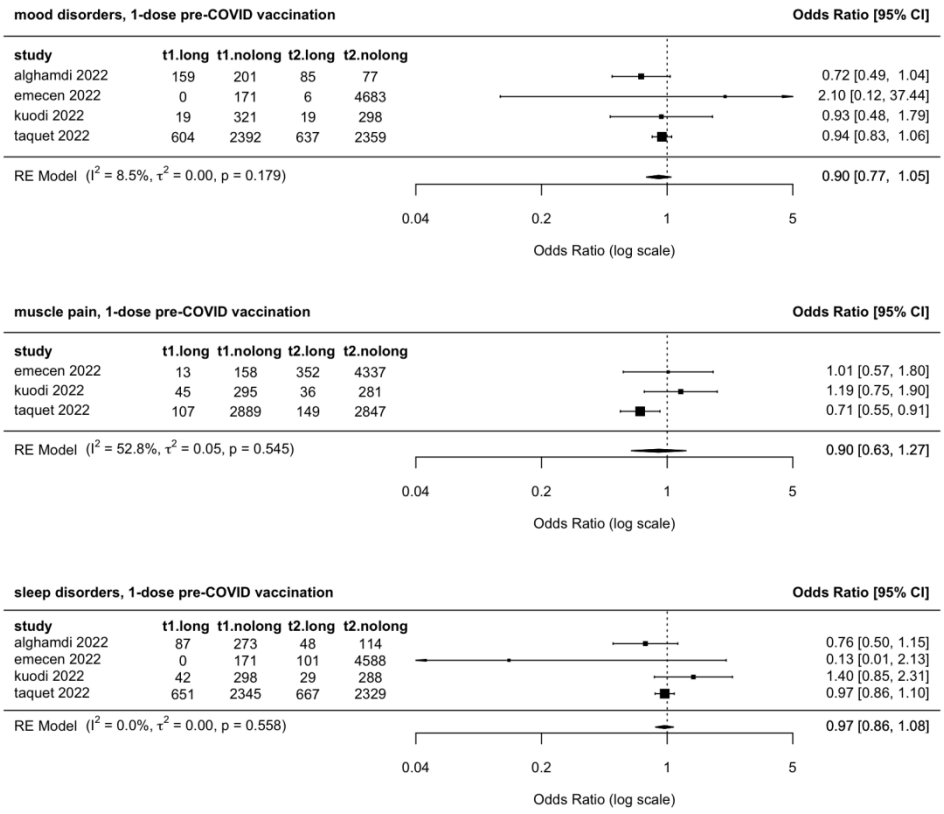
Fig. a-b: Pooled effectiveness of 1-dose and 2-dose pre-COVID vaccination against individual long COVID symptoms. Each bar represents a summary OR of all studies. The estimates were calculated as odd ratios using unvaccinated as the reference group. CI = confidence interval.

Supplementary figures a-j: forest plots for individual symptoms (1-dose pre-COVID vaccination)
The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long COVID; nolong = number of patients without long COVID; CI = confidence interval; NA = not available; RE model = random effects model; I^2 and τ^2 are heterogeneity measures; P = P value of vaccine effectiveness.

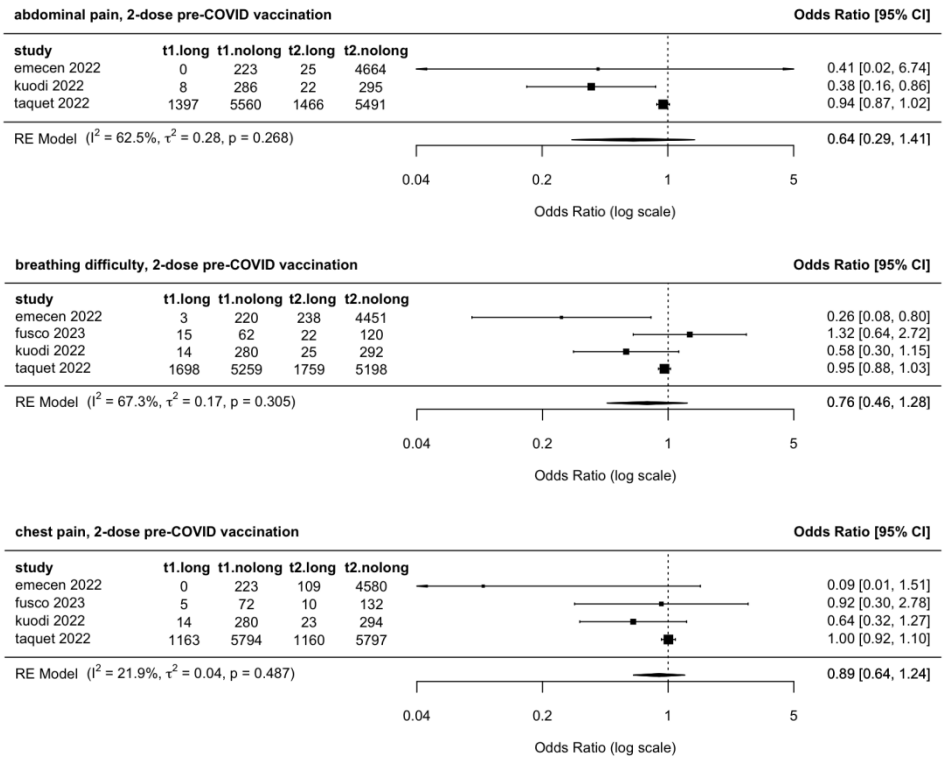
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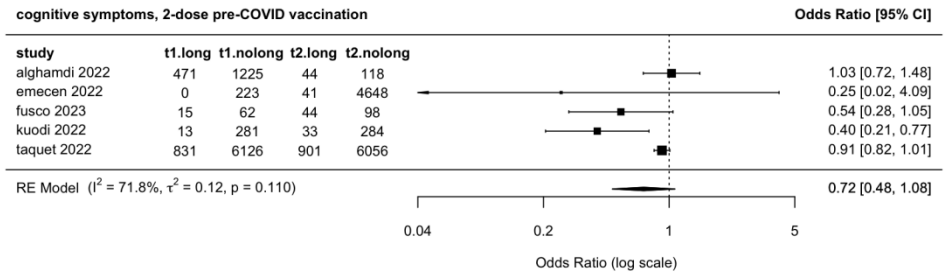
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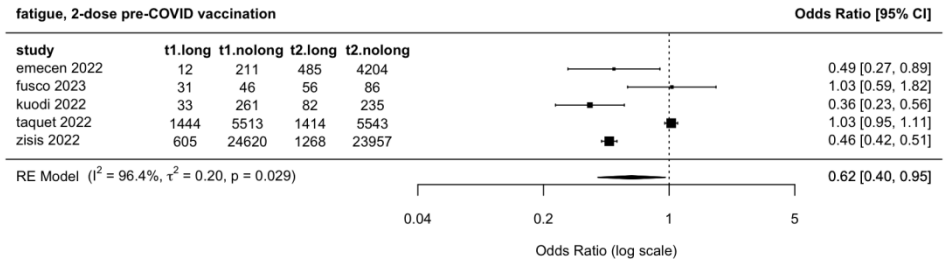


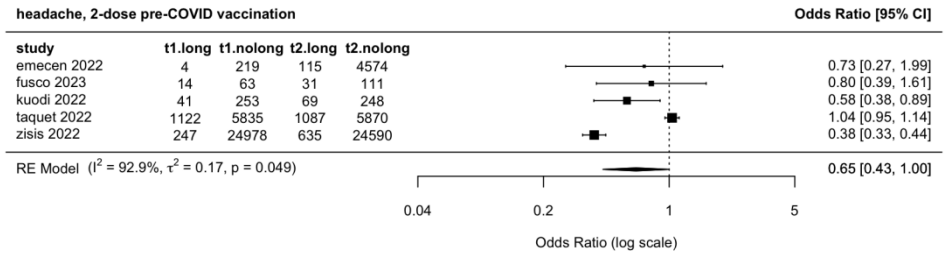
Supplementary figures a-j: forest plots for individual symptoms (2-dose pre-COVID vaccination)
The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long COVID; nolong = number of patients without long COVID; CI = confidence interval; NA = not available; RE model = random effects model; I^2 and τ^2 are heterogeneity measures; P = P value of vaccine effectiveness.

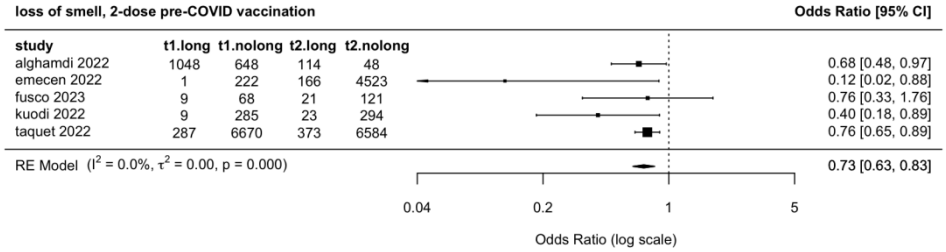


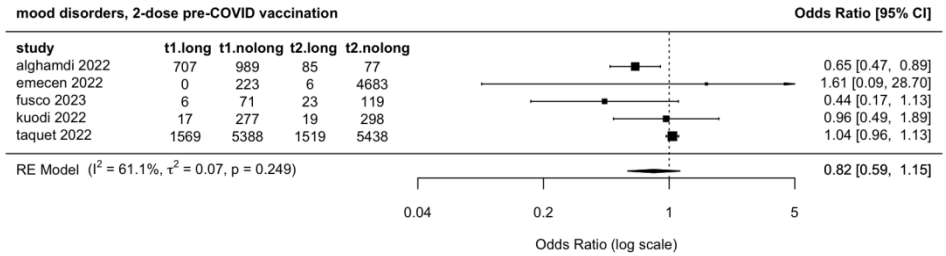
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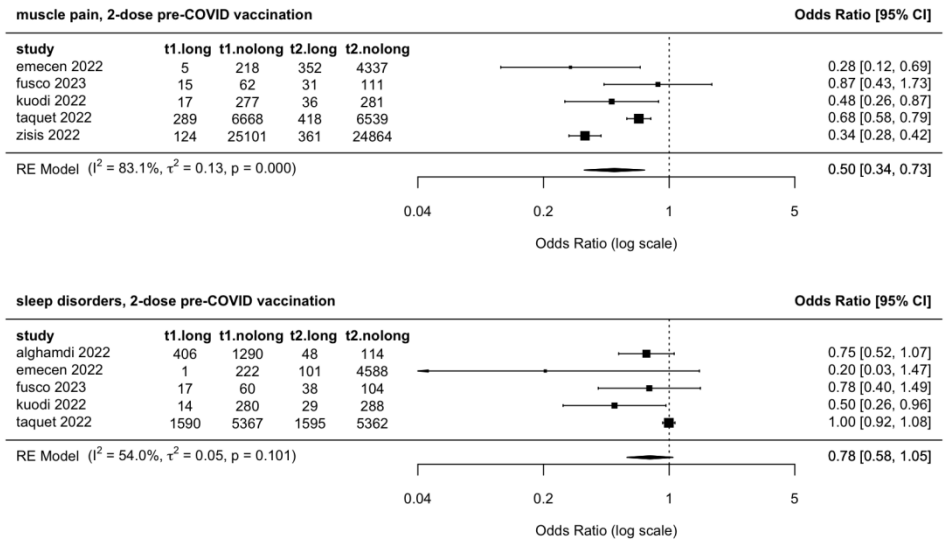




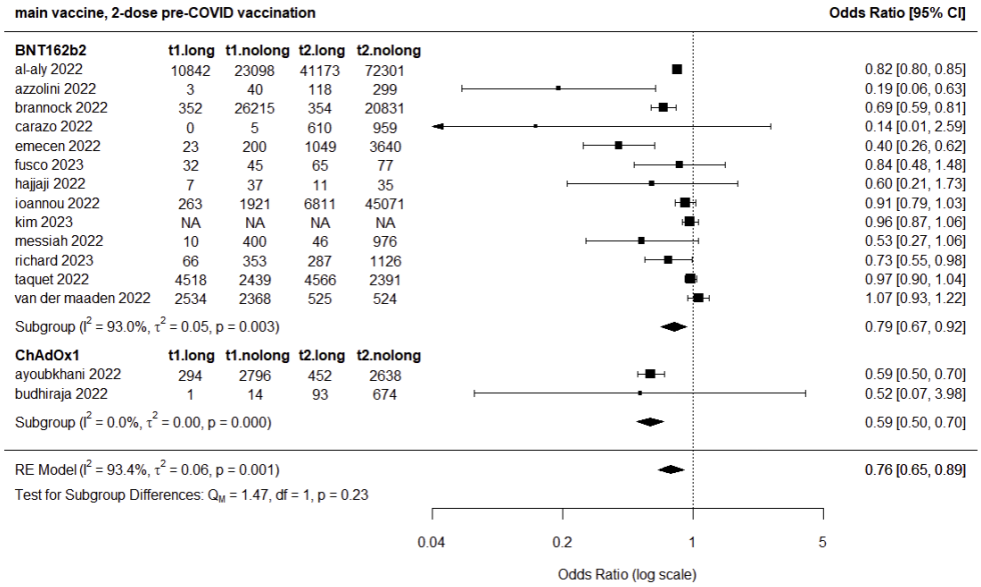




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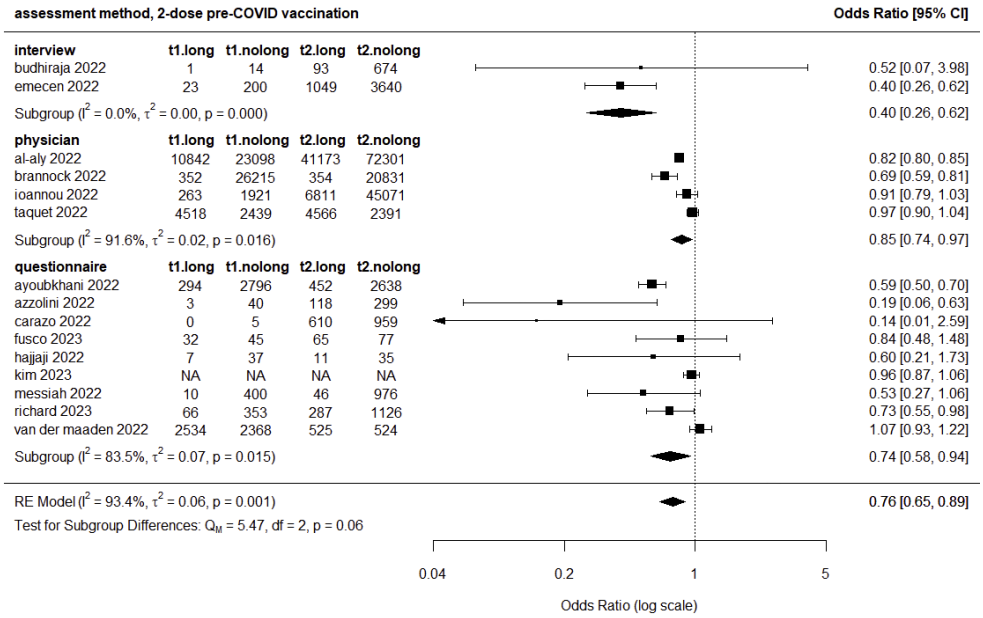


Supplementary figure: forest plot for subgroup analysis (main vaccine, 2-dose pre-COVID vaccination)
The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long Covid; nolong = number of patients without long Covid; CI = confidence interval; NA = not available; I^2 and τ^2 are heterogeneity measures; P = P value of vaccine effectiveness; RE model = random effects model; Q_M = test statistic; df = degree of freedom.



Supplementary figure: forest plot for subgroup analysis (assessment method, 2-dose pre-COVID vaccination)
The estimates were calculated as odd ratios using unvaccinated as the reference group. t1 = vaccinated group; t2 = unvaccinated group; long = number of patients with long Covid; nolong = number of patients without long COVID; CI = confidence interval; NA = not available; I^2 and τ^2 are heterogeneity measures; P = P value of vaccine effectiveness; RE model = random effects model; Q_M = test statistic; df = degree of freedom.

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(5) GRADE evidence profile

Certainty assessment							Effect			Certainty
No. of studies (subjects)	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Odds ratio (95% CI)	Risk of long COVID (infected, specified vaccination status)	Risk of long COVID (infected, unvaccinated)	
10	Observational studies 1-dose pre-COVID vaccination	Not serious [a]	Not serious [b]	Not serious [e]	Serious (-1) [f]	Publication bias (-1) [h] Dose-response gradient (+1) [j]	1.01 (0.88-1.15)	10.1 out of 100	10 out of 100 [k]	Very low ⨁OOO
15	Observational studies 2-dose pre-COVID vaccination	Not serious [a]	Not serious [c]	Not serious [e]	Not serious [g]	Publication bias (-1) [h] Dose-response gradient (+1) [j]	0.76 (0.65-0.89)	7.8 out of 100	10 out of 100 [k]	Low ⨁⨁OO
3	Observational studies 3-dose pre-COVID vaccination	Not serious [a]	Serious (-1) [d]	Not serious [e]	Serious (-1) [f]	Publication bias (-1) [i] Large effect (+1) Dose-response gradient (+1) [j]	0.31 (0.05-1.84)	3.3 out of 100	10 out of 100 [k]	Very low ⨁OOO
5	Observational studies 1-dose post-COVID vaccination	Not serious [a]	Not serious [c]	Not serious [e]	Not serious [g]	Dose-response gradient (+1) [j]	0.85 (0.73-0.98)	8.6 out of 100	10 out of 100 [k]	Moderate ⨁⨁⨁O
3	Observational studies 2-dose post-COVID vaccination	Not serious [a]	Serious (-1) [d]	Not serious [e]	Serious (-1) [f]	Publication bias (-1) [i] Large effect (+1) Dose-response gradient (+1) [j]	0.63 (0.38-1.03)	6.5 out of 100	10 out of 100 [k]	Very low ⨁OOO

CI: Confidence interval; OR: Odds ratio

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a. All studies are non-randomized and evaluated using the ROBINS-1 assessment tool. Some studies are rated with moderate or severe risk but there are no significant differences after excluding those studies in the sensitivity analysis. We rate the risk of bias as 'not serious'.
b. The I^2 is moderate but τ^2 is low. There are overlapping confidence intervals (CI). We rate inconsistency as 'not serious'.
c. The I^2 is high but τ^2 is low. There are overlapping CIs. We rate inconsistency as 'not serious'.
d. The I^2 and τ^2 are high. The CI of one study does not overlap with other studies. We rate inconsistency as 'serious'.
e. Our research question is well addressed by study design: both experimental and control groups are infected, and they only differ in vaccination status. We rate indirectness as 'not serious'.
f. The summary CI crosses the line of no difference (OR = 1). We rate imprecision as 'serious'.
g. The summary CI does not cross the line of no difference, and the number of subjects is large. We rate imprecision as 'not serious'.
h. Publication bias is indicated by Egger's test.
i. Publication bias is indicated by visual interpretation of funnel plot asymmetry.
j. Dose-response gradient is present since the more the vaccine dose, the further the OR is away from 1.
k. A previous review estimates long COVID is present among 10% of people infected by COVID.

PICO question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?

Evidence summary

Our SR included one randomized controlled study [55] involving 155 adult patients with COVID-19, with data collected between January 2020 to November 2024. No SARS-CoV-2 RNA was detected in baseline stool samples from patients with COVID-19 in this study. The trial compared the efficacy and safety of Nirmatrelvir/Ritonavir (NMV/r) versus placebo/Ritonavir (PBO/r). At 10 weeks, there was no statistically significant difference between the NMV/r group and the PBO/r group in the comprehensive severity of six core symptoms of long COVID: fatigue, brain fog, body pain, cardiovascular symptoms, dyspnea, and gastrointestinal symptoms. No significant differences were observed between the two groups in other secondary outcome measures, such as Patient Global Impression of Change, symptom burden and relief, and PROMIS scales. The incidence of adverse events was similar between the NMV/r and PBO/r groups (99% vs. 92.5%), with both groups experiencing low-grade adverse events.

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S5-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	438
	Web of Science Core Collection	Nov 30, 2024	428
	Embase	Nov 30, 2024	581
	The Cochrane Library	Nov 30, 2024	70
	China Biology Medicine (CBM)	Nov 30, 2024	38
	Wanfang Database	Nov 30, 2024	40
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	760
Preprint platform	Medrxiv	Nov 30, 2024	0
	SSRN	Nov 30, 2024	0
	Researchsquare	Nov 30, 2024	3
Clinical trial registries	WHO ICTRP	Nov 30, 2024	297
	ClinicalTrials.gov	Nov 30, 2024	16
	COVID-19 Study Register	Nov 30, 2024	322
Google	-	Nov 30, 2024	28
Reference lists of relevant SRs	-	Nov 30, 2024	2

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Part 2: Flow diagram of the literature search

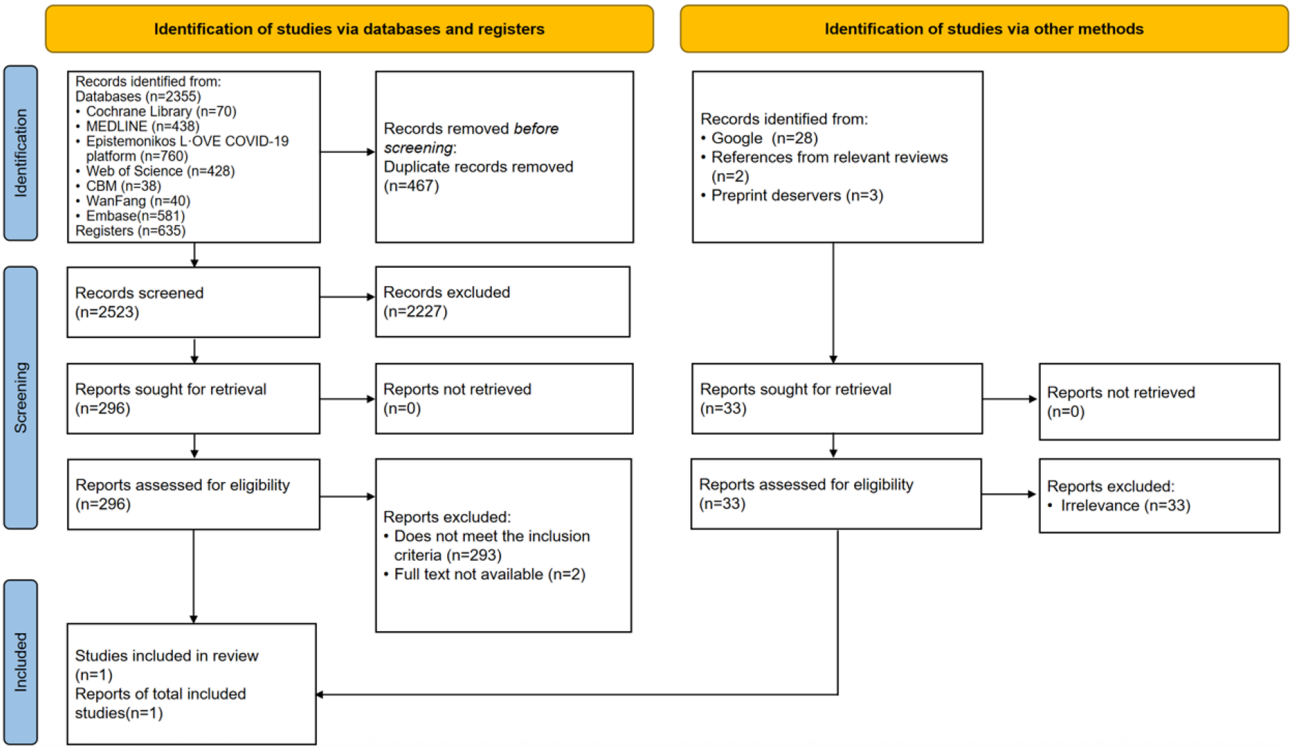


Figure S5-1 Flow diagram of the literature search

Part 3: list of included studies

Table S5-2 list of included studies

No.	Study title	Study design	DOI
1	Nirmatrelvir-Ritonavir and Symptoms in Adults With Postacute Sequelae of SARS-CoV-2 Infection: The STOP-PASC Randomized Clinical Trial	RCT	10.1001/jamainternmed.2024.2007

Part 4: Risk of bias assessment of the included studies

Table S5-3 Risk of bias assessment of included studies

Risk of Bias in the Included RCT Assessed by the Risk of Bias Tool							
Selection bias		Performance bias	Detection bias	Attrition bias	Reporting bias	Other bias	Study
Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Anything else, ideally prespecified	
low	low	low	low	low	low	low	Geng, et al.,2024

Part 5: Evidence syntheses

5.1. Meta-analysis

Table S5-4 Meta-analysis

Outcome	NMV/r	PBO/r		
Participants in group	102	53		
Time to relief of most bothersome symptom	No. (%)		OR (95% CI)	*P* value
Experiencing relief at 10 wk[a]	33 (32.4)	22 (41.5)	0.55 (0.27 to 1.09)	0.09
Experiencing alleviation at 10 wk[b]	7 (6.86)	5 (9.43)	0.72 (0.21 to 2.44)	0.60

5.2. GRADE assessment (Summary of Findings table)

Table S5-5 Summary of Findings table

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No of studies	Sample	Certainty assessment					Study event rates (%)		Relative effect (95%CI)	Overall Certainty of evidence
		Risk of bias	Inconsistency	Indirectness	Imprecision	Publications bias	Nirmatrelvir + ritonavir	Placebo + ritonavir		
Experiencing relief at 10 wk[a]										
1	155	not serious	not serious	not serious	very serious	not serious	33/102(32.4)	22/53(41.5)	RR 0.78 (0.51 to 1.19)	⨁⨁◯◯[c] Low
Experiencing alleviation at 10 wk[b]										
1	155	not serious	not serious	not serious	very serious	not serious	7/102(6.86)	5/53(9.43)	RR 0.73 (0.24 to 2.18)	⨁⨁◯◯[c] Low

[a] Relief was defined as a reduction of severity from moderate to none or severe to mild or none for at least 1 core symptom; time to relief was measured in weeks.

[b] Alleviation was defined as improvement of all core symptoms from none or mild at baseline to none or moderate to severe to none or mild.

[c] Downgraded 2 levels due to imprecision: large confidence interval compatible with clinically important benefit and clinically important harm; also, small sample size.

PICO question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?

Evidence summary

A systematic literature search from January 2020 to November 30, 2024 identified no studies evaluating monoclonal antibody therapy targeting patients who met the WHO definition of long COVID. After removing this definitional restriction, we found one case report.

One case report involved an immunocompromised long COVID patient (thymoma with metastasis to the pleura) [56]. The patient received two non-competing monoclonal antibodies—casirivimab-imdevimab (specific dosages not reported). The report noted symptomatic improvement, including reductions in respiratory system (cough), and general symptoms (including malaise and fever), suggesting potential benefit of monoclonal antibodies in managing immunocompromised individuals.

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S6-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	659
	Web of Science Core Collection	Nov 30, 2024	556
	Embase	Nov 30, 2024	716
	The Cochrane Library	Nov 30, 2024	546
	China Biology Medicine (CBM)	Nov 30, 2024	22
	Wanfang Database	Nov 30, 2024	28
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	18
Preprint platform	Medrxiv	Nov 30, 2024	610
	SSRN	Nov 30, 2024	124
	Researchsquare	Nov 30, 2024	119
Clinical trial registries	WHO ICTRP	Nov 30, 2024	3
	ClinicalTrials.gov	Nov 30, 2024	16
	COVID-19 Study Register	Nov 30, 2024	15
Google	-	Nov 30, 2024	0
Reference lists of relevant SRs	-	Nov 30, 2024	0

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Part 2: Flow diagram of the literature search

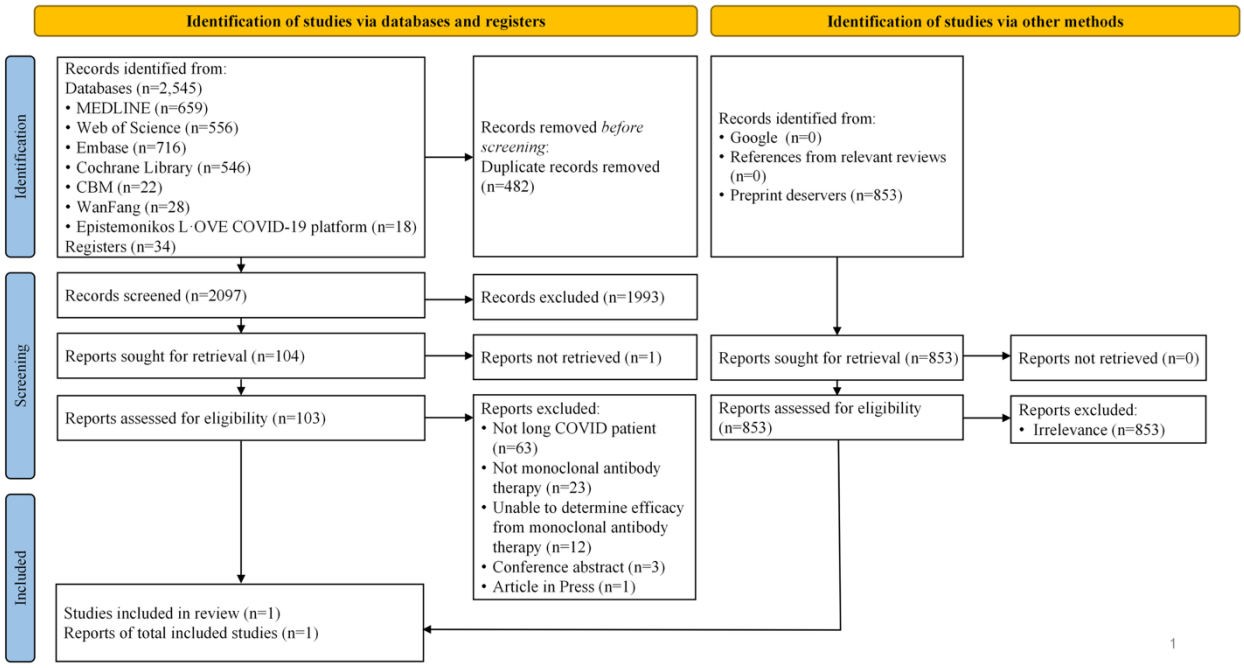


Figure S6-1 Flow diagram of the literature search

Part 3: list of included studies

Table S6-2 list of included studies

No.	Study title		Study design	DOI
1	Effective use of monoclonal antibodies for treatment of persistent COVID-19 infection in a patient on rituximab	Case report	10.1136/bcr-2021-243469	

PICO question 4: In patients with long COVID, should probiotics be used versus no probiotics?

Evidence summary

Our SR included three RCTs (two published in peer-reviewed journals and one available as a preprint) [57-59], involving 587 long COVID adult patients, with data collected between January 1, 2020 and November 30, 2024. All included RCTs investigated multi-species probiotic formulations; however, the reported outcomes varied across studies.

The meta-analysis showed that, compared with placebo, probiotics (oral synbiotic preparation-SIM01, including three bacteria strains) can alleviate several symptoms associated with long COVID, including, respiratory system: shortness of breath (1 RCT, OR: 1.87, 95% CI: 1.16 to 3.03, moderate certainty of evidence) and cough (1 RCT, OR: 1.85, 95% CI: 1.04 to 3.29, moderate certainty of evidence); Musculoskeletal system: joint pain (1 RCT, OR: 2.10, 95% CI: 1.27 to 3.47), inability to exercise (1 RCT, OR: 2.32, 95% CI: 1.23 to 4.38, moderate certainty of evidence), and muscle pain (1 RCT, OR: 1.71, 95% CI: 1.07 to 2.72, moderate certainty of evidence); Neuropsychological system: difficulty in concentration (1 RCT, OR: 2.64, 95% CI: 1.69 to 4.14, moderate certainty of evidence), memory loss (1 RCT, OR: 1.97, 95% CI: 1.27 to 3.04, moderate certainty of evidence), and insomnia (1 RCT, OR: 1.76, 95% CI: 1.12 to 2.77, moderate certainty of evidence); Gastrointestinal system: gastrointestinal upset (1 RCT, OR: 1.99, 95% CI: 1.30 to 3.05, moderate certainty of evidence); General symptoms: fatigue (1 RCT, OR: 2.27, 95% CI: 1.52 to 3.40, moderate certainty of evidence), and general unwellness (1 RCT, OR: 2.36, 95% CI: 1.43 to 3.90, moderate certainty of evidence). No significant adverse events were reported in any of the included RCTs using various probiotic.[57] One RCT examined the efficacy of VSL#3® (comprising Lactobacilli, Bifidobacteria, and Streptococcus thermophilus) in reducing fatigue and improving various aspects of well-being in patients with long COVID syndrome.[58] The results showed that, compared with placebo, VSL#3® significantly reduced LC-associated fatigue, as measured by the CFS Likert score at 4 weeks, with a median

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variation of −24% (IQR: −60.9 to −11.6) compared to −4.4% (IQR: −18.8 to 0.0) in the placebo group (p = 0.0079). Another RCT suggested that a synbiotic mixture (including L. rhamnosus DSM 32550, Humiome® L. plantarum DSM 34532, B. lactis DSM 32269, and B. longum DSM 32946) led to a significant reduction in general fatigue (p = 0.03), physical fatigue (p = 0.03), and mental fatigue (p = 0.03) after 3 months of intake compared with baseline values.[59]

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S7-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	127
	Web of Science Core Collection	Nov 30, 2024	27
	Embase	Nov 30, 2024	64
	The Cochrane Library	Nov 30, 2024	13
	China Biology Medicine (CBM)	Nov 30, 2024	2
	Wanfang Database	Nov 30, 2024	30
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	6
Preprint platform	Medrxiv	Nov 30, 2024	1
	SSRN	Nov 30, 2024	1
	Researchsquare	Nov 30, 2024	11
Clinical trial registries	WHO ICTRP	Nov 30, 2024	1
	ClinicalTrials.gov	Nov 30, 2024	3
	COVID-19 Study Register	Nov 30, 2024	12
Google	-	Nov 30, 2024	100
Reference lists of relevant SRs	-	Nov 30, 2024	13

Part 2: Flow diagram of the literature search

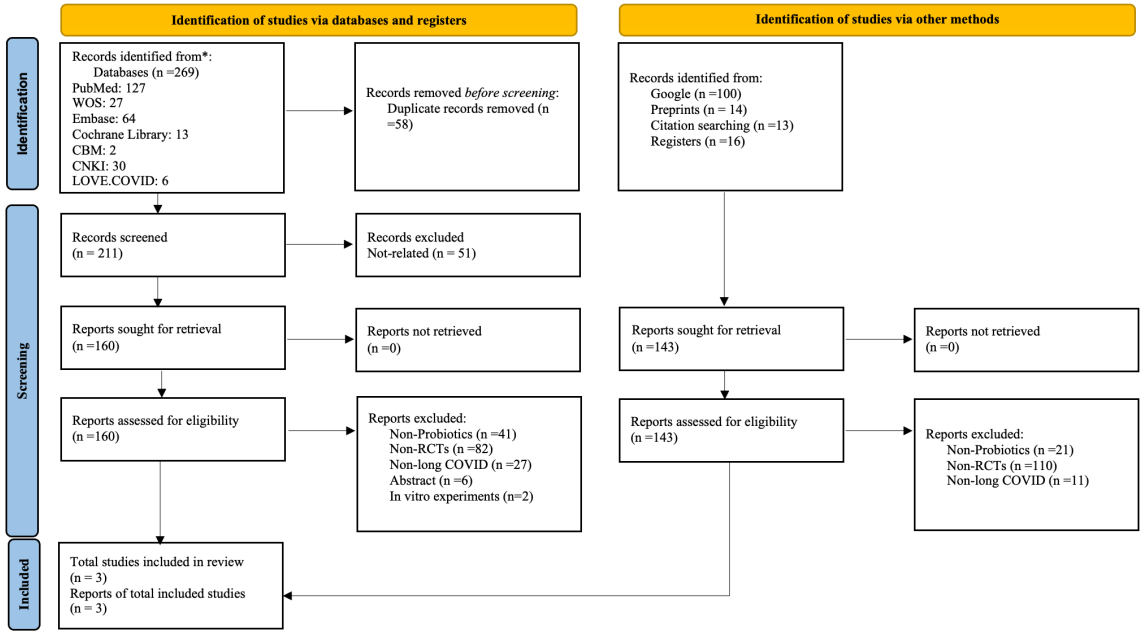


Figure S7-1 Flow diagram of the literature search

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Part 3: List of included studies

Table S7-2 List of included studies

No.	Study title	Study design	DOI
1	A synbiotic preparation (SIM01) for post-acute COVID-19 syndrome in Hong Kong (RECOVERY): a randomised, double-blind, placebo-controlled trial.	RCT	10.1016/S1473-3099(23)00685-0
2	The Role of VSL#3® in the Treatment of Fatigue and Other Symptoms in Long COVID-19 Syndrome: a Randomized, Double-blind, Placebo-controlled Pilot Study (DELong#3)	RCT	10.1101/2023.06.28.23291986
3	The effects of 3-month supplementation with synbiotic on patient-reported outcomes, exercise tolerance, and brain and muscle metabolism in adult patients with post-COVID-19 chronic fatigue syndrome (STOP-FATIGUE): a randomized Placebo-controlled clinical trial	RCT	10.1007/s00394-025-03591-3

Part 4: Risk of bias assessment of the included studies

Table S7-3 Risk of bias assessment of the included studies

Study	ROB							
	Random sequence generation	Allocation concealment	Blinding of participants	Blinding of personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other biases
Lau et al.	Low	Low	Low	Low	Low	Low	Low	Unclear
Marinoni et al.	Low	Low	Low	Low	Unclear	Low	Low	Unclear
Ranisavljev et al.	Low	Low	Low	Low	Unclear	Low	Low	Unclear

Part 5: Evidence syntheses

5.1. Meta-analysis

Table S7-4 Meta-analysis of PACS symptoms alleviation by 6 months

Outcomes	No. of included study	Probiotics		Placebo		OR, MH, Fixed, 95%CI, *P*
		Events	Total	Events	Total	
Difficulty in concentration	1	101	162	62	161	2.64[1.69,4.14, *P*<0.05]
Fatigue	1	123	196	86	202	2.27[1.52,3.40, *P*<0.05]
General unwellness	1	119	154	85	144	2.36[1.43,3.90, *P*<0.05]
Gastrointestinal upset	1	134	191	99	183	1.99[1.30,3.05, *P*<0.05]
Memory loss	1	74	176	52	193	1.97[1.27,3.04, *P*<0.05]
Joint pain	1	69	138	40	124	2.10[1.27,3.47, *P*<0.05]
Inability to exercise	1	73	94	57	95	2.32[1.23,4.38, *P*<0.05]

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Shortness of breath	1	102	149	73	136	1.87[1.16,3.03, *P*<0.05]
Insomnia	1	92	158	65	147	1.76[1.12,2.77, *P*<0.05]
Muscle pain	1	88	152	62	139	1.71[1.07,2.72, *P*<0.05]
Coughing	1	100	126	83	123	1.85[1.04,3.29, *P*<0.05]
Hair loss	1	51	117	30	98	1.75[1.00,3.08, *P*<0.05]
Chest pain	1	60	84	42	72	1.79[0.92,3.48, *P*<0.09]
Mood disturbance	1	74	135	65	137	1.34[0.83,2.16, *P*=0.22]

Table S7-5 Meta-analysis of Adverse events

Outcomes	No. of included study	Probiotics		Placebo		OR, MH, Fixed, 95%CI, *P*
		Events	Total	Events	Total	
Any adverse event	1	22	232	25	231	0.86[0.47,1.58, *P*=0.63]
Diarrhea	1	2	232	5	231	0.39[0.08,2.05, *P*=0.27]
Bloating	1	1	232	2	231	0.50[0.04,5.50, *P*=0.57]
Epigastric pain	1	1	232	1	231	1.00[0.06,16.01, *P*=1.00]

5.2. GRADE assessment (Summary of Findings table)

Table S7-6 Summary of Findings table

Certainty assessment							№ of patients		Effect		Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Probiotics	Placebo	Relative (95% CI)	Absolute (95% CI)	
Proportion of PACS symptoms alleviation by 6 months-Difficulty in concentration											
1	randomised trials	not serious	not serious	not serious	serious[a]	none	101/162 (62.3%)	62/161 (38.5%)	OR 2.64 (1.69 to 4.14)	238 more per 1,000 (from 129 more to 337 more)	⊕⊕⊕◯ Moderate
Proportion of PACS symptoms alleviation by 6 months-Fatigue											
1	randomised trials	not serious	not serious	not serious	serious[a]	none	123/196 (62.8%)	86/202 (42.6%)	OR 2.27 (1.52 to 3.40)	202 more per 1,000 (from 104 more to 290 more)	⊕⊕⊕◯ Moderate
Proportion of PACS symptoms alleviation by 6 months-General unwellness											
1	randomised trials	not serious	not serious	not serious	serious[a]	none	119/154 (77.3%)	85/144 (59.0%)	OR 2.36 (1.43 to 3.90)	182 more per 1,000 (from 83 more to 259 more)	⊕⊕⊕◯ Moderate
Proportion of PACS symptoms alleviation by 6 months-Gastrointestinal upset											
1	randomised trials	not serious	not serious	not serious	serious[a]	none	134/191 (70.2%)	99/183 (54.1%)	OR 1.99 (1.30 to 3.05)	160 more per 1,000 (from 64 more to 241 more)	⊕⊕⊕◯ Moderate
Proportion of PACS symptoms alleviation by 6 months-Memory loss											
1	randomised trials	not serious	not serious	not serious	serious[a]	none	74/176 (42.0%)	52/193 (26.9%)	OR 1.97 (1.27 to 3.04)	151 more per 1,000 (from 50 more to 259 more)	⊕⊕⊕◯ Moderate

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Proportion of PACS symptoms alleviation by 6 months-Joint pain

1	randomised trials	not serious	not serious	not serious	serious[a]	none	69/138 (50.0%)	40/124 (32.3%)	OR 2.10 (1.27 to 3.47)	177 more per 1,000 (from 54 more to 300 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Inability to exercise

1	randomised trials	not serious	not serious	not serious	serious[a]	none	73/94 (77.7%)	57/95 (60.0%)	OR 2.32 (1.23 to 4.38)	177 more per 1,000 (from 49 more to 268 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Shortness of breath

1	randomised trials	not serious	not serious	not serious	serious[a]	none	102/149 (68.5%)	73/136 (53.7%)	OR 1.87 (1.16 to 3.03)	147 more per 1,000 (from 37 more to 242 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Insomnia

1	randomised trials	not serious	not serious	not serious	serious[a]	none	92/158 (58.2%)	65/147 (44.2%)	OR 1.76 (1.12 to 2.77)	140 more per 1,000 (from 28 more to 245 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Muscle pain

1	randomised trials	not serious	not serious	not serious	serious[a]	none	88/152 (57.9%)	62/139 (44.6%)	OR 1.71 (1.07 to 2.72)	133 more per 1,000 (from 17 more to 240 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Coughing

1	randomised trials	not serious	not serious	not serious	serious[a]	none	100/126 (79.4%)	83/123 (67.5%)	OR 1.85 (1.04 to 3.29)	119 more per 1,000 (from 9 more to 197 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Hair loss

1	randomised trials	not serious	not serious	not serious	serious[a]	none	51/117 (43.6%)	30/98 (30.6%)	OR 1.75 (1.00 to 3.08)	130 more per 1,000 (from 0 fewer to 270 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Chest pain

1	randomised trials	not serious	not serious	not serious	serious[a]	none	60/84 (71.4%)	42/72 (58.3%)	OR 1.79 (0.92 to 3.48)	131 more per 1,000 (from 20 fewer to 246 more)	⊕⊕⊕◯ Moderate

Proportion of PACS symptoms alleviation by 6 months-Mood disturbance

1	randomised trials	not serious	not serious	not serious	serious[a]	none	74/135 (54.8%)	65/137 (47.4%)	OR 1.34 (0.83 to 2.16)	73 more per 1,000 (from 46 fewer to 187 more)	⊕⊕⊕◯ Moderate

Any adverse event

1	randomised trials	not serious	not serious	not serious	serious[a]	none	22/232 (9.5%)	25/231 (10.8%)	OR 0.86 (0.47 to 1.58)	14 fewer per 1,000 (from 54 fewer to 53 more)	⊕⊕⊕◯ Moderate

Diarrhea

1	randomised trials	not serious	not serious	not serious	serious[a]	none	2/232 (0.9%)	5/231 (2.2%)	OR 0.39 (0.08 to 2.05)	13 fewer per 1,000 (from 20 fewer to 22 more)	⊕⊕⊕◯ Moderate

Bloating

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1	randomised trials	not serious	not serious	not serious	serious[a]	none	1/232 (0.4%)	2/231 (0.9%)	OR 0.50 (0.04 to 5.50)	4 fewer per 1,000 (from 8 fewer to 37 more)	⊕⊕⊕◯ Moderate

Epigastric pain

1	randomised trials	not serious	not serious	not serious	serious[a]	none	1/232 (0.4%)	1/231 (0.4%)	OR 1.00 (0.06 to 16.01)	0 fewer per 1,000 (from 4 fewer to 61 more)	⊕⊕⊕◯ Moderate

CI: confidence interval; OR: odds ratio

Explanations

a. small sample size.

PICO question 5: In patients with Long COVID, should glucocorticoids be used versus no glucocorticoids?

Evidence summary

Our SR included 10 studies (four RCTs, three cohort studies and three before-and-after studies), involving 3,414 adult patients with long COVID. Data were retrieved from January 1, 2020, to November 30, 2024.

Effect of glucocorticoids on cardiopulmonary function among long COVID patients with persistent respiratory symptoms

Six studies (one RCT, three cohort studies and two before-and-after studies) assessed the effect of glucocorticoids on cardiopulmonary function among long COVID adult patients with persistent respiratory symptoms. Significant heterogeneity exists among the glucocorticoids administration methods, dosages, and follow-up periods, preventing conducting the meta-analysis. We listed the glucocorticoids dosages of the six studies in Appendix 3 Table S2. Based on clinical relevance and core group discussion, the primary outcomes selected for evaluation were 6-minute walk distance (6MWD), peripheral oxygen saturation (SpO_2), FEV_1, FVC, and DLCO.

Of the six studies, five observational studies compared glucocorticoids use versus non-use with different follow-up time. Four studies reported changes in 6MWD: two studies reported statistically significant improvements, mean difference (MD) = +23m [60,61], while the other two studies found no significant improvement [62,63]. None of the studies reported SpO_2 outcomes. Regarding FEV_1, one study [64] reported a significant improvement in FEV_1 (L) (baseline 2.33 ± 0.67, follow-up 2.66 ± 0.90; *P*=0.008), whereas another study [62] found no significant change in FEV_1 (% predicted). Three studies [60-62] reported improvement in FVC (% predicted), and two studies [60,64] showed glucocorticoids could improve FVC (L) from the baseline of 2.93 ± 0.82 to a follow-up of 3.28 ± 1.11; *P*=0.007; MD: 0.42, 95%CI: 0.28 to 0.56). Regarding DLCO and DLCO [%], one study [64] reported the use of steroid was associated with a significant improvement in DLCO (DLCO [mL/min/mmHg]: baseline 14.63 ± 5.50, follow-up 18.09 ± 7.31; *P* = 0.07) and DLCO [%] (baseline 56 ± 17, follow-up 64 ± 18; *P* = 0.04). In the non-steroid group, the changes compared to baseline were not statistically significant. And another study [62] found a significant improvement in DLCO (%) (MD ± standard deviation [SD]=10.18 ± 12.55) in the corticosteroids group. Therefore, the magnitude of improvement across studies was relatively modest, and findings were inconsistent, which is similar in other respiratory outcomes, including CT score, total lung capacity (TLC), residual volume (RV), transfer factor of the lung for carbon monoxide (TLCO), etc, making it difficult to draw definitive conclusions.

One RCT [65] compared effect of different glucocorticoids doses in 130 long COVID adults with persistent dyspnea or hypoxemia. The high-dose group received an initial prednisone dose of 40 mg/day for one week, followed by gradual tapering (30 mg/day for one week, 20 mg/day for two weeks, and 10 mg/day for two weeks). The low-dose group received 10 mg/day of prednisone for six weeks. There were no significant differences between the two groups in complete radiological response (*P*=0.39), complete or good radiological response (*P*=0.49), FVC (*P*=0.21), resting SpO_2 (*P*=0.91) and mMRC score (*P*=0.52) at six weeks.

Effect of glucocorticoids on olfactory disorders in long COVID

Four studies (three RCTs and one before-and-after study) assessed the effect of glucocorticoids on olfactory disorders in adult patients with long COVID. Results of meta-analysis [66-68] (involving 138 adult patients with anosmia or olfactory disorders from three RCTs, glucocorticoids was given by inhalation, local injection and systemic administration) showed, when compared to no glucocorticoids, the use of glucocorticoids can improve the olfactory score (MD: 1.66, 95% CI: 1.10 to 2.23, I^2 = 0%, low certainty of evidence), but had no effect on patients' self-reported olfactory recovery (risk ratio [RR]:

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1.33, 95% CI: 0.90 to 1.99, I^2 = 0%, very low certainty of evidence). One 2024 before-and-after study [69] involving 87 adults with persistent olfactory and gustatory disorders, showed that combination therapy with glucocorticoids, glycosaminoglycans-based antithrombotic drugs, and diuretics can effectively improve the VAS score. At 12 months, mean olfactory score increased from a baseline of 2.21 to 6.99, with mean gustatory score increased from a baseline of 3.17 to 7.14. Subgroup analysis showed that the glucocorticoids was more effective when initiated within 180 days post SARS-CoV-2 infection. Under the same treatment, olfactory symptoms improved more substantially than gustatory symptoms.

Safety of glucocorticoids use in long COVID patients

Across all included studies, no serious adverse events related to glucocorticoids use were reported during follow-up. Only the local injection group in long COVID patients with olfactory disorders reported controllable bleeding and local pain.

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S8-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	592
	Web of Science Core Collection	Nov 30, 2024	976
	Embase	Nov 30, 2024	1783
	The Cochrane Library	Nov 30, 2024	72
	China Biology Medicine (CBM)	Nov 30, 2024	21
	Wanfang Database	Nov 30, 2024	24
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	59
Preprint platform	Medrxiv	Nov 30, 2024	630
	SSRN	Nov 30, 2024	21
	Researchsquare	Nov 30, 2024	1612
Clinical trial registries	WHO ICTRP	Nov 30, 2024	0
	ClinicalTrials.gov	Nov 30, 2024	7
	COVID-19 Study Register	Nov 30, 2024	48
Google	-	Nov 30, 2024	4
Reference lists of relevant SRs	-	Nov 30, 2024	52

Part 2: Flow diagram of the literature search

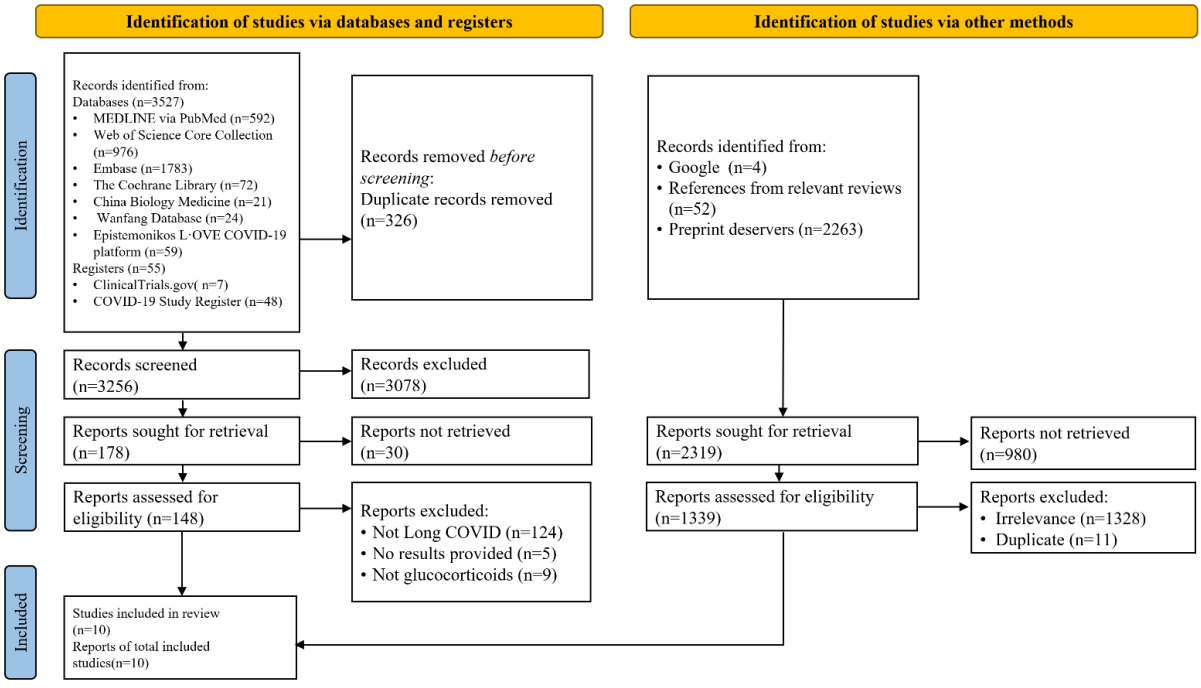


Figure S8-1 Flow diagram of the literature search

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Part 3: list of included studies

Table 5-2 List of included studies

No.	Study title	Study design	DOI
1	Olfactory mucosa steroid injection in treatment of post-COVID-19 olfactory dysfunction: a randomized control trial	RCT	10.1186/s43163-023-00478-0
2	Intranasal Corticosteroid Treatment on Recovery of Long-Term Olfactory Dysfunction Due to COVID-19	RCT	10.1002/lary.30353
3	Efficacy of corticosteroid therapy in the treatment of long-lasting olfactory disorders in COVID-19 patients	RCT	10.4193/rhin20.515
4	High-dose versus low-dose prednisolone in symptomatic patients with post-COVID-19 diffuse parenchymal lung abnormalities: an open-label, randomised trial (the COLDSTER trial)	RCT	10.1183/13993003.02930-2021
5	Corticosteroid treatment for persistent pulmonary infiltrates following COVID-19 infection: Clearing the fog?	Cohort study	10.4103/atm.atm_121_23
6	The effectiveness of glucocorticoid treatment in post-COVID-19 pulmonary involvement	Cohort study	10.1186/s41479-023-00123-7
7	Outpatient management of Post-COVID syndrome – single center experience	Cohort study	10.1016/j.hrtlng.2024.05.004
8	Recovery of Smell and Taste in Patients With Persistent COVID-19-Related Hyposmia and Dysgeusia by Targeting Inflammation and Endothelial Dysfunction	Follow-up study	10.7759/cureus.54925
9	Persistent Post–COVID-19 Interstitial Lung Disease: An Observational Study of Corticosteroid Treatment	Follow-up study	10.1513/AnnalsATS.202008-1002OC
10	Systemic corticosteroids for management of 'long-COVID': an evaluation after 3 months of treatment	Follow-up study	10.4081/monaldi.2021.1981

Part 4 Risk of bias assessment of the included studies

Table S8-3 Risk of Bias assessment of the included RCTs assessed by the Risk of Bias Tool

Selection bias		Performance bias	Detection bias	Attrition bias	Reporting bias	Other bias	Study
Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Anything else, ideally pre-specified	
Unclear	Low	Unclear	Unclear	Low	Low	Low	Lasheen 2023
Low	Unclear	Low	Low	Low	Low	Low	Hosseinpoor 2022
Low	Unclear	High	Unclear	Low	Unclear	Unclear	Vaira 2021
Low	Low	High	High	Partly reporting	Low	Low	Dhooria 2022

Table S8-4 Risk of bias assessment of the included non-randomized studies (Newcastle-Ottawa Scale)

Selection				Comparability	Outcome				Study
Representativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Total Score	
1	1	1	1	1	1	1	1	7	Mizera 2023
1	0	1	1	1	1	1	1	7	Torres 2024
1	1	1	1	1	1	1	1	8	Ntiamoah 2024

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1	0	1	1	1	1	1	1	7	Vento 2024
1	0	1	1	1	1	1	1	7	Myall 2021
1	0	1	1	1	1	1	1	7	Goel 2022

Part 5: Evidence syntheses

5.1. Meta-analysis

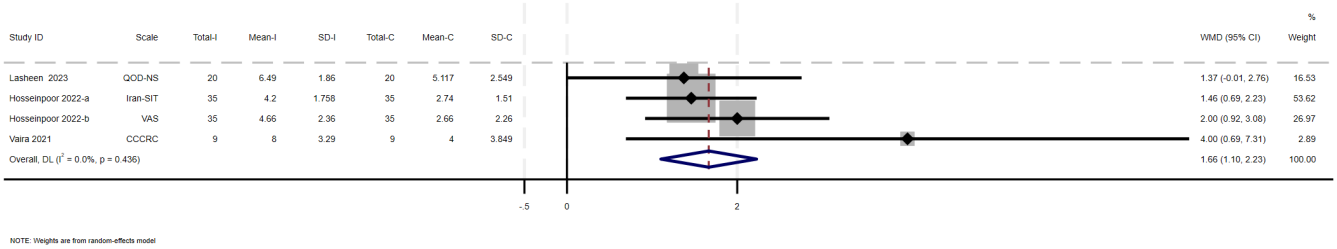


Figure S8-2 Meta-analysis forest plot of olfactory score changes with glucocorticoid vs. non-glucocorticoid treatment (all scales are uniformly standardized into a 10-point system with the same measurement direction)

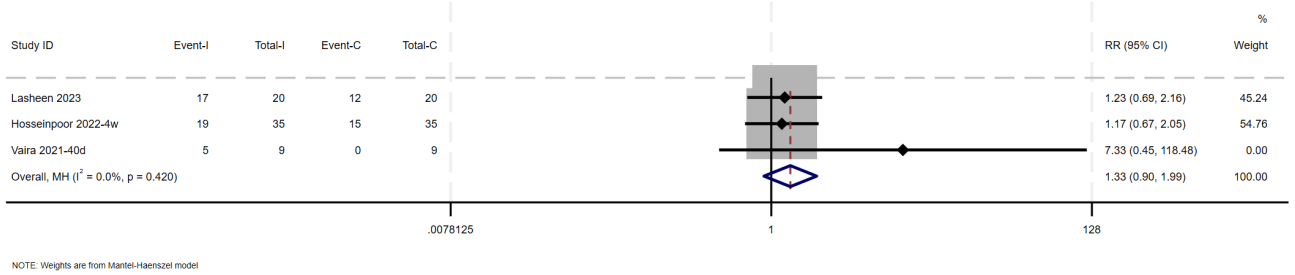


Figure S8-3 Meta-analysis forest plot of self-reported olfactory improvement with glucocorticoid vs. non-glucocorticoid treatment

5.2. GRADE assessment (Summary of Findings table)

Table S8-5 Summary of Findings table for persistent respiratory symptoms

Certainty assessment							№ of patients		Effect		Certainty	Comment
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	glucocorticoid	non-glucocorticoid	Relative (95% CI)	Absolute (95% CI)		
6MWD (m)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	28	-	-	MD 23 higher (3.3 lower to 49.3 higher)	⊕◯◯◯ Very low	Post-COVID ILD: who received MMF and steroids
6MWD (m)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	30	-	-	MD 213 higher	⊕◯◯◯ Very low	Post–SARS-CoV-2 ILD: who received corticosteroids
6MWD (% pred)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	28	-	-	MD 1 higher	⊕◯◯◯ Very low	Post-COVID ILD: who received MMF and steroids
FEV1 (L)												
1	Cohort study	not serious	not serious	not serious	serious[b]	none	16	6	-	MD 0.33 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
FEV1 (% pred)												

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Certainty assessment							№ of patients		Effect		Certainty	Comment
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	glucocorticoid	non-glucocorticoid	Relative (95% CI)	Absolute (95% CI)		
1	Cohort study	not serious	not serious	not serious	serious[b]	none	16	6	-	MD 0.02 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
FVC (L)												
1	Cohort study	not serious	not serious	not serious	serious[b]	none	16	6	-	MD 0.35 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
FVC (L)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	28	-	-	MD 0.3 higher	⊕◯◯◯ Very low	Post-COVID ILD: who received MMF and steroids
FVC (L)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	30	-	-	MD 0.42 higher (0.28 higher to 0.56 higher)	⊕◯◯◯ Very low	Post–SARS-CoV-2 ILD: who received corticosteroids
FVC (% pred)												
1	Cohort study	not serious	not serious	not serious	serious[b]	none	16	6	-	MD 4 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
FVC (% pred)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	28	-	-	MD 8 higher	⊕◯◯◯ Very low	Post-COVID ILD: who received MMF and steroids
FVC (% pred)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	30	-	-	MD 9.63 higher (4.49 higher to 14.7 higher)	⊕◯◯◯ Very low	Post–SARS-CoV-2 ILD: who received corticosteroids
DLCO (mL/min/mmHg)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	12	-	-	MD 3.46 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
DLCO (% pred)												
1	Before and after study	not serious	serious[a]	not serious	serious[b]	none	12	-	-	MD 8 higher	⊕◯◯◯ Very low	Post-COVID persistent pulmonary infiltrates: who received systemic corticosteroids
DLCOc (% pred)												

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Certainty assessment							№ of patients		Effect		Certainty	Comment
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	glucocorticoid	non-glucocorticoid	Relative (95% CI)	Absolute (95% CI)		
1	Before and after study	not serious	serious[a]	not serious	not serious	none	131	-	-	MD 10.18 higher (8.03 higher to 12.33 higher)	⊕◯◯◯ Very low	The treatment with glucocorticoids (CS) was considered in patients with persistent pathological findings on CXR and/or HRCT and/or impaired pulmonary function parameters, namely decreased DLCO.

ILD: interstitial lung disease; MMF: mycophenolate mofetil; CI: confidence interval; MD: mean difference
Explanations
a. There is no control group.
b. The sample size is insufficient.

Table 5-6 Summary of Findings table for olfactory disorders

Certainty assessment							№ of patients		Effect		
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Glucocorticoid	No-Glucocorticoid	Relative (95% CI)	Absolute (95% CI)	Certainty
Scale score											
3	randomised trials	serious[a]	not serious	not serious	not serious	none	64	64	-	MD 1.66 higher (1.10 higher to 2.23 higher)	⊕⊕⊕◯ Moderate[a]
Improvement in self-perceived symptoms											
3	randomised trials	serious[a]	not serious	not serious	serious[b]	none	39/64 (60.9%)	27/64 (42.2%)	RR 1.33 (0.90 to 1.99)	139 more per 1,000 (from 42 fewer to 418 more)	⊕⊕◯◯ Low[a,b]

CI: confidence interval; RR: risk ratio; SMD: standardised mean difference
Explanations
a. The risk of bias is high due to the blinding of participants and personnel.
b. The range of the confidence interval includes the null line.

PICO question 6: In patients with long COVID, should immunomodulators be used versus no immunomodulators?

Evidence summary

Our SR included one cohort study, one case-control study, one case series, and three case reports, involving 95 adult patients with long COVID [70-75]. Data were retrieved from January 1, 2020, to November 30, 2024. Due to significant heterogeneity across the studies, meta-analysis was not conducted. The findings are as follows:

A cohort study published in 2023 [70] involved 48 long COVID adult patients with interstitial lung disease. Of these, 21 received methylprednisolone, 11 received methylprednisolone combined with mycophenolate mofetil (MMF), and one received MMF alone. After six months of follow-up, all patients demonstrated significant clinical improvements in key outcomes, including mMRC score, SpO_2, FVC, FVC (% of predicted), FEV_1 (% of predicted), and DLCO (% of predicted) (P<0.005). Patients tolerated MMF well, with no serious adverse events reported. A case-control study published in 2023 [74] included 41 long COVID patients with pulmonary involvement (hypoxia and tachypnea with radiological evidence of active pulmonary inflammation) persisting for over four weeks. All patients received corticosteroid upon hospitalization. Among the 21 patients treated with JAK inhibitors (20% with baricitinib and 80% tofacitinib), corticosteroid doses were rapidly tapered within 7-10 days after initiation. Other 20 patients who did not receive JAK inhibitors required continued corticosteroid for 3-4 weeks, with duration adjusted based on clinical improvement. The results showed no significant difference in oxygen support requirements between the two groups (P=0.269). Patients not receiving JAK inhibitors experienced significantly higher rates of adverse events compared to those treated with JAK inhibitors (P=0.003). Two

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case reports from 2022 and 2024 described tacrolimus treatment for ILD in long COVID adult patients [73,75]. Both reports showed favorable outcomes, with two patients discontinuing oxygen therapy by day 12 and day 56 post-treatment respectively. Chest X-ray revealed improved pulmonary involvement, and no serious adverse events were reported.

Other immunomodulators have been discussed in case series or case reports. A 2023 case series [72] reported the use of methotrexate or leflunomide to treat three adult patients who developed persistent joint pain after recovery from acute SARS-CoV-2 infection and presented abnormal joint imaging. However, the treatments were discontinued in two patients due to significant adverse events, and one patient did not report the final results. Another 2023 case report [71] described the use of anakinra in adult long COVID patients with persistent steroid-dependent multisystem inflammatory syndrome, showing good tolerability over 12-month period, with no immunosuppression-related symptoms or adverse events. The use of anakinra allowed tapering of the prednisone does with no recurrences of fever or increase in C-reactive protein levels.

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S9-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	383
	Web of Science Core Collection	Nov 30, 2024	268
	Embase	Nov 30, 2024	1689
	The Cochrane Library	Nov 30, 2024	348
	China Biology Medicine (CBM)	Nov 30, 2024	30
	Wanfang Database	Nov 30, 2024	475
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	0
Preprint platform	Medrxiv	Nov 30, 2024	137
	SSRN	Nov 30, 2024	412
	Researchsquare	Nov 30, 2024	245
Clinical trial registries	WHO ICTRP	Nov 30, 2024	8
	ClinicalTrials.gov	Nov 30, 2024	100
	COVID-19 Study Register	Nov 30, 2024	325
Google	-	Nov 30, 2024	490
Reference lists of relevant SRs	-	Nov 30, 2024	0

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Part 2: Flow diagram of the literature search

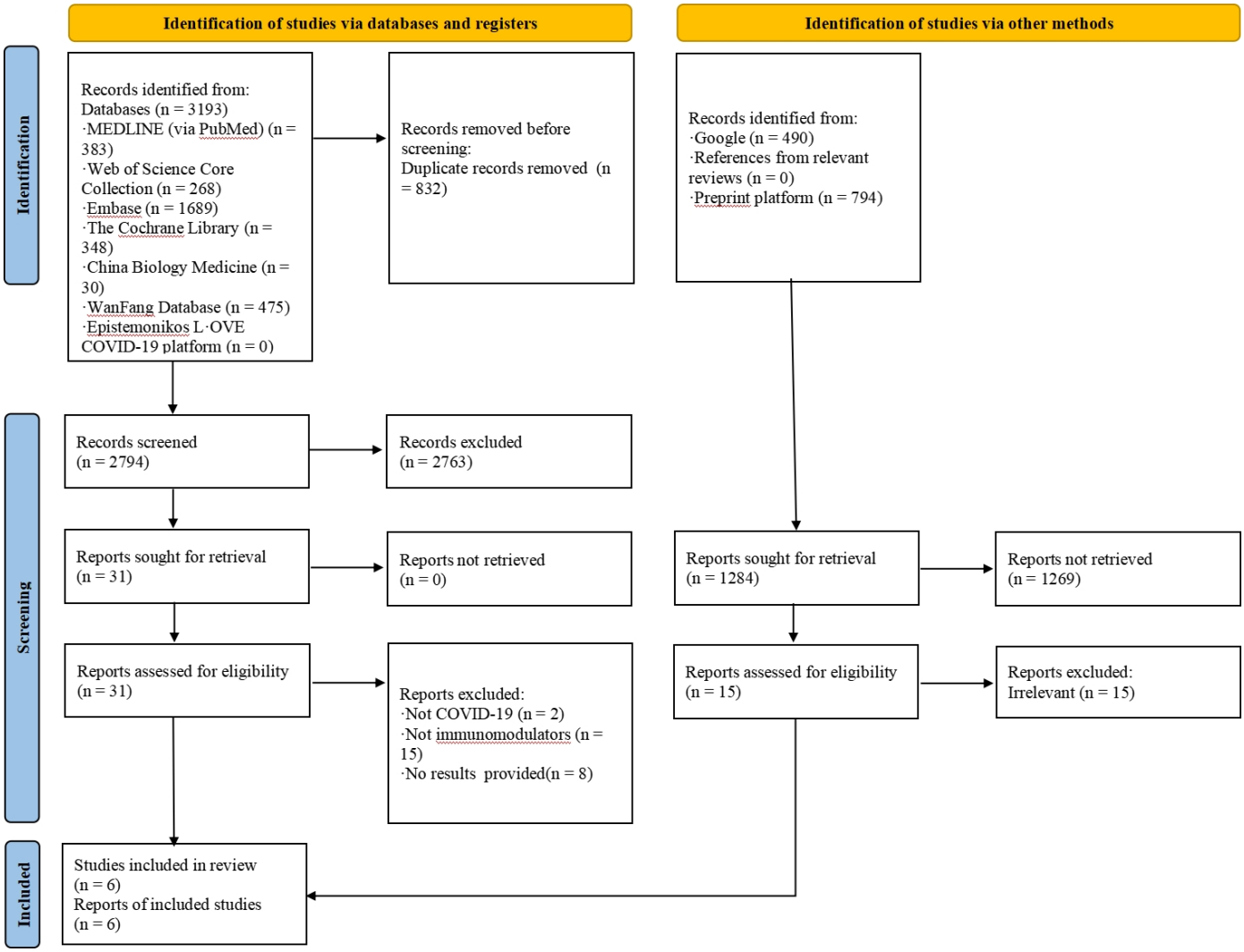


Figure S9-1 Flow diagram of the literature search

Part 3: list of included studies

Table S9-2 List of included studies

No.	Study title	Study design	DOI
1	The clinical, functional, and radiological effect of long-term used immunosuppressive therapy for post-COVID-19 interstitial lung disease	Cohort study	10.36141/svdld.v40i4.15055
2	Role of Janus Kinase inhibitors in the management of pulmonary involvement due to Long COVID-19 disease: A case control study	Case control study	10.4103/tjem.tjem_363_22
3	Interstitial lung changes and persistent COVID-19 in a patient with follicular lymphoma: A case report	Case report	10.1002/rcr2.1298
4	Tacrolimus Treatment for Post-COVID-19 Interstitial Lung Disease	Case report	10.2169/internalmedicine.7971-21
5	Inflammatory Arthritis After COVID-19: A Case Series	Case series	10.12659/AJCR.939870
6	Efficacy of interleukin-1 blockade with anakinra in the management of post-COVID-19 steroid-dependent multisystem inflammatory syndrome: a case report	Case report	10.1080/03009742.2022.2140485

Part 4: Risk of bias assessment of the included studies

Table S9-3 Risk of bias assessment of included cohort studies

Study	Selection	Comparability	Outcome

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	Representativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure to implants	Demonstration that outcome of interest was not present at start of study	Study controls the most important factor	Study controls for any additional factor	Assessment of outcome	Was follow up long enough for outcomes to occur	Adequacy of follow up of cohorts
Gül et al., 2023	Yes	Yes	No	No	No	No	Yes	Yes	Yes

Table S9-4 Risk of bias assessment of included case control studies

Study	Selection				Comparability		Exposure		
	Is the Case Definition Adequate	Representativeness of the Cases	Selection of Controls	Definition of Controls	Study controls the most important factor	Study controls for any additional factor	Ascertainment of Exposure	Same method of ascertainment for cases and controls	Non-Response Rate
Singh et al., 2023	Yes	Yes	No	No	No	No	Yes	Yes	Yes

Table S9-5 Risk of bias assessment of included case series and case report

Study	Section	Ascertainment		Causality				Reporting
	Does the patient(s) represent(s) the whole experience of the investigator (centre) or is the selection method unclear to the extent that other patients with similar presentation may not have been reported?	Was the exposure adequately ascertained?	Was the outcome adequately ascertained?	Were other alternative causes that may explain the observation ruled out?	Was there a challenge/rechallenge phenomenon?	Was there a dose–response effect?	Was follow-up long enough for outcomes to occur?	Is the case(s) described with sufficient details to allow other investigators to replicate the research or to allow practitioners make inferences related to their own practice?
Yomota et al., 2024	Unclear	Yes	Yes	Yes	No	No	No	Yes
Ohgushi et al., 2022	Unclear	Yes	Yes	Yes	No	No	Yes	Yes
Yadav et al., 2023	Unclear	Yes	Yes	No	Yes	No	Unclear	No
Rinaldi et al., 2023	Unclear	Yes	Yes	Yes	No	No	Unclear	No

Part 5: Evidence syntheses

The available evidence for this PICO question is not suitable for meta-analysis, only part of the original text is presented below.

5.1. Study characters

Table S9-6 Results of immunomodulators for long-COVID in cohort studies

Treatment				before the treatment		1st months of treatment		6th months of treatment		*P*	sample size
NO.	Study ID	therapeutic drug	outcomes	Mean-1	SD-1	Mean-2	SD-2	Mean-3	SD-3		
1	Gül1 et al.2023	methylprednisolone;Mycophenolate mofetil (MMF); MMF	SpO2 %	90.09	5.22	94.1	3.13	95.88	2	<0.001	33
			FVC	2.43	0.98	2.85	1	2.75	1.01	0.006	

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	(cohort)		FVC %	68.06	19.09	73.95	16.76	77.48	18.11	0.001	
			FEV1 %	73.81	19.37	79.5	20.56	81.16	20.62	0.005	
			DLCO %	45.1	18.7	47.95	16.3	54.87	23.46	0.011	

Table S9-7 Results of immunomodulators for long-COVID in case control studies

Treatment			corticosteroid		Janus Kinase inhibitors (JAKi)		*P*
NO.	Study ID	outcomes	Event-1	Total-1	Event-2	Total-2	
1	Singh et al.2023 (case control)	Hyperglycemia	12	20	1	21	0.002
		Oral candidiasis	12		1		<0.001
			Median-1	Total-1	Median-2	Total-2	
		Time to liberation from oxygen therapy (days)	19	20	9	21	<0.001

Table S9-8 Results of immunomodulators for long-COVID in case series and case report

NO.	Study	Age	Gender	Disease/ symptom	Duration of Long COVID	Immunomodulators	Start time of intervention	Duration of intervention	Treatment details	Outcomes
1	Yomota et al., 2024	69	female	Interstitial lung changes and persistent COVID-19	⩾ 11months	Tacrolimus	8 months and 3 days after COVID-19	6 months	Eight months after COVID-19 onset, the patient was re-admitted. Oral tacrolimus 2 mg twice daily for a total dosage of 0.078 mg/kg/day was therefore start from the third day of admission. She is currently receiving PSL 6 mg, which will gradually be tapered, and tacrolimus 4 mg 14 months after COVID-19 onset.	The reticular shadow has decreased
2	Ohgushi et al., 2022	72	male	Interstitial Lung Disease	⩾3 months	Tacrolimus	On the 20th day of admission	>58 days	①On day 20, tacrolimus treatment was started. ②On day 56, Doses of prednisolone (PSL) were further tapered after starting tacrolimus. ③On day 78, he was discharged with 15 mg/day of PSL and 4 mg/day of tacrolimus.	His respiratory condition had further improved, Chest X-ray showed improved pulmonary infiltration in general, but follow-up CT showed that lung fibrosis had progressed in some areas

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3	Yadav et al., 2023	a. 45 b. 46 c. 61	All are female	Inflammatory arthritis	a. ⩾ 3months b. ⩾ 2months c. /	Methotrexate, Leflunomide,	a. about 5weeks after COVID-19 b. 14 months after first episode of COVID-19 c. 1month after COVID-19	a. / b. / c. /	a. She was trialed on methotrexate (unknown dosage); however, she had significant adverse effects, so she was transitioned to golimumab infusions (intravenously every 8 weeks) in February 2022. b. She was started on methotrexate 25 mg/mL injection weekly, along with folic acid 1 mg daily by our rheumatologists. c. A trial of methotrexate and leflunomide as steroid-sparing agents was unsuccessful due to intolerable adverse effects.	a. She had significant adverse effects. b. / c. Intolerable adverse effectsof methotrexate and leflunomide; Hydroxychloroquine was tolerated.
4	Rinaldi et al., 2023	82	male	Multisystem inflammatory	⩾2months	anakinra	/	About 12 months	Anakinra was weltolerated and no adverse events were recorded. At thelast evaluation, the patient was not receiving any immu-nosuppressive treatment and was free of any symptoms,with no laboratory infammatory features	Anakinra was welltolerated and no adverse events were recorded. At thelast evaluation, the patient was not receiving any immu-nosuppressive treatment and was free of any symptoms,with no laboratory infammatory features.

5.2. GRADE assessment (Summary of Findings table)

Table S9-9 Summary of effectiveness and safety of immunomodulators for long COVID

Certainty assessment							№ of patients		Effect		Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	immunomodulator	without immunomodulator	Relative (95% CI)	Absolute (95% CI)	
SpO2 % (follow-up: mean 6 months)											
1	non-randomised studies	not serious	not serious	serious[a]	not serious	none	12	37	-	see comment	⊕◯◯◯ Very low[a]
FVC (follow-up: mean 6 months)											
1	non-randomised studies	not serious	not serious	serious[a]	not serious	none	12	37	-	see comment	⊕◯◯◯ Very low[a]
FVC % (follow-up: mean 6 months)											
1	non-randomised studies	not serious	not serious	serious[a]	not serious	none	12	37	-	see comment	⊕◯◯◯ Very low[a]
FEV1 % (follow-up: mean 6 months)											

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1	non-randomised studies	not serious	not serious	serious[a]	not serious	none	12	37	-	see comment	⊕◯◯◯ Very low[a]

DLCO % (follow-up: mean 6 months)

1	non-randomised studies	not serious	not serious	serious[a]	not serious	none	12	37	-	see comment	⊕◯◯◯ Very low[a]

Respiratory support (follow-up: range 3 weeks to 4 weeks)

1	non-randomised studies	not serious	not serious	not serious	not serious	none	21 participants	20 participants	-	see comment	⊕⊕◯◯ Low
							-	0.0%		not pooled	

Adverse effects-Hyperglycemia (follow-up: range 3 weeks to 4 weeks)

1	non-randomised studies	not serious	not serious	not serious	not serious	none	21 participants	20 participants	-	see comment	⊕⊕◯◯ Low
							-	0.0%		not pooled	

Adverse effects-Oral candidiasis (follow-up: range 3 weeks to 4 weeks)

1	non-randomised studies	not serious	not serious	not serious	not serious	none	21 participants	20 participants	-	see comment	⊕⊕◯◯ Low
							-	0.0%		not pooled	

Interstitial Lung Disease

2	non-randomised studies	serious[b]	not serious	not serious	not serious	publication bias strongly suspected[b]	2 participants		-	see comment	⊕◯◯◯ Very low[b]
							-	0.0%		not pooled	

CI: confidence interval

Explanations

a. No comparisons were made between the immunomodulator and other groups

b. Selective reporting of results

PICO question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?

Evidence summary

Our SR included four studies (one RCT and three before-after studies)[76-79], involving 247 adult patients with long COVID. Data were retrieved from January 1, 2020, to November 30, 2024.

Two studies (involving 183 adult patients with long COVID) assessed the effect of cognitive behavioral therapy (CBT) on fatigue. Results of meta-analysis showed that CBT significantly alleviated general fatigue symptoms compared to no CBT (or baseline) (standardized mean difference [SMD]: 4.01, 95% CI: 1.17 to 6.85; $I^2 = 99\%$, low certainty of evidence) with substantial heterogeneity. One study (using the CIS-fatigue scale) found significant improvement in fatigue symptoms among CBT recipients (SMD: 12.54, 95% CI: 10.86 to 14.22) [77]. The other study used two different scales [79]: The SSD-12 scale showed no statistically significant difference (SMD: 0.08, 95% CI: -0.27 to 0.43), and the FSS scale showed statistically significant improvement (SMD: 0.29, 95% CI: -0.06 to 0.64).

Two studies (involving 93 adult patients with long COVID) assessed the effect of CBT on anxiety. Results of meta-analysis showed that CBT had no significant improvement in anxiety symptoms compared to no CBT (or baseline) (SMD: 0.32, 95% CI: -0.19 to 0.83, $I^2 = 62.7\%$, very low certainty of evidence). One study (using the GAD-7 scale) showed a small and non-statistically significant effect (SMD: 0.10, 95% CI: -0.26 to 0.44) [78]. The other study (using the Self-rating Anxiety Scale) showed significant reduction in anxiety symptoms (SMD: 0.62, 95% CI: 0.10 to 1.14) [78].

Two studies (involving 93 adult patients with long COVID) assessed the effect of CBT on depression. Results of meta-analysis showed that CBT had no significant improvement in depressive compared to no CBT (or baseline) (SMD: 0.25, 95% CI: -0.10 to 0.60, $I^2 = 27.5\%$, low certainty of evidence). One study (using the PHQ-9 scale) showed a small and non-statistically significant effect (SMD: 0.12, 95% CI: -0.23 to 0.47) [76]. The other study (using the Self-rating Depression Scale) showed CBT had a significant improvement in depressive symptoms (SMD: 0.49, 95% CI: -0.03 to 1.00) [78].

Two studies (involving 87 adult patients with long COVID) assessed the effect of CBT on social functioning. Results of meta-analysis showed that CBT had no significant improvement in social functioning compared to no CBT (or baseline)

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(SMD: 5.63, 95% CI: -3.91 to 15.17; $I^2 = 99.4\%$, very low certainty of evidence) with substantial heterogeneity. One study (using the WSAS scale) showed significant improvement in social functioning after CBT (SMD: 10.52, 95% CI: 9.10 to 11.95) [77]. The other study using the SF-36 scale) showed CBT had a moderate improvement in social functioning (SMD: 0.79, 95% CI: 0.26 to 1.31) [78].

Two studies assessed the effect of CBT on self-efficacy [79] and stress [76] in adult patients with long COVID. Due to the limited number of studies for each outcome, meta-analysis was not feasible. The previous results showed that CBT reduced stress levels and increased self-efficacy, with both improvements being statistically significant.

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S10-1 Identification of studies from databases and additional sources

Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	1555
	Web of Science Core Collection	Nov 30, 2024	238
	Embase	Nov 30, 2024	1471
	The Cochrane Library	Nov 30, 2024	398
	PsycINFO	Nov 30, 2024	2140
	China Biology Medicine (CBM)	Nov 30, 2024	4
	Wanfang Database	Nov 30, 2024	42
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	2140
Preprint platform	Medrxiv	Nov 30, 2024	4
	SSRN	Nov 30, 2024	145
	Researchsquare	Nov 30, 2024	137
Clinical trial registries	WHO ICTRP	Nov 30, 2024	141
	ClinicalTrials.gov	Nov 30, 2024	0
	COVID-19 Study Register	Nov 30, 2024	10
Google	-	Nov 30, 2024	6870
Reference lists of relevant SRs	-	Nov 30, 2024	0

Part 2: Flow diagram of the literature search

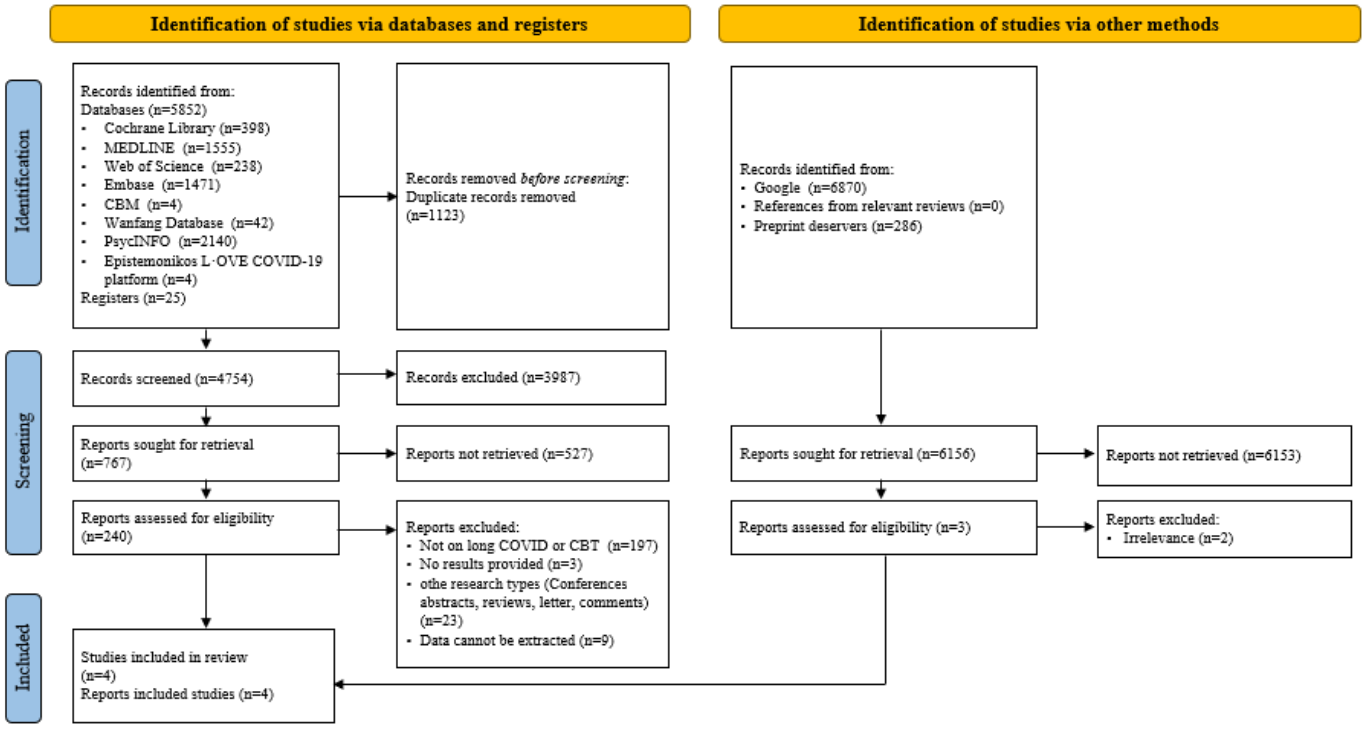


Figure S10-1 Flow diagram of the literature search

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Part 3: list of included studies

Table S10-2 List of included studies

No.	Study title	Study design	DOI
1	Efficacy of Cognitive-Behavioral Therapy Targeting Severe Fatigue Following Coronavirus Disease 2019: Results of a Randomized Controlled Trial	RCT	10.1093/cid/ciad257
2	Cognitive-behavioral therapy for patients with post-COVID-19 condition (CBT-PCC): a feasibility trial	CCT	10.1017/S0033291723002921
3	Effect of cognitive therapy based on positive thinking on depression, stress, and anxiety of mothers after recovering from COVID-19	CCT	10.30514/icss.25.3.95
4	Physical and psychological reconditioning in long COVID syndrome: Results of an out-of-hospital exercise and psychological - based rehabilitation program	CCT	10.1016/j.ijcha.2022.101080

Part 4: Risk of bias assessment of the included studies

Table S10-3 Risk of bias assessment of included studies (RCT)

Random sequence generation	Allocation concealment	Blinding of participants	Blinding of personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other biases	Study
Low	Low	High	High	Unclear	Low	Low	Low	Kuut TA (2019)

Table S10-4 Risk of bias assessment of included studies (CCT)

Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviations from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	overall riak of bias	Study
High	Moderate	Moderate	Moderate	High	Moderate	Unclear	High	Huth D (2024)
High	High	Moderate	Moderate	Moderate	Moderate	Moderate	High	Younesi BS (2023)
High	High	Low	Low	Low	Moderate	Unclear	High	Compagno S (2022)

Part 5: Evidence synthesis

5.1. Meta-analysis

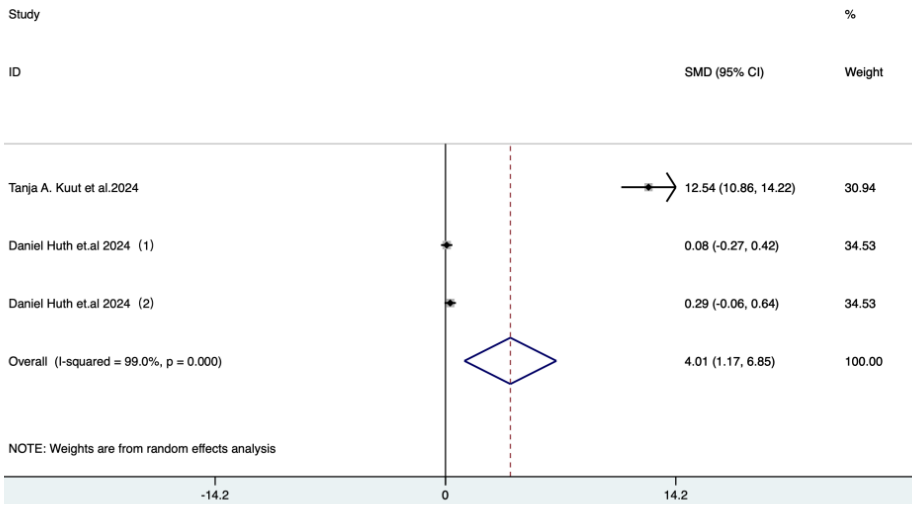


Figure S10-2 Forest plot with summary effect sizes for the impact of CBT on patients with long COVID (fatigue)

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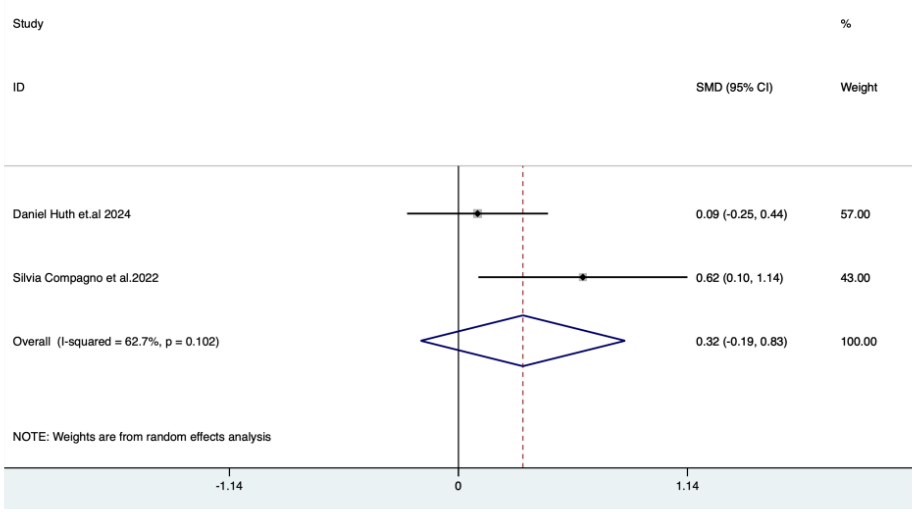


Figure S10-3 Forest plot with summary effect sizes for the impact of CBT on patients with long COVID (anxiety)

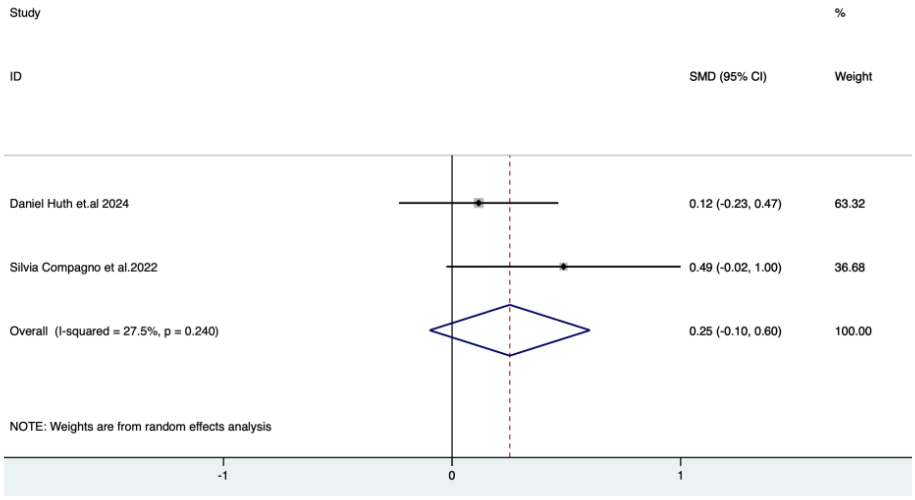


Figure S10-4 Forest plot with summary effect sizes for the impact of CBT on patients with long COVID (depression)

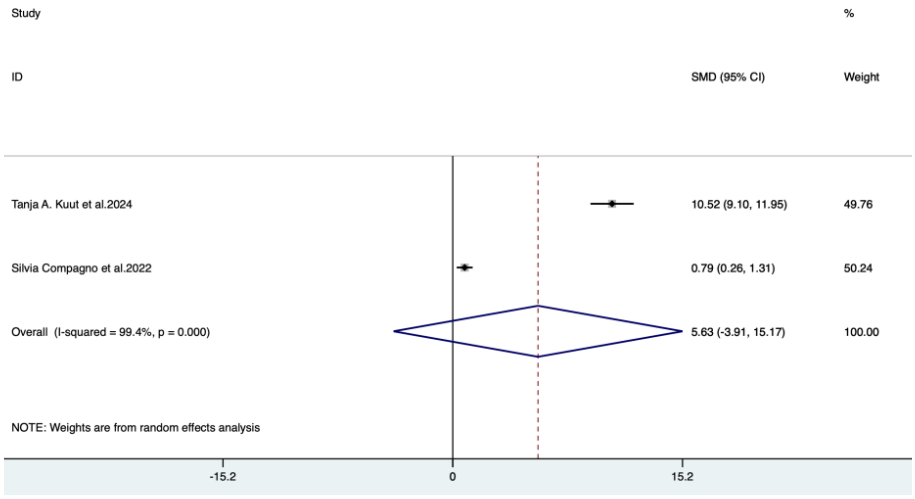


Figure S10-5 Forest plot with summary effect sizes for the impact of CBT on patients with long COVID (social functioning)

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5.2. GRADE assessment (Summary of Findings table)

Table S10-5 Summary of Findings table

Certainty assessment							No. of patients		Effect		Evidence quality
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	other Risk of bias	Intervention group	Control group	Relative (95% CI)	Absolute (95% CI)	
Fatigue											
2	1RCT ,1CCT	serious[a]	serious[b]	not serious	not serious	none	183/366	183/366	SMD 4.01 (1.17, 6.85)	-	⊕⊕◯◯ low
Anxiety											
3	3CCT	serious[a]	serious[b]	not serious	serious[c]	none	113/226	113/226	SMD 0.32 (-0.19, 0.83)	-	⊕◯◯◯ Very low
Depression											
3	3CCT	serious[a]	not serious	not serious	serious[c]	none	113/226	113/226	SMD 0.25 (-0.10, 0.60)	-	⊕⊕◯◯ low
Social Functioning											
2	1RCT ,1CCT	serious[a]	serious[b]	not serious	serious[c]	none	107/214	107/214	SMD 5.63 (-3.91, 15.17)	-	⊕◯◯◯ Very low

a. The studies had a risk of bias.
b. There was substantial heterogeneity.
c. The pooled effect estimate crossed the line of null effect.

PICO question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?

Evidence summary

One SR [80] (data collected from inception to February 2023) included 14 RCTs involving 1,244 adult patients with long COVID (more than a third have been hospitalized for COVID-19) and investigating effect of rehabilitation interventions. Compared with usual care, rehabilitation interventions (breathing exercises [six trials, 815 participants], breathing exercises in combination with resistance and/or aerobic training [five trials, 298 participants], strengthening and aerobic exercises [two trials, 71 participants], or aerobic exercises alone [one trial, 60 participants]) may improve exercise capacity (SMD: -0.56, 95% CI: -0.87 to -0.22, moderate certainty of evidence), dyspnea (SMD: -1.00, 95% CI: -1.94 to -0.10, low certainty of evidence), and quality of life (SMD: -0.41, 95% CI: -0.73 to -0.06, moderate certainty of evidence). No significant differences were observed between rehabilitation interventions and usual care in lower limb muscle strength and endurance (SMD: -0.77, 95% CI: -1.94 to 0.44, very low certainty of evidence), FEV_1 (SMD: -0.16; 95% CI: -0.42 to 0.11, low certainty of evidence), and FVC (SMD: -0.13, 95% CI: -0.38 to 0.12, low certainty of evidence). Adverse events were reported in five trials, however, there was no compelling evidence for a difference in the odds of adverse events occurring between rehabilitation interventions and control group. The wide 95% CI indicates a high level of uncertainty and imprecision in this estimate (OR: 1.68, 95% CI: 0.32 to 9.94, low certainty of evidence).

One living SR [81] (data collected from inception to December 2023, update search in February 2024 through the Epistemonikos covid-19 repository) included eight RCTs involving 985 adult patients with long COVID and investigating effect of rehabilitation interventions. Compared with usual care, inspiratory muscle training may improve dyspnea (MD: 1.10, 95% CI: 0.44 to 1.76, low certainty of evidence), and have little or no important effect on breathlessness and activities (MD: 2.40, 95% CI: -2.66 to 7.46, low certainty of evidence), and quality of life (MD: -1.3, 95% CI: -5.9 to 3.3, low certainty of evidence). Compared with standard physiotherapy, multicomponent exercise of progressively increasing intensity may improve physical function (MD: 6.96, 95% CI: 2.7 to 11.22, low certainty of evidence), dyspnea (mMRC dyspnea scale: MD: -0.76, 95% CI: -1.2 to -0.32, low certainty of evidence; multidimensional dyspnea profile: MD: -18.61, 95% CI: -27.4 to -9.82, low certainty of evidence), breathing discomfort (MD: -1.74, 95% CI: -2.79 to -0.69, low certainty of evidence), emotional response to dyspnea (MD: -6.95, 95% CI: -12.44 to -1.46, low certainty of evidence), and sensory dimension of dyspnea (MD: -9.93, 95% CI: -14.56 to -5.3, low certainty of evidence), however, may have little or no important effect on mental health (MD: 2.06, 95% CI: -3.52 to 7.64, low certainty of evidence). Compared with standard

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physiotherapy, the active breathing techniques may improve fatigue (MD: -9.97, 95% CI: -11.17 to -8.77, very low certainty of evidence). Compared with continuous aerobic exercise, intermittent aerobic exercise probably improves physical function (MD: 3.80, 95% CI: 1.12 to 6.48, moderate certainty of evidence), and may have little or no important effect on mental health (MD: 0, 95% CI: -3.69 to 3.69, low certainty of evidence). Compared with high-intensity aerobic strength training, low-intensity aerobic strength training may improve quality of life (8 weeks, MD: 8.50, 95% CI: 8.08 to 8.92, low certainty of evidence; 24 weeks, MD: 10.40, 95% CI: 9.97 to 10.83, low certainty of evidence).

One SR and network meta-analysis [82] (data collected from 2020 to January 2024) included 10 RCTs involving 765 adult patients with long COVID and investigating effect of telerehabilitation and face-to-face pulmonary rehabilitation interventions. Compared with usual care, face-to-face rehabilitation (MD: 29.06, 95% CI: 2.76 to 55.37, moderate certainty of evidence) and telerehabilitation (MD: 3.13, 95% CI: -26.57 to 32.82, low certainty of evidence) may both improve physical function. Indirect comparison results showed that face-to-face rehabilitation was more effective than telerehabilitation in terms of physical function evaluated by 6MWT or 30 seconds sit-to-stand test. (MD: -25.94, 95% CI: -65.61 to 13.73, low certainty of evidence). Compared with usual care, face-to-face rehabilitation and telerehabilitation may both improve quality of life [physical domain: MD: 6.58, 95%CI: -0.14 to 13.30, low certainty of evidence (face-to-face rehabilitation); MD: 5.17, 95%CI: -1.52 to 11.86, low certainty of evidence (telerehabilitation); Mental domain: MD: 2.14, 95%CI: -4.36 to 8.64, low certainty of evidence (face-to-face rehabilitation); MD: 7.12, 95%CI: 0.61 to 13.64, moderate certainty of evidence (telerehabilitation)]. Indirect comparison results showed that face-to-face rehabilitation was more effective than telerehabilitation in the physical domain of quality of life (MD: 1.41, 95%CI: -8.07 to 10.9, low certainty of evidence), whereas face-to-face rehabilitation was less effective than telerehabilitation in the mental domain of quality of life (MD: -4.98, 95%CI -14.19 to 4.22, low certainty of evidence).

Evidence retrieval and synthesis

Part 1: Search resources and results

Table S11-1 Identification of studies from databases and additional sources

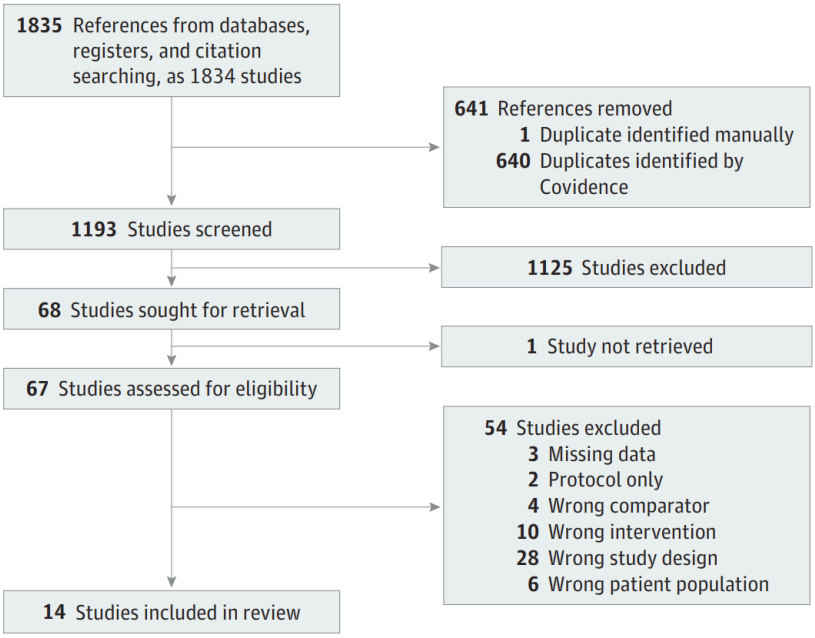
Category	Name	Search time	Number
Database	MEDLINE (via PubMed)	Nov 30, 2024	301
	Web of Science Core Collection	Nov 30, 2024	332
	Embase	Nov 30, 2024	196
	The Cochrane Library	Nov 30, 2024	4
	China Biology Medicine (CBM)	Nov 30, 2024	280
	Wanfang Database	Nov 30, 2024	424
	Epistemonikos L·OVE COVID-19 platform	Nov 30, 2024	711
Preprint platform	Medrxiv	Nov 30, 2024	10
	SSRN	Nov 30, 2024	5
	Researchsquare	Nov 30, 2024	352
Clinical trial registries	WHO ICTRP	Nov 30, 2024	502
	ClinicalTrials.gov	Nov 30, 2024	535
	COVID-19 Study Register	Nov 30, 2024	1104
Google	-	Nov 30, 2024	12
Reference lists of relevant SRs	-	Nov 30, 2024	2

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Part 2: SRs used to support the PICO question 11

PICO question 11 included three SRs to support the recommendations. The detailed information on three SRs is presented below:

Title: Rehabilitation Interventions for Physical Capacity and Quality of Life in Adults With Post-COVID-19 Condition: A Systematic Review and Meta-Analysis.
Journal: JAMA Network Open
Search date: from January 2020 to February 2023
Flow diagram of the literature search:



Characteristics of the Included Studies:

Table 1. Characteristics of the Included Studies

Source	Primary outcome	Population	Age, mean (SD) y	Participants, No.	Women, No. (%)	Men, No. (%)	Intervention	Control	Length of inpatient stay, d	Level of dyspnea
Teixeira DO Amaral et al,[18] 2022	6MWT, FEV_1,[a] and FVC[a]	Hospitalized patients due to COVID-19 (6.2% ICU patients)	52.0 (10.8)	32	17 (53.1)	15 (46.9)	Clinician supported, home-based aerobic and resistance exercise program	Usual care	6.8	NI
Capin et al,[20] 2022	30-s STS, and dyspnea[b]	Hospitalized patients due to COVID-19 (with and without ICU stay)	53.0 (10)	41	18 (44.0)	23 (56.0)	Supervised breathing and clearance techniques, high-intensity strength training, aerobic and cardiovascular exercise, balance exercises, functional activities, stretching, coaching, and motivational interviewing	Usual care	6.0	Moderate to severe
Del Corral et al,[21] 2023	30-s STS, health-related quality of life,[c] FEV_1,[a] and FVC[a]	Patients reporting persistent fatigue and dyspnea following PCR diagnosis of COVID-19 infection (32.0% hospital admission rate; 6.0% ICU patients)	46.5 (10.2)	88	14 (16.0)	74 (84.0)	Supervised inspiratory muscle training and respiratory muscle training	Sham	NI	NI
De Souza et al,[19] 2021	30-s STS	Nonhospitalized patients	NI	196	NI	NI	Supervised low-intensity pulmonary rehabilitation	Usual care	NA	NI
Jimeno-Almazán et al,[22] 2023 and Jimeno-Almazán et al,[23] 2022	Dyspnea,[b] quality of life,[d] FEV_1,[a] and FVC[a]	Nonhospitalized patients with a PCR diagnosis of COVID-19 presenting with persistent symptoms	45.3 (9.7)	39	29 (74.3)	10 (25.7)	Supervised resistance training combined with aerobic training	Usual care	NA	Mild and moderate
Li et al,[16] 2022	6MWT, quality of life,[d] FEV_1,[a] and FVC[a]	Patients reporting persistent moderate dyspnea (mMRC >2-3) after inpatient treatment for COVID-19	50.6 (11.0)	119	66 (55.5)	53 (44.5)	Unsupervised breathing control and thoracic expansion, aerobic exercise, and lower limb muscle strengthening exercises	Usual care	26.2	Moderate
Liu et al,[17] 2020	6MWT	Hospitalized patients due to COVID-19 aged ≥65 y	67.5 (7.8)	72	23 (32.0)	49 (68.0)	Supervised respiratory muscle training, cough exercise, diaphragmatic training, stretching exercise, and unsupervised home exercise	Usual care	NI	NI
McNarry et al,[24] 2022	Dyspnea[e] and quality of life[f]	Patients reporting persistent dyspnea following COVID-19 infection	46.6 (12.2)	281	247 (88.0)	34 (12.0)	Unsupervised inspiratory muscle training	Usual care	NI	Moderate
Okan et al,[25] 2022	6MWT, dyspnea,[b] quality of life,[g] FEV_1,[a] and FVC[a]	Patients presenting with persistent dyspnea following COVID-19–induced pneumonia	50.0 (12.8)	52	25 (48.0)	27 (52.0)	Supervised breathing exercise	Usual care	9.5	Moderate
Phillip et al,[26] 2022	Dyspnea[h] and quality of life[i]	Patients reporting persistent dyspnea following COVID-19 infection (17.0% hospital admission rate)	49.5 (12.0)	150	26 (17.5)	124 (82.5)	Supervised posture and breathing exercises	Usual care	NI	Mild
Rodriguez-Blanco et al,[30] 2023	6MWT, 30-s STS, dyspnea[h]	Patients with symptoms attributed to COVID-19 by medical services	40.7 (13.4)	48	26 (54.2)	22 (45.8)	Breathing and strength-based exercises	Usual care	NI	Mild

(continued)

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Table 1. Characteristics of the Included Studies (continued)

Source	Primary outcome	Population	Age, mean (SD) y	Participants, No.	Women, No. (%)	Men, No. (%)	Intervention	Control	Length of inpatient stay, d	Level of dyspnea
Romanet et al,[27] 2022	Dyspnea[b] and quality of life[d]	Patients reporting persistent dyspnea (mMRC >1) following COVID-19-related acute respiratory distress syndrome (ICU admission)	58.2 (12.5)	60	23 (38.1)	37 (61.9)	Supervised endurance training rehabilitation	Usual care	26.0	Moderate
Sari et al,[28] 2023	6MWT	PCC with pulmonary involvement (83.0% hospitalized, 17.0% ICU)	56.2 (4.5)	24	8 (33.3)	16 (66.7)	Nonsupervised breathing exercises, resistance exercises, and inspiratory muscle training	Usual care	NI	Mild and moderate
Şahin et al,[29] 2023	6MWT, dyspnea,[b] FEV_1,[a] FVC,[a] and quality of life[g]	Patients hospitalized with PCC (ICU and ward for >10 d)	60.7 (8.2)	42	14 (33.3)	28 (66.7)	Clinician supported breathing exercises, strength exercises, and walking program	Usual care	11.5	Moderate

Abbreviations: 6MWT, 6-minute walking-test; FEV_1, forced expiratory volume in the first second; FVC, forced vital capacity; ICU, intensive care unit; mMRC, modified Medical Research Council Dyspnea Scale; NA, not applicable; NI, no information; PCC, post-COVID-19 condition; PCR, polymerase chain reaction; STS, sit-to-stand test.

[a] Calculated as estimated percentages.

[b] Measured with the mMRC.

[c] Measured with the Euroqol-5 Dimension, 5-Level (EQ-5D-5L) Questionnaire.

[d] Measured with the 12-Item Short Form Survey (SF-12) mental component score and physical component score.

[e] Measured with the Transition Dyspnea Index.

[f] Measured with The King's Brief Interstitial Lung Disease Questionnaire.

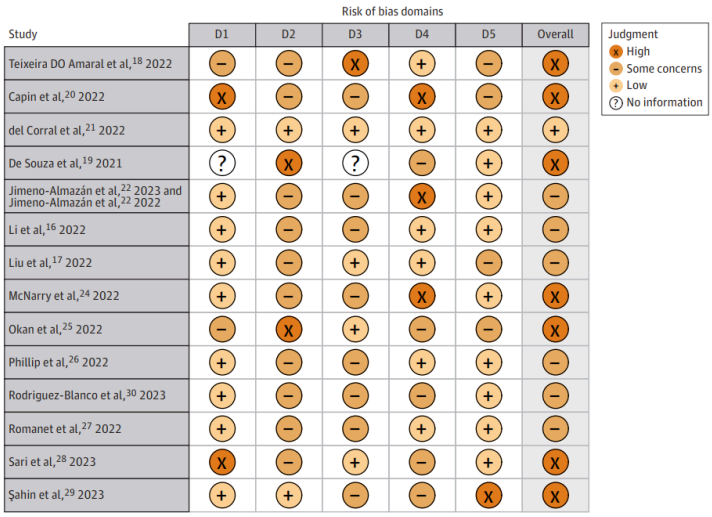
[g] Measured with the St George Respiratory Questionnaire.

[h] Measured with the Dyspnea-12 Questionnaire.

[i] Measured with the physical health component and mental health component of the 36-Item Short Form Survey (SF-36).

Risk of Bias Assessment:

Figure 2. Risk of Bias Assessment



Domains include bias arising from the randomization process (D1), bias due to deviations from intended intervention (D2), bias due to missing outcome data (D3), bias in measurement of the outcome (D4), and bias in selection of the reported result (D5).

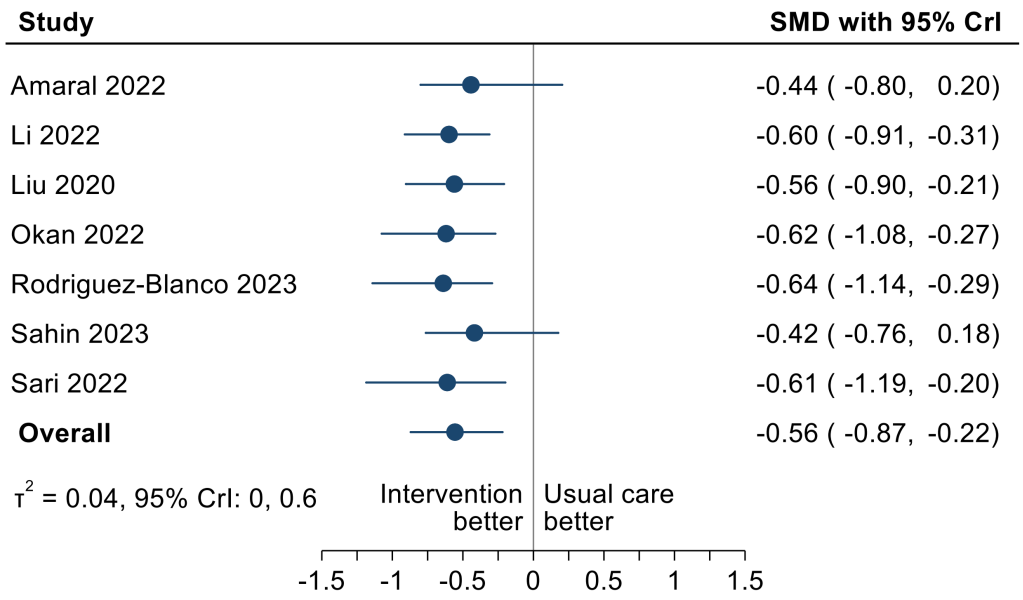
Relevant result:

Functional Aerobic Capacity: 6-minute walking test (7 RCTs, 389 participants)

Rehabilitation interventions were associated with a greater improvement in functional exercise capacity compared with usual care (SMD, −0.56; 95% CrI, −0.87 to −0.22). Rehabilitation interventions demonstrated a 99.6% posterior probability of superiority and were associated with 95% probability of reaching the MID threshold when compared with controls. Specifically, patients in the intervention group were able to cover a mean (SD) of 35.84 (6.55) m more than patients in the usual care group during the 6-minute walking test (95% CI, 34.97 m to 36.71 m). The value of τ2 (0.04; 95% CrI, 0.00 to 0.60) indicated low heterogeneity for exercise capacity. The confidence in the certainty of evidence was moderate and was rated down due to high risk of bias.

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eFigure 1. Caterpillar plot for exercise capacity



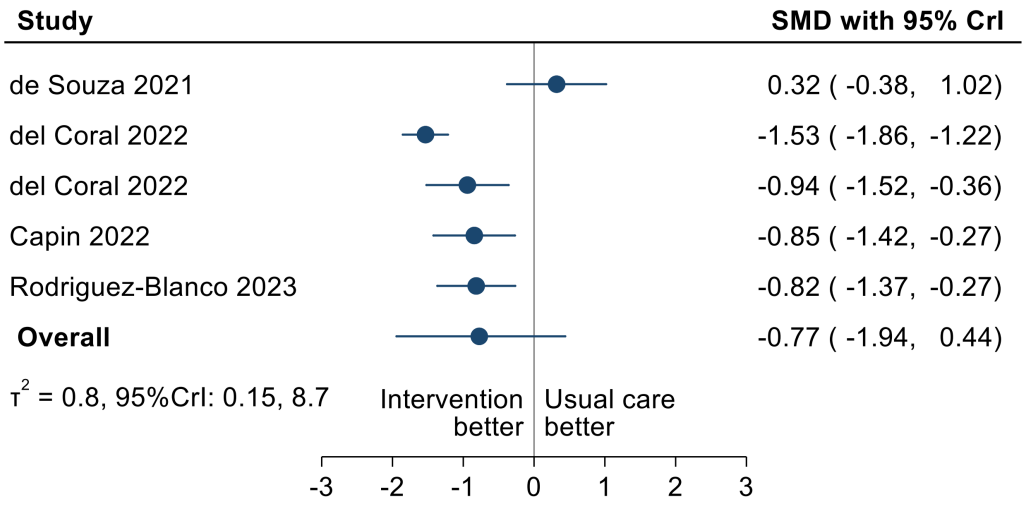
Fatigue

The available evidence on fatigue outcomes was limited and could not be synthesized. A single study measured the presence or absence of fatigue during everyday activities as a binary outcome without quantifying it. A total of 2 studies assessed only exercise-induced fatigue levels, and 3 studies (4 records) did not adequately report the circumstances under which fatigue was measured. Additionally, 2 studies recorded the presence of exercise-induced fatigue as an adverse event.

Functional Lower Limb Muscle Strength and Endurance (4 RCTs, 371 participants)

The summary point estimate suggested that rehabilitation interventions could be associated with improvements in lower limb muscle function when compared with usual care. However, given that the CrI included both positive and negative values, the magnitude and direction of the association remain uncertain (SMD, −0.77; 95% CrI, −1.94 to 0.44). Rehabilitation interventions demonstrated a 93% posterior probability of superiority and were associated with 84% probability of reaching the MID threshold when compared with usual care. The value of τ2 (0.8; 95% CrI, 0.15 to 8.70) indicated substantial statistical heterogeneity for the 30-second sit to stand test. The confidence in the certainty of evidence of the treatment outcomes was very low and was rated down due to high risk of bias, imprecision, and indirectness.

eFigure 2. Caterpillar plot for lower limb muscle function



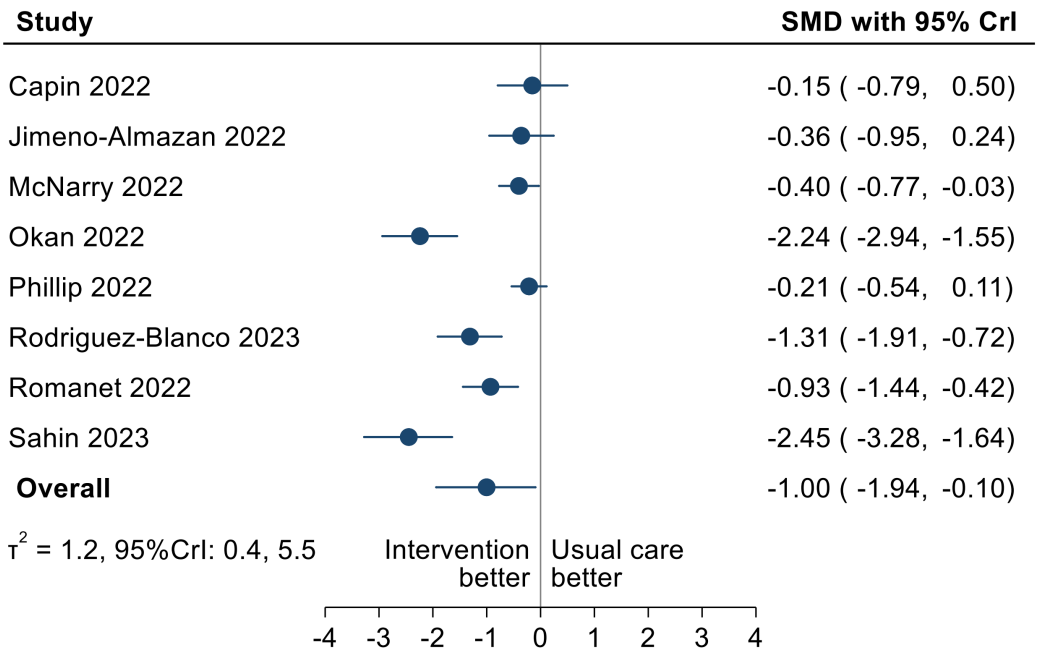
Dyspnea (8 RCTs, 573 participants)

Rehabilitation interventions were associated with a greater improvement in functional exercise capacity compared with usual care (SMD, −1.00; 95% CrI, −1.94 to −0.10). Rehabilitation interventions were associated with a 98% posterior probability of superiority when compared with usual care and were associated with a 95% probability to reach the MID threshold. The

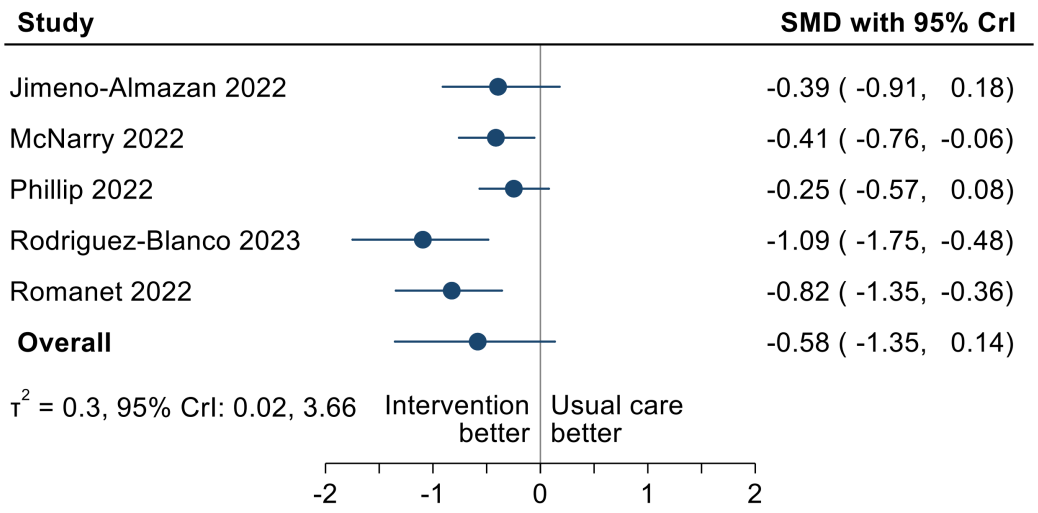
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magnitude of τ2 indicated substantial statistical heterogeneity (τ2, 1.18; 95% CrI, 0.36 to 5.50). Heterogeneity was partially explained when adjusting for allocation concealment. The confidence in the certainty of evidence was low and rated down due to high risk of bias and inconsistency.

eFigure 3. Caterpillar plot for dyspnea



eFigure 4. Sensitivity analysis for dyspnea (allocation concealment)

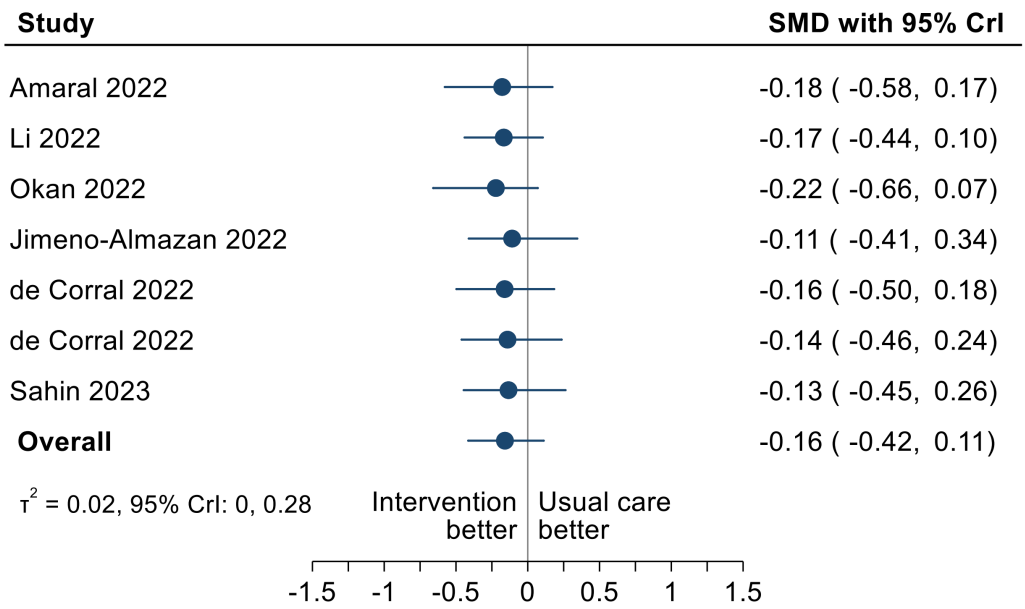


Respiratory Function: FEV_1 and FVC (6 RCTs, 363 participants)

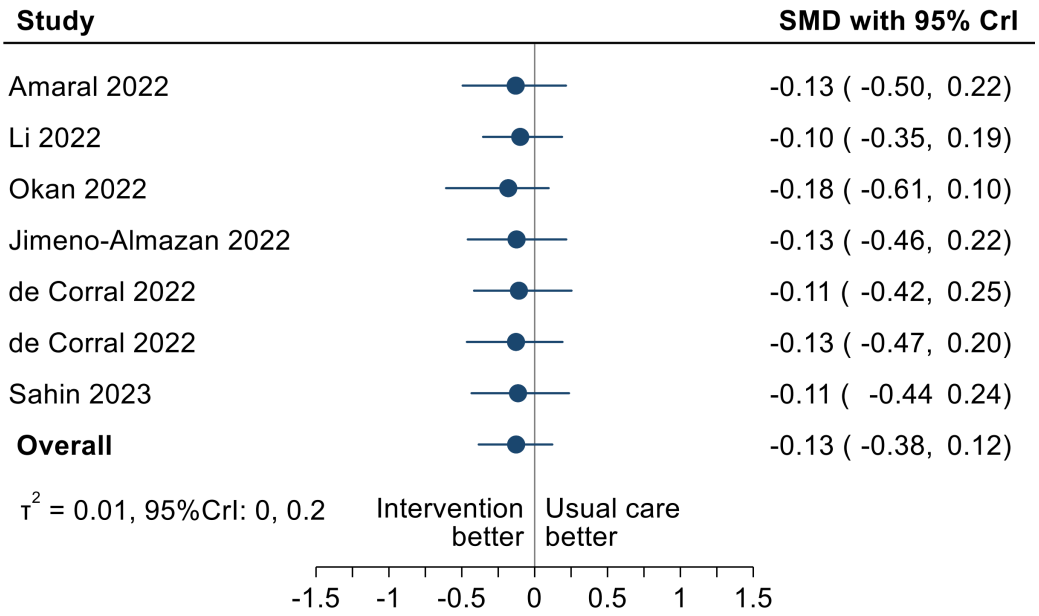
They found no difference between rehabilitation interventions and usual care in either FEV1 (SMD, −0.16; 95% CrI, −0.42 to 0.11) or FVC (SMD, −0.13; 95% CrI, −0.38 to 0.12). Rehabilitation interventions demonstrated 89% and 85% posterior probability of superiority when compared with usual care for FEV1 and FVC respectively. The magnitude of τ2 indicated very low statistical heterogeneity for both FEV1 (τ2, 0.02; 95% CrI, 0.00 to 0.28) and FVC (τ2, 0.01; 95% CrI, 0.00 to 0.23). The confidence in the certainty of evidence was low for both outcomes and downgraded for high risk of bias and indirectness.

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eFigure 5. Caterpillar plot for respiratory function (FEV1)



eFigure 6. Caterpillar plot for respiratory function (FVC)

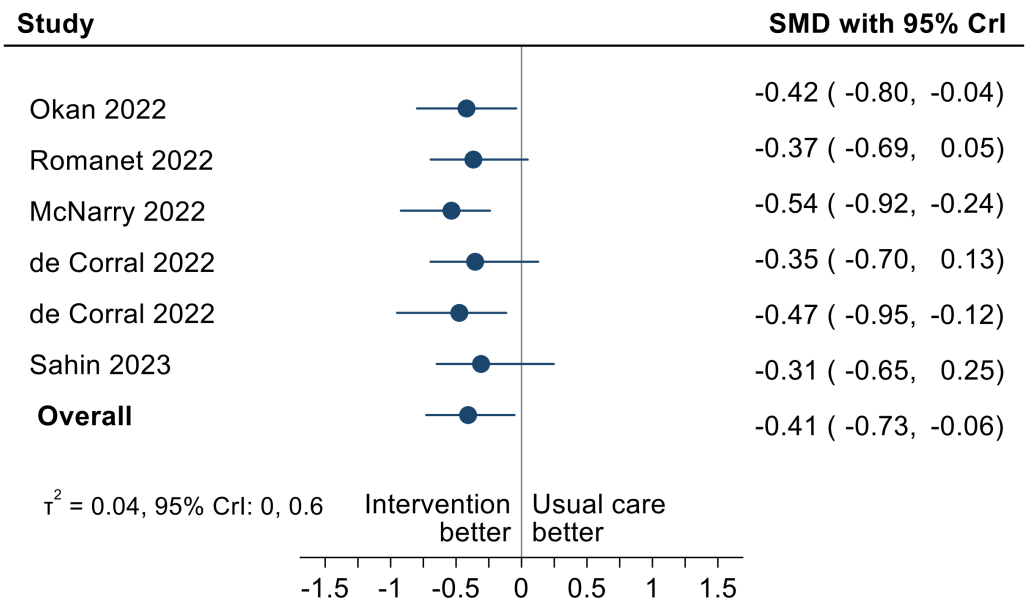


Quality of Life (5 RCTs, 366 participants)

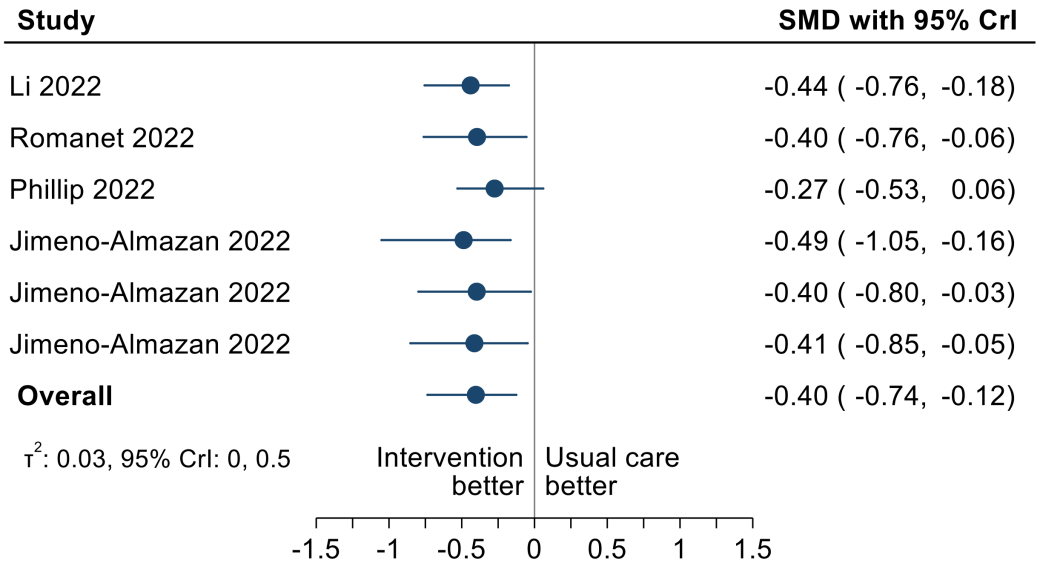
Rehabilitation interventions were associated with larger improvement in quality of life compared with the comparison group (SMD, −0.41; 95% CrI, −0.73 to −0.06). Rehabilitation interventions were associated with a 98.4% posterior probability of being superior to the comparison group and were associated with a 76.9% probability to reach the MID threshold. The magnitude of τ2 indicated very low statistical heterogeneity (τ2, 0.04; 95% CrI, 0.00 to 0.63). The confidence in the certainty of evidence was moderate and rated down for high risk of bias.

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eFigure 7. Caterpillar plot for quality of life outcomes

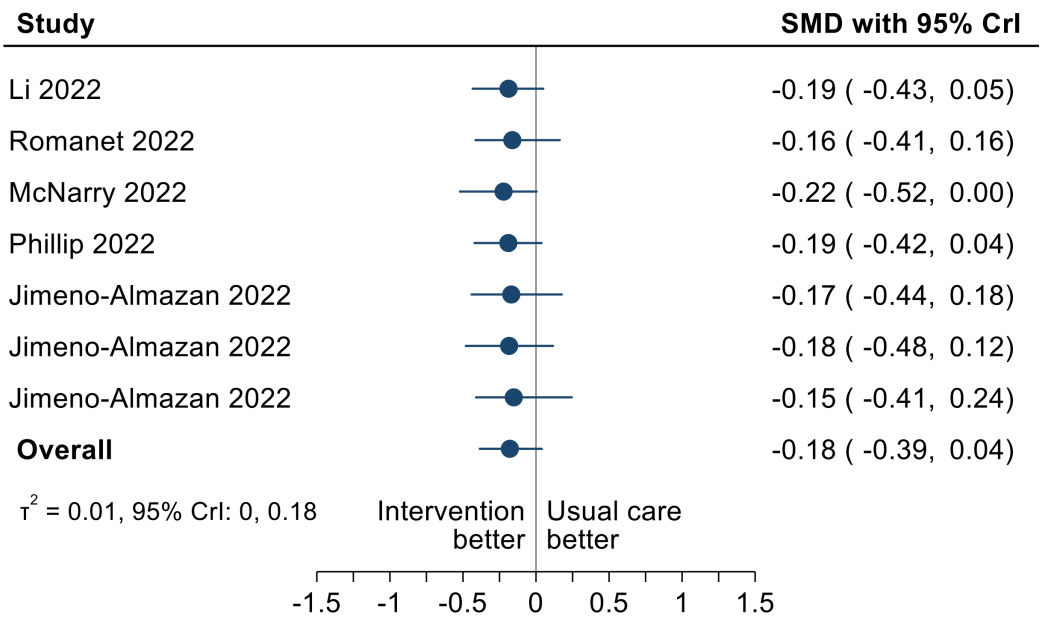


eFigure 8. Caterpillar plot of physical health outcomes



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eFigure 9. Caterpillar plot for mental health outcomes



Adverse Events

Five trials did not provide any information on adverse events, and 4 trials (5 records) 0 reported no adverse events related to the intervention. Adverse events were reported in 5 trials. However, we did not find compelling evidence for a difference in the odds of adverse events occurring. The wide 95% CrI indicates a high level of uncertainty and imprecision in this estimate (OR, 1.68; 95% CrI, 0.32 to 9.94).

eTable 4. Adverse events Results

	Odds Ratio			Heterogeneity	
Model	**Median**	**95% CrI**	**Probability of AEs**	**Median**	**95% CrI**
Fixed Effect	1.45	0.65, 3.40	82%	-	-
Random Effects	1.68	0.32, 9.94	75%	0.30	0.07, 117.70

eTable 5: Model fit and model comparison for adverse events

Summary	Fixed effect model	Random effects model
Posterior mean residual deviances	35.18	25.58
Posterior mean deviances	40.91	33.17
Effective number of parameters	5.73	7.60
Deviance information criteria (DIC)	46.65	40.77

GRADE assessment results

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Table 2. Grade Rating of Studies

Outcomes	Randomized clinical trials, No.	Participants, No.	Effect size estimate, (95% CrI)	Certainty of evidence (GRADE)[a]	Reason for downgrade
Functional exercise capacity (6MWT)	7	389	−0.56 (−0.87 to −0.22)[b]	Moderate	Downgraded for high risk of bias
Functional leg strength and endurance (STS)	4	371	−0.77 (−1.94 to 0.44)[b]	Very low	Downgraded for high risk of bias and imprecision
Dyspnea	8	573	−1.00 (−1.94 to −0.10)[b]	Low	Downgraded for high risk of bias and inconsistency
Respiratory function (FEV_1, %)	6	363	−0.16 (−0.42 to 0.11)[b]	Low	Downgraded for high risk of bias and indirectness
Respiratory function (FVC, %)	6	363	−0.13 (−0.38 to 0.12)[b]	Low	Downgraded for high risk of bias and indirectness
Quality of life (overall)	5	366	−0.41 (−0.73 to −0.06)[b]	Moderate	Downgraded for high risk of bias
Quality of life (mental health)	5	568	−0.18 (−0.39 to 0.04)[b]	Moderate	Downgraded for high risk of bias
Quality of life (physical health)	4	444	−0.40 (−0.74 to −0.12)[b]	Moderate	Downgraded for high risk of bias
Adverse events	5	544	1.68 (0.32 to 9.94)[c]	Low	Downgraded for high risk of bias

Abbreviations: 6MWT, 6-minute-walking-test; CrI, credible interval; FEV_1, forced expiratory volume in the first second; FVC, forced vital capacity; GRADE, grading of recommendations, assessment, development, and evaluations; STS, sit-to-stand test.

[a] The certainty of evidence was rated using the GRADE system, with a rating of high indicating we are very confident that the true effect lies close to the effect size estimate; moderate indicating we are moderately confident in the effect size estimate and that the true effect is likely to be close to the estimate of effect size, but there is a possibility that it is substantially different; low indicating our confidence in the effect size estimate is limited and that the true effect may be substantially different from the size estimate of effect; and very low indicating we have very little confidence in the effect size estimate and that the true effect is likely to be substantially different from the size estimate of effect.

[b] Reported as a standardized mean difference.

[c] Reported as an odds ratio.

Title: Interventions for the management of long COVID (post-COVID condition): living systematic review
Journal: The BMJ
Search date: from inception to December 2023
Flow diagram of the literature search: Eight of the 239 registered trials investigated physical activity or rehabilitation (n=985).

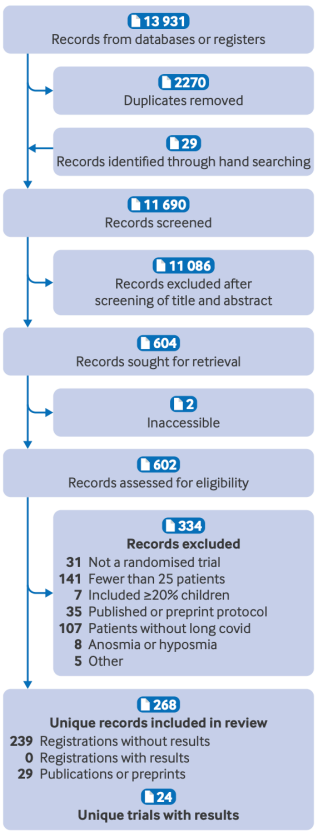


Fig 1 | Selection of trials for inclusion in systematic review

Characteristics of the Included Studies:
Only six studies that included outcomes of interest were recorded.

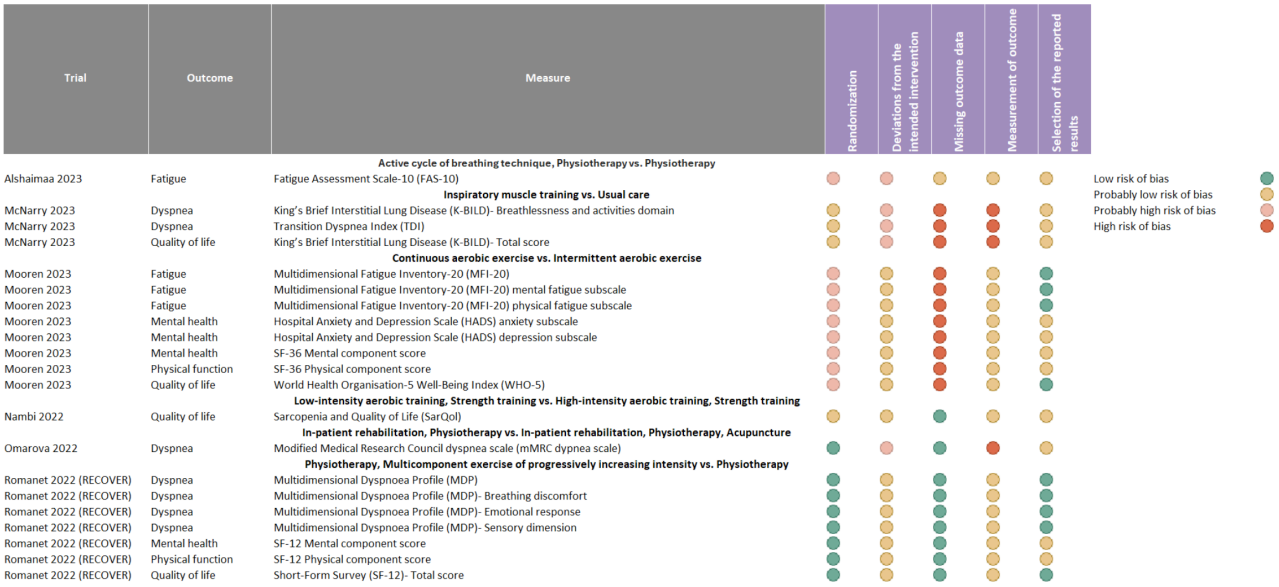
Author	Year	Trial name	Registration	Publication status	Funding	Trial design	Country	Recruitment	Method for recruitment	Subtype of long COVID	% Male	Age	Laboratory confirmed infection	Hospitalized	ICU	Fully vaccinated (according to CDC)	Duration of long COVID	Number of participants randomized
Romanet	2022	RECOVER	NCT04569266	Peer-reviewed, Preprint	Institutional	Parallel	France	July 2020 to January 2022	Other (Hospital)	Respiratory	61.67	58.1	NR	100	100	NR	NR	27
Mooren	2023	NR	NCT06016192	Peer-reviewed	NR	Parallel	Germany	August 2021 to not reported	Other (Inpatient rehabilitation clinic)	General	61.8	49.3	100	28.2	NR	NR	≥3	69/70
Nambi	2022	NR	NCT04796064	Peer-reviewed	Institutional	Parallel	Saudi Arabia	March 2020 to not reported	Other (Local and	General (sarcopenia)	100	63.65	NR	NR	NR	NR	NR	38/38

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									government hospitals)									
McNarry	2022	NR	NR	Peer-reviewed	Government	Parallel	United Kingdom	NR	(Social)media, Other (Hospital, Online support groups)	Respiratory	12.84	46.6	NR	NR	NR	NR	NR	224/69
Alshaimaa	2023	NR	NR	Peer-reviewed	NR	Parallel	Egypt	NR	Other (Hospital)	Respiratory	41.67	45.67	NR	NR	NR	NR	NR	30/30
Omarova	2023	NR	NR	Peer-reviewed	None	Parallel	Kazakhstan	March 2022 to not reported	Other (Rehabilitation center)	General	24.4	61.3	NR	NR	NR	NR	NR	80/80

Risk of Bias Assessment:

Supplement 5. Risk of bias of trials reporting on physical activity and rehabilitative interventions



Relevant result:

Two trials (n=209 patients) compared rehabilitation programmes involving physical activity against usual care or general education about COVID-19 and activities of daily living. Physical activity programmes involved two or three 60-minute exercise sessions incorporating aerobic exercise and strength training for 12 or 15 weeks, one of which was delivered online. These trials did not report on our outcomes of interest.

Moderate certainty evidence from one trial (n=110 patients) suggests that intermittent aerobic exercise 3-5 times weekly for 4-6 weeks probably improves physical function compared with continuous exercise (mean difference 3.8, 95% CI 1.12 to 6.48); SF-36 physical component score; range 0-100; higher scores indicate less impairment).

Other trials compared a programme of multicomponent exercise of increasing intensity combined with physiotherapy against physiotherapy alone, low versus high intensity aerobic and strength training, a programme of in-patient rehabilitation combined with acupuncture against in-patient rehabilitation alone, inspiratory muscle training (a form of respiratory training to strengthen the muscles involved in inhalation) against usual care, and a combination of physiotherapy and active cycle of breathing (breathing exercises intended to improve dyspnoea) against physiotherapy alone. The effects of these interventions were supported by only low or very low certainty evidence.

	Effect estimates. Mean difference or risk difference per 1000 people (95% CI)						
Comparison	**Recovery or important improvement**	**Fatigue**	**Physical function**	**Cognitive function**	**Mental health**	**Quality of life or wellbeing**	**Serious adverse events**
Intermittent aerobic exercise *v* continuous aerobic exercise			3.8 (1.12 to 6.48) SF-36 physical component score (range 0-100; higher scores indicate less impairment)		0 (-3.69 to 3.69) SF-36 mental component score (range 0-100; higher scores indicate less impairment)		
Multicomponent exercise of progressively increasing intensity physiotherapy *v* physiotherapy			6.96 (2.70 to 11.22) SF-36 physical component score (range 0-100; higher scores indicate less impairment)		2.06 (-3.52 to 7.64) SF-36 mental component score (range 0-100; higher scores indicate less impairment)		

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Inspiratory muscle training *v* usual care						-1.3 (-5.90 to 3.30) KBILD total score (range 0-100; higher scores indicate less impairment)	

GRADE ratings and interpretation				
	High certainty	Definitely more effective	Definitely worse	Definitely no different
	Moderate certainty	Probably more effective	Probably worse	Probably no different
	Low certainty	May be more effective	May be worse	May be no different
	Very low certainty	Very uncertain		

GRADE assessment results

Supplement 14: Summary of findings table comparing Physiotherapy, Multicomponent exercise of progressively increasing intensity and Physiotherapy

Patients: Patients with long COVID
Intervention: Physiotherapy, Multicomponent exercise of progressively increasing intensity
Comparator: Physiotherapy

Outcome	Trials (patients)	Relative Risk (95% CI)	Absolute effect (per 1000 patients)	Certainty of the evidence	Plain language summary
Physical function 12.85 weeks	1 (60 patients)	MD: 6.96 (95% CI: 2.7 to 11.22) SF-36 Physical Component Score (Range: 0 to 100; higher scores indicate less impairment)		Low due to very serious imprecision[a]	May improve physical function.
Mental health 12.85 weeks	1 (60 patients)	MD: 2.06 (95% CI: -3.52 to 7.64) SF-36 Mental Component Score (Range: 0 to 100; higher scores indicate less impairment)		Low due to very serious imprecision[a]	May have little or no important effect on mental health.
Dyspnea 12.85 weeks	1 (60 patients)	MD: -0.76 (95% CI: -1.2 to -0.32) Modified Medical Research Council Dyspnea Scale (Range: 0 to 4; higher scores indicate greater impairment)		Low due to very serious imprecision[a]	May improve dyspnea.
Dyspnea 12.85 weeks	1 (60 patients)	MD: -18.61 (95% CI: -27.4 to -9.82) Multidimensional Dyspnoea Profile (Range: 0 to 10; higher scores indicate greater impairment)		Low due to very serious imprecision[a]	May improve dyspnea.
Dyspnea 12.85 weeks	1 (60 patients)	MD: -1.74 (95% CI: -2.79 to -0.69) Multidimensional Dyspnoea Profile - Breathing Discomfort (Range: 0 to 10; higher scores indicate greater impairment)		Low due to very serious imprecision[a]	May improve breathing discomfort.
Dyspnea 12.85 weeks	1 (60 patients)	MD: -6.95 (95% CI: -12.44 to -1.46) Multidimensional Dyspnoea Profile - Emotional Response (Range: 0 to 10; higher scores indicate greater impairment)		Low due to very serious imprecision[a]	May improve emotional response to dyspnea.
Dyspnea 12.85 weeks	1 (60 patients)	MD: -9.93 (95% CI: -14.56 to -5.3) Multidimensional Dyspnoea Profile - Sensory Dimension (Range: 0 to 10; higher scores indicate greater impairment)		Low due to very serious imprecision[a]	May improve dyspnea.

RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval

a Likely much too few participants to achieve prognostic balance.

Romanet C, Wormser J, Fels A, Lucas P, Prudat C, Sacco E, et al. Effectiveness of exercise training on the dyspnoea of individuals with long COVID: A randomised controlled multicentre trial. Ann Phys Rehabil Med. 2023;66(5):101765.

Supplement 15: Summary of findings table comparing Intermittent aerobic exercise and Continuous aerobic exercise

Patients: Patients with long COVID
Intervention: Intermittent aerobic exercise
Comparator: Continuous aerobic exercise

Outcome	Trials (patients)	Relative Risk (95% CI)	Absolute effect (per 1000 patients)	Certainty of the evidence	Plain language summary
Physical function 5 weeks	1 (110 patients)	MD: 3.8 (95% CI: 1.12 to 6.48) SF-36 Physical component score (Range: 0 to 100; higher scores indicate less impairment)		Moderate due to serious risk of bias[a]	Probably improves physical function.
Mental health 5 weeks	1 (110 patients)	MD: 0 (95% CI: -3.69 to 3.69) SF-36 Mental component score (Range: 0 to 100; higher scores indicate less impairment)		Low due to serious risk of bias[a] and serious imprecision[b]	May have little or no important effect on mental health.

RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval

a Concerns related to bias due to the randomization process and missing data.

b The confidence interval include both appreciable benefit and harm.

Mooren JM, Garbsch R, Schafer H, Kotewitsch M, Waranski M, Teschler M, et al. Medical Rehabilitation of Patients with Post-COVID-19 Syndrome-A Comparison of Aerobic Interval and Continuous Training. J Clin Med. 2023;12(21).

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Supplement 16: Summary of findings table comparing Low-intensity aerobic exercise, Strength training and High-intensity aerobic exercise, Strength training

Patients: Men with long COVID and sarcopenia **Intervention:** Low-intensity aerobic training, Strength training **Comparator:** High-intensity aerobic training, Strength training		**NOTE:** CONCERNS WITH TRIAL INTEGRITY. The trial is registered in March 2021. The trial, however, was conducted between March 2020 to April 2021. The standard deviations of baseline characteristics and outcome measures are remarkably small.			
Outcome	**Trials (patients)**	**Relative Risk (95% CI)**	**Absolute effect (per 1000 patients)**	**Certainty of the evidence**	**Plain language summary**
Quality of life 8 weeks	1 (73 patients)	MD: 8.5 (95% CI: 8.08 to 8.92) Sarcopenia and Quality of Life (SarQol) (Range: 0 to 100; higher scores indicate less impairment)		Low due to serious indirectness[a] and serious imprecision[b]	May improve quality of life.
Quality of life 24 weeks	1 (69 patients)	MD: 10.4 (95% CI: 9.97 to 10.83) Sarcopenia and Quality of Life (SarQol) (Range: 0 to 100; higher scores indicate less impairment)		Low due to serious indirectness[a] and serious imprecision[b]	May improve quality of life.
RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval					
a The trial was conducted in men with sarcopenia.					
b Likely too few participants to achieve prognostic balance.					
Nambi GA-O, Abdelbasset WA-OX, Alrawaili SM, Elsayed SH, Verma A, Vellaiyan A, et al. Comparative effectiveness study of low versus high-intensity aerobic training with resistance training in community-dwelling older men with post-COVID 19 sarcopenia: A randomized controlled trial. 2021(1477-0873 (Electronic))					

Supplement 17: Summary of findings table comparing Inspiratory muscle training and Usual care

Patients: Patients with long COVID **Intervention:** Inspiratory muscle training **Comparator:** Usual care					
Outcome	**Trials (patients)**	**Relative Risk (95% CI)**	**Absolute effect (per 1000 patients)**	**Certainty of the evidence**	**Plain language summary**
Quality of life 8 weeks	1 (148 patients)	MD: -1.3 (95% CI: -5.9 to 3.3) King's Brief Interstitial Lung Disease - Total score (Range: 0 to 100; higher scores indicate less impairment)		Low due to serious risk of bias[a] and serious imprecision[b]	May have little or no important effect on quality of life.
Dyspnea 8 weeks	1 (148 patients)	MD: 2.4 (95% CI: -2.66 to 7.46) King's Brief Interstitial Lung Disease - Breathlessness and activities domain (Range: 0 to 100; higher scores indicate less impairment)		Low due to serious risk of bias[a] and serious imprecision[c]	May have little or no important effect on breathlessness and activities.
Dyspnea 8 weeks	1 (148 patients)	MD: 1.1 (95% CI: 0.44 to 1.76) Transition Dyspnea Index (TDI) (Range: -9 to 9; higher scores indicate less impairment)		Low due to serious risk of bias[a] and serious imprecision[c]	May improve dyspnea.
RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval					
a Concerns related to bias due to deviations from the intended intervention, missing outcome data, and measurement of outcome.					
b The confidence interval includes both appreciable benefit and harm.					
c The confidence interval includes both appreciable benefit and no important effect.					
McNarry MA, Berg RMG, Shelley J, Hudson J, Saynor ZL, Duckers J, et al. Inspiratory muscle training enhances recovery post-COVID-19: a randomised controlled trial. Eur Respir J. 2022;60(4).					

Supplement 18: Summary of findings table comparing Active cycle of breathing technique, Physiotherapy and Physiotherapy

Patients: Patients with long COVID **Intervention:** Active cycle of breathing technique, Physiotherapy **Comparator:** Physiotherapy		**NOTE:** CONCERNS WITH TRIAL INTEGRITY. The trial reports minimal baseline characteristics that appear remarkably similar across trial arms with little variability and 0 patients lost to follow-up. Equal numbers of participants are randomized to each trial arm though the trial report does not describe block randomization. The trial reports even, round numbers (e.g., age of participants is reported to range between 40 and 50 years). The p value reported corresponding to the difference in age between arms at baseline appears incorrect. The trial is unregistered.			
Outcome	**Trials (patients)**	**Relative Risk (95% CI)**	**Absolute effect (per 1000 patients)**	**Certainty of the evidence**	**Plain language summary**
Fatigue 12 weeks	1 (60 patients)	MD: -9.97 (95% CI: -11.17 to -8.77) Fatigue Assessment Scale-10 (Range: 10 to 50; higher scores indicate greater impairment)		Very Low due to serious risk of bias[a] and very serious imprecision[b]	Uncertain of the effect on fatigue.
RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval					
a Concerns related to bias due to the randomization process and deviations from the intended intervention.					
b Likely much too few participants to achieve prognostic balance.					
Alshaimaa A. Ali NGE, Samir A. Algazzar , Abdel Wahab M. Lotfy , Emad M. Taha. Impact Of Active Cycle Of Breathing Technique On Selected Pulmonary Outcomes In Post-COVID Syndrome Patients. Journal of Pharmaceutical Negative Results. 2023:710-7.					

Supplement 19: Summary of findings table comparing In-patient rehabilitation, Physiotherapy, Acupuncture and In-patient rehabilitation, Physiotherapy

Patients: Patients with long COVID **Intervention:** In-patient rehabilitation, Physiotherapy, Acupuncture **Comparator:** In-patient rehabilitation, Physiotherapy		**NOTE:** CONCERNS WITH TRIAL INTEGRITY. The trial reports an equal number of participants randomized to each arm though the trial does not describe block randomization. The trial reports implausibly small measures of variability for outcome measures. The trial is unregistered.			
Outcome	**Trials (patients)**	**Relative Risk (95% CI)**	**Absolute effect (per 1000 patients)**	**Certainty of the evidence**	**Plain language summary**
Dyspnea 1.7 weeks	1 (160 patients)	MD: -0.6 (95% CI: -0.74 to -0.46) Modified Medical Research Council dyspnea scale (mMRC dyspnea scale) (Range: 0 to 4; higher scores indicate greater impairment)		Low due to serious risk of bias[a] and serious indirectness[b]	May improve dyspnea.
RR: Relative risk; RD: Risk difference; MD: Mean difference; CI: Confidence interval					
a Concerns related to bias due to deviations from the intended intervention and the measurement of outcome.					
b In-patient setting.					
Omarova I, Akanova A, Kurmanova A, Kurmanova G, Glushkova N, Seidanova A, et al. Acupuncture as an Additional Method of Rehabilitation Post-COVID-19: a randomized controlled trial. J Pharmacopuncture. 2023;26(3):238-46.					

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Title: Effectiveness of telerehabilitation versus face-to-face pulmonary rehabilitation on physical function and quality of life in people with post COVID-19 condition: a systematic review and network meta-analysis.
Journal: European Journal of Physical and Rehabilitation Medicine
Search date: from 2020 to January 5th, 2024
Flow diagram of the literature search:

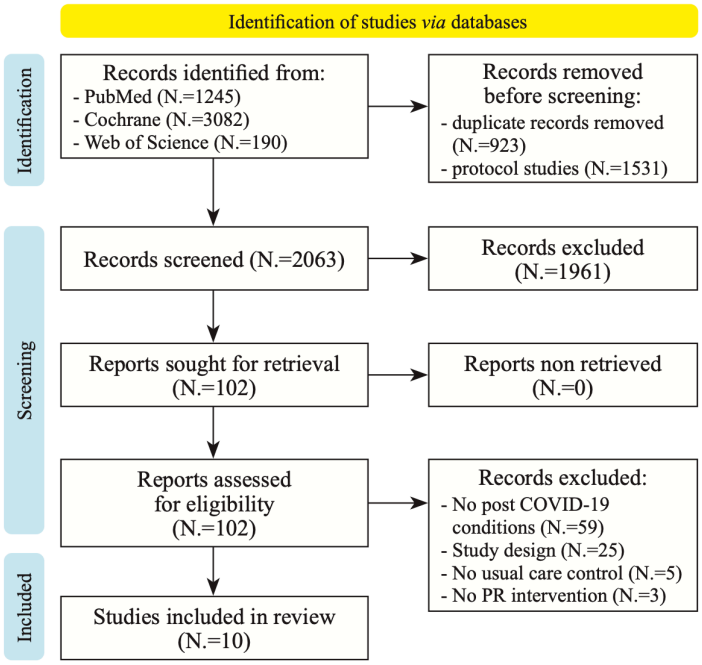


Figure 1.—PRISMA flow chart of included studies.

Characteristics of the Included Studies:

Supplementary Table I.—Characteristics of the included studies.[33, 34, 43-50]

Author (year)	Study design	Population	Sample size	Intervention	Control group	Outcomes	Results
Face-to-face pulmonary rehabilitation							
Elhamrawy 2023[33]	RCT	Adults with PCC	G1: n=18 (33% female) Mean Age: 65.7 y/o G2: n=18 (39% female) Mean Age: 66.2 y/o Con: n=18 (33% female) Mean Age: 66.3y/o	Face-to-face pulmonary rehabilitation	Usual Care	Physical function (30STST)	Both intervention groups improved physical function compared to baseline, without differences in the control group. Differences between groups were significant when G1 and G2 were compared to the control group, but no differences existed.
Jimeno-Almanzán et al 2022[43]	RCT	Adults with PCC	Int: n=19 (68% female) Mean Age: 44.6 y/o Con: n=20 (80% female) Mean Age: 46y/o	Face-to-face pulmonary rehabilitation	Usual care	Quality of life (SF-12)	Both groups improved their physical and mental quality of life after the intervention. Significant differences between groups were found for the physical domain (p=0.024) but not for the mental domain (p=0.444).
Jimeno-Almanzán et al 2022b[34]	RCT	Adults with PCC	G1: n=20 (70% female) Mean Age: 44.3 y/o G2: n=23 (74% female) Mean Age: 44.1y/o G3: n=17 (59% female) Mean Age: 43y/o Con: n=20 (70% female) Mean Age: 47.8 y/o	Face-to-face pulmonary rehabilitation	Usual Care	Quality of life (SF-12)	Quality of life improved in all intervention groups, without changes in the control group after intervention. Combined training resulted in greater improvements in quality of life compared to respiratory or resistance training alone.
Kogel 2023[44]	RCT	Adults with PCC	Int: n=22 (59.05% female) Mean Age: 40.4y/o Con: n=22 (63.64% female) Mean Age: 43.6y/o	Face-to-face pulmonary rehabilitation	Usual care	Quality of life (MQOL)	Both groups improved their quality of life after the intervention. However, only the psychological quality of life subdomain was significantly better in the intervention group compared to control (P<0.05).
Longobardi 2023[45]	RCT	Adults with PCC	Int: n=25 (52% female) Mean Age: 60.8y/o Con: n=25 (48% female) Mean Age: 61.2y/o	Face-to-face pulmonary rehabilitation	Usual Care	Quality of life (SF-36) Physical function (30STST)	The intervention resulted in significant differences in the physical component summary of SF-36 (P<0.05), without differences in the mental component (P>0.05).

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							In addition, significant differences were found in 30STST after treatment compared to the control group.
Telemedicine pulmonary rehabilitation							
Alsharidah 2023[46]	RCT	Adults with PCC	Int: n=24 (100% female) Mean Age: 23.33 y/o Con: n=24 (100% female) Mean Age: 22.58 y/o	Telerehabilitation intervention	Usual Care	Physical function (6MWT) Quality of life (SF-36)	Although both groups improved outcomes compared to baseline, significant differences between groups in terms of 6MWT and SF-36 improvements were found favoring the intervention group (p=0.001).
McNarry 2022[47]	RCT	Adults with PCC	Int: n=111 (86% female) Mean Age: 46.76 y/o Con: n=37 (95% female) Mean Age: 46.13 y/o	Telerehabilitation intervention	Usual care	Quality of life (K-BILD)	Although quality of life improved within the group, statistically significant differences were found between groups favoring the intervention group (P<0.05).
Okan 2022[48]	RCT	Adults with PCC	Int: n= 26 (42.3% female) Mean Age: 48.85 y/o Con: n=26 (53.8% female) Mean Age: 52.19 y/o	Telerehabilitation intervention	Usual care	Physical performance (6MWT)	Physical performance improved with statistically significant differences in the intervention group compared to the control (P<0.001).
Philip 2022[49]	RCT	Adults with PCC	Int: n=74 (78% female) Mean Age: 49 y/o Con: n=76 (83% female) Mean Age: 50 y/o	Telerehabilitation intervention	Usual care	Quality of life (SF-36)	The intervention group improved the mental component of SF-36 with statistical differences compared to the control (p=0.047), with no differences in the physical component (p=0.54).
Samper-Pardo 2023[50]	RCT	Adults with PCC	Int: n=52 (84.5% female) Mean Age: 48.25 y/o Con: n=48 (68.75% female) Mean Age: 48.31 y/o	Telerehabilitation intervention	Usual care	Quality of life (SF-36) Physical function (30STST)	The intervention group had greater quality of life and physical function improvements compared to usual care, but there were no significant differences between groups.

RCT: randomised controlled trial; PCC: Post COVID-19 condition; G: group; y/o: years old; 30STST: 30-second sit-to-stand test; Int: intervention group; Con: control group; SF-12: short-form 12 questionnaire; MQOL: McGill quality of life questionnaire; SF-36: short-form 36 questionnaire; 6MWT: 6-Minute Walking Test; K-BILD: King's Brief Interstitial Lung Disease Questionnaire.

Risk of Bias Assessment:

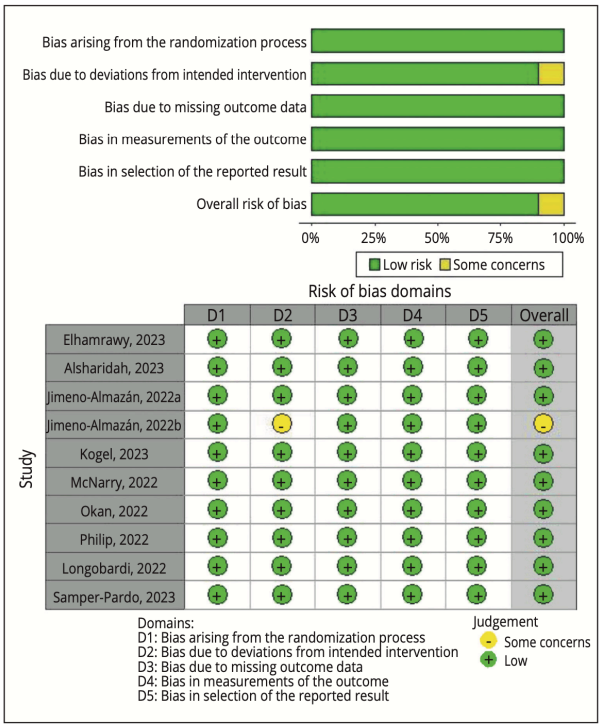


Figure 2.—Risk of bias of included studies.

Relevant result

Two different comparisons were performed: face-to- face pulmonary rehabilitation (FTF) against usual care and telerehabilitation (TL) against usual care.

In the multivariate NMA, significant differences are evident in the FTF vs. TL (21.979 [17.233, 27]) interventions with greater increases in physical function and physical and mental quality of life in the FTF. Compared to usual care, there is a significant improvement in FT, QoL-P and Qol-M in the FTF group, but not in the TL group compared to usual care.

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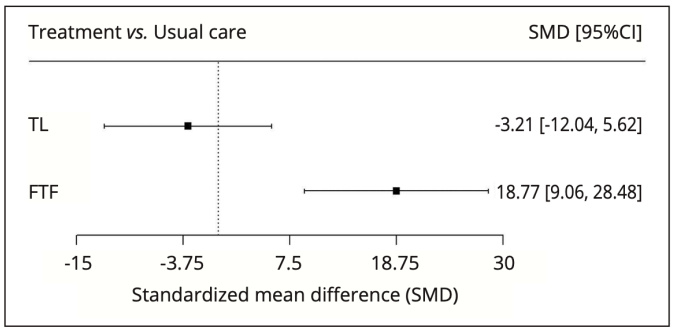


Figure 4.—Physical tests and live physical and mental multivariate forest plot quality.

In the univariate NMA, in the FT and the Qol-M, the league table shows significant differences between the TL vs. Usual care with improvements in physical function and perceived level of quality of life in the TL group, while in QoL-P, no significant differences are evident.

Supplementary Table VI.—League table.

Physical tests		
FTF		
-25.939 (-65.611, 13.733)	TL	
3.125 (-26.574, 32.824)	29.064 (2.762, 55.367)	UC
Quality of live physical		
FTF		
1.413 (-8.072, 10.899)	TL	
6.583 (-0.138, 13.304)	5.169 (-1.524, 11.863)	UC
Quality of live mental		
FTF		
-4.984 (-14.188, 4.221)	TL	
2.140 (-4.363, 8.642)	7.123 (0.609, 13.638)	UC

FTF: Face to face treatment; TL: Tele-rehabilitation; UC: Usual care.
Significant differences shown in red.

GRADE assessment result

In multivariate analysis, moderate quality of evidence concluded that FTF was superior to usual care or TL, improving all outcomes. In addition, low-quality evidence suggests that TL was superior to usual care.

Supplementary Table VII.—Certainty of evidence of multivariate analysis.

Comparison	Mean difference (95% CI)	Quality of evidence	Reasons to downgrade
Multivariate analysis			
FTF vs UC	18.77 (9.06, 28.48)	Moderate	Inconsistency *
TL vs UC	-3.21 (-12.04, 5.62)	Low	Inconsistency *, Imprecision ‡
FTF vs TL	21.979 (17.233, 27)	Moderate	Inconsistency *

*Downgraded due to high heterogeneity in univariate analysis, I^2>90% in all outcomes
‡Confidence intervals include values favouring either treatment

In univariate analysis, for physical function and the mental domain of quality of life, moderate quality of evidence concluded that TL was superior to usual care, while low quality of evidence concluded that FTF was superior to usual care and TL. for the physical domain of quality of life, all comparisons (FTF vs. usual care, TL vs. usual care and FTF vs. TL) showed low quality of evidence.

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Supplementary Table VIII.—Certainty of evidence of univariate analysis.

Comparison	Mean difference (95% CI)	Nature of evidence	Quality of evidence	Reasons to downgrade
Outcome: Physical function				
FTF vs UC	3.125 (-26.574, 32.824)	Direct	Low	Inconsistency *, Imprecision ‡
TL vs UC	29.064 (2.762, 55.367)	Direct	Moderate	Inconsistency *
FTF vs TL	-25.939 (-65.611, 13.733)	Indirect	Low	Inconsistency *, Imprecision ‡
Outcome: Quality of life (Physical domain)				
FTF vs UC	6.583 (-0.138, 13.304)	Direct	Low	Inconsistency *, Imprecision ‡
TL vs UC	5.169 (-1.524, 11.863)	Direct	Low	Inconsistency *, Imprecision ‡
FTF vs TL	1.413 (-8.072, 10.899)	Indirect	Low	Inconsistency *, Imprecision ‡
Outcome: Quality of life (Mental domain)				
FTF vs UC	2.140 (-4.363, 8.642)	Direct	Low	Inconsistency *, Imprecision ‡
TL vs UC	7.123 (0.609, 13.638)	Direct	Moderate	Inconsistency *
FTF vs TL	-4.984 (-14.188, 4.221)	Indirect	Low	Inconsistency *, Imprecision ‡
*Downgraded due to high heterogeneity in univariate analysis, I^2>90% in all outcomes ‡Confidence intervals include values favoring either treatment				

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Appendix 3 Evidence-to-Decision (EtD) frameworks for clinical question

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PICO Question 1

PICO Question 1.1: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID? (Can vaccine prevent long COVID among those with or without confirmed SARS-CoV-2 infection?)	
POPULATION:	Individuals before or after SARS-CoV-2 infection
INTERVENTION:	COVID-19 vaccine
COMPARISON:	Unvaccinated individuals
MAIN OUTCOMES:	Incidence of long COVID; prevalence of individual symptoms of long covid; adverse events
SETTING:	Hospital and community
PERSPECTIVE:	Clinical and public health perspective, aiming to reduce long covid burden
BACKGROUND:	The incidence of long COVID is reduced in vaccinated individuals compared with those who are unvaccinated.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒Probably yes	vaccination prior to SARS-CoV-2 infection could reduce the risk of long COVID post-infection vaccination may decrease the risk of long COVID	The role of vaccination in preventing long COVID remains controversial, as reflected in existing guidelines

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☐ Yes ☐ Varies ☐ Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Trivial ☐ Small ☐ Moderate ☒ Large ☐ Varies ☐ Don't know	Multiple systematic reviews have confirmed that vaccination reduces the risk of developing long COVID.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT ☐ Large ☐ Moderate ☐ Small ☒ Trivial ☐ Varies ☐ Don't know	RESEARCH EVIDENCE Across multiple systematic reviews and observational studies, no significant safety concerns were reported for covid-19 vaccination in relation to long covid prevention. Specifically, none of the included studies identified serious adverse events attributable to vaccination, regardless of timing (before or after infection), vaccine type, or number of doses.	ADDITIONAL CONSIDERATIONS
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☒Very low ☐Low (RCT) ☐ Moderate ☐ High ☐ No included studies	Irrespective of prior SARS-CoV-2 infection, vaccination against COVID-19 is suggested to prevent the development of long COVID A total of three high-quality systematic reviews (44 observational studies, 14,842,052 participants) evaluated the preventive effect of covid-19 vaccination on the risk of long COVID. All three SRs showed that vaccination prior to SARS-CoV-2 infection could reduce the risk of long COVID. Prior to SARS-CoV-2 infection vaccination: two vaccine doses prior to infection significantly lowered the risk of long COVID (OR=0.76, 95% CI:0.65 to 0.89, I^2=93.4%, very low certainty of evidence). one dose (OR: 1.01, 95% CI:0.88 to 1.15, I^2=63.0%, very low certainty of evidence) and three doses (OR: 0.31, 95% CI:0.05 to 1.84, I^2=97.3%, very low certainty of evidence) showed no statistically significant effect. Post-infection vaccination： one dose after infection significantly reduced the risk of long COVID (OR: 0.85, 95% CI:0.73 to 0.98, I^2=77.9%, moderate certainty of evidence), while two doses did not have statistical significance (OR: 0.63, 95% CI:0.38 to 1.03, I^2=88.4%, very low certainty of evidence). Vaccine types: BNT162b2 (OR: 0.79, 95% CI:0.67 to 0.92, I^2=93.0%) and ChAdOx1 nCoV-19 (OR: 0.59, 95% CI:0.50 to 0.70, I^2=0.0%) significantly reduced the risk of long COVID, regardless of vaccination timing relative to infection.	There was an overall very low quality of evidence for vaccination due to 1) substantial heterogeneity in study populations, definitions of long covid, and timing/doses of vaccination, 2) reliance on observational studies with inherent risk of bias, 3) the lack of high-quality randomized controlled trials.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☒Important uncertainty or variability ☐Possibly important uncertainty or variability ☐ Probably no important uncertainty or variability	Not assessed.	Studies assessing vaccination included individuals vaccinated before or after SARS-CoV-2 infection and compared with unvaccinated individuals, with outcomes focusing on the incidence of long COVID and specific symptoms, without evidence of

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☐ No important uncertainty or variability		substantial variability in how patients value these outcomes.
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☐Probably favors the intervention ☒ Favors the intervention ☐ Varies ☐ Don't know	Regarding specific dosage, one systemic review found that two vaccine doses prior to infection significantly lowered the risk of long COVID (OR=0.76, 95% CI:0.65 to 0.89, I^2=93.4%, very low certainty of evidence). The reduction was particularly evident for fatigue, headache, olfactory dysfunction, and myalgia. However, one dose (OR: 1.01, 95% CI:0.88 to 1.15, I^2=63.0%, very low certainty of evidence) and three doses (OR: 0.31, 95% CI:0.05 to 1.84, I^2=97.3%, very low certainty of evidence) showed no statistically significant effect. No adverse events were reported.	No RCTs found that vaccine can prevent the long COVID.
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large costs ☐Moderate costs ☐ Negligible costs and savings ☐Moderate savings	Cost effectiveness was not assessed.	Long COVID strains individual financial healthand has wide and deep ramifications on national economies

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☐Large savings ☐Varies ☒ Don't know		
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate ☐ High ☒No included studies	Cost effectiveness was not assessed.	Although no included study directly evaluated the resource requirements, existing research consistently indicates that long COVID imposes a substantial economic burden. It increases direct healthcare costs, strains disability and social support systems, and reduces labor force participation, productivity, and employment rates.
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Favors the comparison ☐ Probably favors the comparison ☐Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Cost effectiveness was not assessed.	Estimates from the United States place the total economic cost of long COVID at USD 3.7 trillion—comparable to the 2008 global financial crisis—driven by lost quality of life, reduced earnings, and medical expenses. Given the high economic toll, even moderate preventive efficacy may translate into favorable cost-effectiveness.
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐ Probably reduced ☐ Probably no impact ☒Probably increased ☐Increased ☐ Varies ☐ Don't know	Not formally assessed.	Regions with limited healthcare capacity may face challenges in implementing widespread COVID-19 vaccination. Capacity needs to be assessed before implementation
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ No ☐ Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	Vaccine is widely accepted by the public and healthcare.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ No ☐ Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	Despite the recognized role of vaccination against COVID-19, its effectiveness in preventing long COVID may vary with different SARS-CoV-2 variants, which should be considered in implementation planning.

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☒ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☒ Trivial	☐ Varies	☐ Don't know	/

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	JUDGEMENT						
CERTAINTY OF EVIDENCE	☒ Very low	☐ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☒ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☐ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☒ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☐ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☒ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☒ Probably increased	☐ Increased	☐ Varies	☐ Don't know

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	JUDGEMENT						
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☒ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

Irrespective of prior SARS-CoV-2 infection, vaccination against COVID-19 is suggested to prevent the development of long COVID. The specific vaccination dosage and timing should be determined by taking into account the availability, requirements of the local department, and patient's condition, while closely monitoring any potential adverse events.

Justification

Evidence from multiple systematic reviews suggests that COVID-19 vaccination, given before or after infection, may reduce the incidence of long COVID and the symptoms. The certain vaccine types (e.g., BNT162b2, ChAdOx1 nCoV-19) and for two doses given before infection, with minimal reported adverse events.

Subgroup consideration

Two doses before infection reduced risk of long COVID.

One or three doses before infection showed no significant effect.

One dose after infection reduced risk; two doses not significant.

Both BNT162b2 and ChAdOx1 nCoV-19 showed protective effect.

Implementation considerations

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	Requires established vaccination infrastructure and equitable access across regions. Effectiveness in preventing long COVID may vary by SARS-CoV-2 variant. Regions with limited healthcare capacity may face challenges in delivering timely and widespread vaccination.
Monitoring and evaluation	
	Ongoing surveillance of long COVID incidence among vaccinated and unvaccinated populations, and monitoring of vaccine safety and effectiveness against different variants.
Research priorities	
	High-quality RCTs assessing vaccination for long COVID prevention; standardized definitions of long COVID; evaluation of optimal vaccine type, dosing, and timing; and studies on special populations.

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PICO Question 1.2: What vaccines or treatments (antiviral drugs, anti-inflammatories, monoclonal antibodies, metformin, etc.) can prevent long COVID? (Can antiviral drugs prevent long COVID among those with confirmed SARS-CoV-2 infection?)	
POPULATION:	Individual with confirmed acute SARS-CoV-2 infection
INTERVENTION:	Antiviral drugs (molnupiravir, remdesivir, nirmatrelvir, nirmatrelvir/ritonavir, favipiravir, ensitrelvir, azvudine)
COMPARISON:	No acute-phase treatment or alternative standard care
MAIN OUTCOMES:	Long COVID; specific long COVID symptoms—fatigue, dyspnea, myalgia, sleep disturbances, and cognitive impairment
SETTING:	Hospital and community
PERSPECTIVE:	Population perspective
BACKGROUND:	Given the limited research on treatments for established long COVID, it is important to investigate whether early treatment during acute SARS-CoV-2 infection could help prevent its development.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes ☒ Yes	Identifying effective strategies to prevent long COVID is of considerable clinical importance. Current research primarily focuses on vaccination and the use of pharmacologic interventions during the acute phase following SARS-CoV-2 infection.	

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☐Varies ☐Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☐ Small ☒ Moderate ☐Large ☐Varies ☐Don't know	Pharmacologic treatment during the acute phase (within 4 weeks of infection) may reduce the risk of long COVID. Patients with different risk profiles may derive different levels of benefit from antiviral therapies	Existing evidence suggests that administering antiviral drugs during the acute phase of SARS-CoV-2 infection may help prevent the development of long COVID.
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☐ Small ☐Trivial ☒Varies ☐Don't know	Overall, antiviral drugs used in the acute phase were associated with few serious adverse events. One study reported a higher number of serious adverse events with favipiravir (14 events vs 4 in the control group). Ritonavir-containing regimens require careful management of drug–drug interactions, and dosing adjustments are necessary in patients with renal or hepatic impairment. No significant safety concerns were reported for other antivirals.	No severe adverse events were reported in monoclonal antibodies, corticosteroids and metformin.
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Very low ☒Low ☐ Moderate ☐ High ☐ No included studies	Acute-phase antiviral therapy (administered within the first 4 weeks post-infection) is suggested to reduce the risk of long COVID. A total of **33** studies evaluated the effect of antiviral drugs (molnupiravir, remdesivir, nirmatrelvir, nirmatrelvir/ritonavir, favipiravir, ensitrelvir, azvudine) on the risk of long COVID; 19 studies were included in the meta-analysis. Overall, antiviral drugs significantly reduced the risk of long COVID compared with control (OR: 0.69, 95% CI: 0.64 to 0.74, I^2=98.0%, low certainty of evidence). antiviral therapy significantly reduced fatigue, cognitive impairment, and sleep disorders; but had no significant effect on dyspnea or myalgia.	For critical outcomes, the certainty of evidence was low in RCTs, mainly due to high heterogeneity, variable definitions of long COVID and imprecision in effect estimates. The certainty was very low in observational studies, largely because of inherent study design limitations, high risk of bias, inconsistent follow-up durations, and substantial between-study variability
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☒Important uncertainty or variability ☐Possibly important uncertainty or variability ☐ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not assessed.	High-risk groups, such as individuals with obesity, older age, or comorbidities, may focus more on acute adverse outcomes, while low-risk groups may be more concerned about the risk of long COVID; therefore, variability exists in how different populations perceive its value.
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☒Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know	Different drugs may be associated with adverse events. Only one study reported 14 serious adverse events in the favipiravir group (compared with 4 in the control group), while no adverse reactions have been reported for other antiviral agents. No serious adverse events have been reported for other treatment.	
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☐Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☒ Don't know	Cost effectiveness were not assessed.	Treatment costs depend on the drug or chemical used.

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Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate ☐ High ☒ No included studies	Cost effectiveness were not assessed.	Treatment costs depend on the drug or chemical used.
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Cost effectiveness were not assessed.	The certainty of evidence remains low to very low, and the observed preventive efficacy differs by drug type and target symptoms. Considering the balance of benefits, harms, and costs, the use of specific antiviral agents during the acute phase of COVID-19 may help reduce the risk of long COVID; however, routine use of other pharmacologic agents is not currently supported.

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Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☒ Varies ☐ Don't know	Not formally assessed.	In settings with limited economic resources and healthcare capacity, antiviral drugs may not be readily available.
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Although some other guidelines also suggest early treatment for SARS-CoV-2 infection with antiviral drugs to prevent long COVID, the antiviral drugs was primarily used for treatment of acute infection. It may be not that acceptable for key stakeholders among some regions, especially those with limited resources for antiviral drugss.	
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no	Not formally assessed.	

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☐ Probably yes ☐ Yes ☐ Varies ☒ Don't know		

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☒ Moderate	☐ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☐ Trivial	☒ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☒ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☐ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know

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	JUDGEMENT						
RESOURCES REQUIRED	☐ Large costs	☒ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
FEASIBILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☐ Varies	☒ Don't know	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☒ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

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We suggest using antiviral drugs during the acute phase of COVID-19 to prevent the development of long COVID, especially those who are at higher risk.

Justification
Pharmacologic treatment during the acute phase (within 4 weeks of infection) may also reduce the risk of long COVID. Our updated SR supports this potential benefit, although the certainty of evidence remains low to very low. The observed preventive efficacy varies by drug type and target symptoms. Notably, patients with different risk profiles may derive different levels of benefit from antiviral therapies. However, the available evidence does not support subgroup analyses. Considering the balance of benefits, harms, and costs, we suggest the use of specific antiviral agents in the acute phase of COVID-19 to reduce long COVID risk; however, routine use of other agents is not currently supported. Clinical decision-making should integrate individual risk profiles, drug availability, and practical contextual considerations to ensure optimal outcomes. Notably, dosing strategies for antiviral drugs also varied, with nine studies reporting adjustments for special populations, including those with renal impairment.
Subgroup considerations
Nirmatrelvir/ritonavir, molnupiravir, and favipiravir significantly reduced the risk of long COVID. Remdesivir, ensitrelvir, and azvudine showed no significant effect. Antiviral therapy reduced fatigue, cognitive impairment, and sleep disorders, but not dyspnea or myalgia.
Implementation considerations
Selection of drug type, dosage, timing of initiation, and target patient populations.
Monitoring and evaluation
Track antiviral use, timing of administration, and follow-up long COVID outcomes.
Research priorities
Standardization of outcome measures and evaluation of antiviral efficacy in diverse populations.

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PICO Question 2

PICO Question 2: In patients with long COVID, should antiviral drugs be used versus no antiviral drugs?	
POPULATION:	Adult patient with long COVID
INTERVENTION:	Antiviral drug
COMPARISON:	No antiviral drug
MAIN OUTCOMES:	Long COVID symptoms
SETTING:	Hospital and Community
PERSPECTIVE:	Population perspective
BACKGROUND:	The hypothesis of long COVID mechanism is not diverse, including persistent virus or viral fragments, immune dysregulation, inflammation, microvascular problems, and neuroinflammation. This makes treatment challenging. Antiviral drugs, such as nirmatrelvir/ritonavir (Paxlovid), are being studied to see if they can reduce symptoms in long COVID by targeting persistent virus. Some clinical trials are ongoing or recently completed. Currently, evidence from randomized trials is limited and inconsistent, and the best timing, safety, and patient groups for antiviral use are still uncertain.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐No ☐Probably no ☐Probably yes ☒ Yes ☐Varies ☐Don't know	Viral persistence is one of the major mechanism hypothesis for long COVID. Exploring the potential effect of antiviral treatment for long COVID patients is one of the most important ways to find therapeutic methods for long COVID.	
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☒Trivial ☐ Small ☐ Moderate ☐Large ☐Varies ☐Don't know	Evidence from one RCT (n=155) shows no significant improvement in long COVID symptoms (fatigue, brain fog, body pain, cardiovascular, dyspnea, GI symptoms) with NMV/r versus placebo at 10 weeks. Desirable effects are therefore minimal or uncertain.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☒ Small ☐Trivial ☐Varies ☐Don't know	Adverse events were similar between NMV/r and placebo (99% vs. 92.5%), mostly low-grade. Serious adverse events were rare. Overall undesirable effects are small.	

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Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Very low ☒Low ☐ Moderate ☐ High ☐ No included studies	Evidence was based on a single RCT, with small sample size, limited follow-up, and patients having negative stool SARS-CoV-2 RNA at baseline.	
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☒Possibly important uncertainty or variability ☐ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Uncertainty exists regarding how patients value improvements in long COVID symptoms; variability may exist across populations, but no formal studies quantify this.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the	Minimal or no symptom improvement observed; undesirable effects are minor, but benefits do not outweigh lack of efficacy.	

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comparison ☒ Does not favor either the intervention or the comparison ☐Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know		
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☒Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☐ Don't know	NMV/r requires prescription, monitoring for drug interactions, and potential costs of antiviral drugs. Marginal resources required relative to no antiviral, but no demonstrated benefit.	
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Very low ☐ Low ☐ Moderate ☐ High ☒No included studies	Costs and resource use are not formally evaluated, with no high-quality economic evaluation specifically for long COVID treatment.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☒Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ No included studies	Given low certainty of effectiveness, net benefits are unlikely to justify cost and resource use.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☐ Varies ☒ Don't know	Not assessed due to the non-significant improvement.	Antiviral treatment could increase access disparities if provided without demonstrated benefit; no evidence of differential effect in disadvantaged groups.
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐ Probably yes ☐ Yes ☐ Varies ☒ Don't know	Not assessed due to the non-significant improvement.	
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no ☐Probably yes ☐Yes ☐ Varies	Not assessed due to the non-significant improvement.	Drug is available and feasible to prescribe, but inappropriate use for unproven indications is a concern. Implementation feasible but not recommended due to lack of benefit.

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☒ Don't know		

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☒ Trivial	☐ Small	☐ Moderate	☐ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☒ Small	☐ Trivial	☐ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☒ Possibly important uncertainty or variability	☐ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☒ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know

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	JUDGEMENT						
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☒ Moderate	☐ High	☐ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☐ Varies	☒ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☐ Varies	☒ Don't know	/
FEASIBILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☐ Varies	☒ Don't know	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☒ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☐ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation
We suggest against the use of nirmatrelvir-ritonavir for the treatment of long COVID in adult patients.

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Justification
The panel issued a conditional recommendation against the use of nirmatrelvir-ritonavir for the treatment of long COVID in adult patients. This recommendation applies to the current evidence base, which consists of a single randomized controlled trial showing no significant clinical benefit over placebo in symptom improvement, quality of life, or functional outcomes, and similar rates of low-grade adverse events. Given the low certainty of evidence, the absence of demonstrated benefit, and the potential for resource use and access disparities, the panel judged that the undesirable effects and resource implications outweigh the uncertain benefits in most patients. The recommendation is provisional and should be reconsidered if future studies—especially those including patients with detectable SARS-CoV-2 persistence—demonstrate clear benefit.
Subgroup considerations
Patients with confirmed or suspected persistent SARS-CoV-2 RNA/protein in tissues might plausibly benefit more from antivirals, though this remains unproven. Those with contraindications or high risk of drug-drug interactions (e.g., patients on multiple cardiovascular or psychiatric medications) may experience greater harm. Disadvantaged populations may have reduced access to testing and follow-up, which could influence both safety and efficacy monitoring.
Implementation considerations
If antivirals are used off-label for long COVID, a shared decision-making process should be conducted, ensuring patients understand the current evidence and uncertainties. Careful review of potential drug-drug interactions (especially with ritonavir) is essential. Use should ideally be limited to research settings or well-monitored clinical programs until more robust evidence emerges. Cost considerations and supply chain stability should be evaluated before any wider adoption.
Monitoring and evaluation
Active pharmacovigilance for adverse events, especially hepatotoxicity and drug interactions. Monitoring for symptom progression or resolution using standardized patient-reported outcome measures (PROMs). If used in research, consider testing for SARS-CoV-2 persistence at baseline and follow-up to explore potential subgroups who might benefit.
Research priorities
Large, well-powered RCTs including patients with documented SARS-CoV-2 persistence to evaluate antiviral efficacy in this subgroup. Studies assessing duration and dosing of antiviral therapy for long COVID. Trials of other antiviral agents (amantadine, ensitrelvir, etc.) with different mechanisms of action. Pharmacokinetic, safety, and efficacy studies in special populations

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(elderly, immunocompromised, pregnant women). Investigation of biomarkers (e.g., viral RNA/protein detection, immune activation markers) to predict treatment response. Longitudinal studies assessing cost-effectiveness and real-world acceptability.

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PICO Question 3

PICO Question 3: In patients with long COVID, should monoclonal antibody be used versus no monoclonal antibody?	
POPULATION:	Adult patients with long COVID
INTERVENTION:	Neutralizing monoclonal antibody
COMPARISON:	No neutralizing monoclonal antibody
MAIN OUTCOMES:	Improvement of symptoms such as fever, cough and malaise
SETTING:	Hospital
PERSPECTIVE:	Population perspective
BACKGROUND:	Long COVID represents a range of symptoms persisting for weeks, months, or even years after the initial infection. In the context of long COVID, therapeutic monoclonal antibodies are neutralizing agents designed to inhibit viral entry and replication. However, the effectiveness and safety of monoclonal antibody for treating long COVID patients is unclear.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no	This question is probably a priority due to the urgent need for effective long COVID treatments and the potential of monoclonal antibodies to target persistent viral reservoirs or immune dysfunction.	

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☒Probably yes ☐Yes ☐Varies ☐Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☐Small ☐Moderate ☐Large ☐Varies ☒Don't know	One case report of an immunocompromised patient with long COVID described symptomatic improvement after casirivimab-imdevimab use, suggesting possible benefit in selected populations.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☒ Small ☐Trivial ☐Varies ☐Don't know	The undesirable anticipated effects of monoclonal antibodies for treating long COVID are likely small, including infusion reactions and potential immune-related adverse events, but serious harms appear uncommon. No adverse events were reported in the case report.	

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Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☒Very low ☐Low ☐ Moderate ☐ High ☐ No included studies	Very limited clinical evidence, with only one case report available.	
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☒Possibly important uncertainty or variability ☐ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not formally assessed. There is probably uncertainty and variability in how much people value symptom resolution in long COVID, as patients' priorities and tolerances for persistent symptoms differ widely.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the	The desirable effects of monoclonal antibodies have not been established based on the current evidence.	

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comparison ☒ Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know		
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☒ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☐ Don't know	The resource requirements for using monoclonal antibodies to treat long COVID are substantial, given their high drug costs, need for specialized infusion facilities, and potential long-term monitoring needs.	
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Very low ☐ Low ☐ Moderate ☐ High ☒No included studies	The certainty of the evidence on the resource requirements for monoclonal antibody use in long COVID cannot be assessed due to lack of studies on this topic.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☒Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ No included studies	Despite high drug cost, the uncertain effectiveness, due to a lack of available evidence, does not favor either the intervention or the comparison.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Reduced ☒ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☐ Varies ☐ Don't know	The use of monoclonal antibodies for long COVID treatment could probably exacerbate health inequities due to high costs, geographic disparities in access to specialized care, and potential underrepresentation of marginalized groups in clinical trials, unless deliberately implemented with equitable distribution strategies.	
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐ Probably yes ☐ Yes ☐ Varies ☒ Don't know	The acceptability of monoclonal antibodies for long COVID treatment among key stakeholders, such as patients, clinicians, and payers remains uncertain due to limited efficacy data, high costs, and variable risk-benefit perceptions.	
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no ☒Probably yes ☐Yes ☐ Varies	Implementing monoclonal antibody treatment for long COVID is probably feasible, given specialized facilities, trained staff, and reliable supply chains, which may limit scalability in some settings.	

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☐ Don't know		

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☐ Large	☐ Varies	☒ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☒ Small	☐ Trivial	☐ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☒ Very low	☐ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☒ Possibly important uncertainty or variability	☐ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☒ Large costs	☐ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know

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	JUDGEMENT						
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ No included studies
EQUITY	☐ Reduced	☒ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☐ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☐ Varies	☒ Don't know	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☐ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation
We do not have recommendation favor or against using neutralizing monoclonal antibodies in adult patients with long COVID. (very low certainty of evidence)

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Justification
Evidence base: No studies specifically enrolling patients who met the WHO definition of long COVID were found. After relaxing that criterion, evidence consists of one case report only. The case report described symptomatic improvement after casirivimab-imdevimab use in an immunocompromised patient with long COVID, suggesting possible benefit in selected populations. Harms and costs: Direct safety signals were not reported; however, monoclonal antibodies are costly, variably available, and may be affected by viral evolution, with uncertain benefit in long COVID pathophysiology. Overall certainty: Very low, due to indirectness (population not strictly WHO-defined long COVID) and lack of data.
Subgroup considerations
Immunocompromised hosts: The single case report described symptomatic improvement (fever, cough, malaise) after casirivimab-imdevimab administration in a patient with thymoma and pleural metastasis. This suggests possible benefit where ongoing viral persistence is suspected.
Implementation considerations
● Monoclonal antibody use requires specialized facilities, trained staff, and reliable supply chains, ● Do not implement monoclonal antibodies for long COVID outside clinical studies. Exceptional cases can include confirmed immunocompromised status with concern for persistent viral replication. ● Discuss uncertainty of benefit, costs, and access; document shared decision-making.
Monitoring and evaluation
● Symptom scales across domains (constitutional, ENT, cardiopulmonary, GI, neurological, sleep), functional status (e.g., WHODAS 2.0), fatigue scales (e.g., FACIT-F), cognitive measures (e.g., MoCA or digital cognitive tests), QoL (e.g., EQ-5D-5L), and PCR/Ct if persistence suspected. ● Follow-up windows can include 2, 4, and 12 weeks post-infusion. ● Safety monitoring including infusion reactions, hypersensitivity and thromboembolic events etc.
Research priorities

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- Randomized controlled trials in patients meeting WHO long COVID definitions, with stratification by immunocompromised status and evidence of viral persistence.
- Biomarker-enriched designs: virologic markers (PCR/Ct, antigenemia, viral culture, sequencing), immune profiling (e.g., B-/T-cell phenotypes), and correlation with response.
- Domain-specific outcomes including prespecified neurological, cardiopulmonary, and fatigue/cognition endpoints.
- Comparative effectiveness vs. antiviral small molecules and vs. multimodal rehabilitation/usual care.
- Safety and cost-effectiveness analyses, including variant susceptibility and real-world implementation barriers.
- Pediatric studies and longer-term follow-up to assess durability of any benefit or harms.

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PICO Question 4

PICO Question 4: In patients with long COVID, should probiotics be used versus no probiotics?	
POPULATION:	Adult patient with long COVID
INTERVENTION:	Probiotics (multi-species)
COMPARISON:	No probiotics / placebo
MAIN OUTCOMES:	Long COVID symptoms
SETTING:	Hospital and Community
PERSPECTIVE:	Population perspective
BACKGROUND:	The mechanism of long COVID remains unclear with limited treatments. The gut microbiome has been implicated in immune regulation and systemic inflammation, with emerging evidence suggesting microbiota dysbiosis in long COVID. Probiotics have been investigated as a potential therapeutic option to restore microbiome balance and improve long COVID symptoms.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐No ☐Probably no ☒Probably yes ☐ Yes ☐Varies ☐Don't know	Three RCTs evidence indicates that probiotics may safely alleviate fatigue, cognitive impairment, and multiple systemic symptoms in patients with long COVID and improve patients' quality of life. However, recommendations for treatment of probiotics in long COVID patients is limited.	
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☐ Small ☐ Moderate ☐Large ☒Varies ☐Don't know	RCT evidence shows that probiotics significantly improved multiple long COVID symptoms, with SIM01 yielding favorable odds ratios (e.g., fatigue OR 2.27, 95% CI 1.52–3.40) and VSL#3® reducing fatigue at 4 weeks ($p = 0.0079$); additional formulations also produced significant improvements in fatigue ($p = 0.03$), with no major adverse events reported. However, current evidence is limited and varies in study population and interventions, without RCTs of large sample size and clearly measurable outcomes.	Although intestinal flora imbalance is a potential mechanism of contracting COVID-19, the specific probiotics that can intervene are still unclear, so the effects of current probiotic products may be variable.
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☒ Small ☐Trivial ☐Varies ☐Don't know	No significant adverse events were reported in any of the included RCTs using various probiotic formulations.	Probiotics are generally well tolerated in adults.

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Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Very low ☐Low ☒ Moderate ☐ High ☐ No included studies	We included three RCTs, all of which showed benefits. Due to the variance in the reported outcomes, we conducted a meta-analysis including only one RCT. This particular RCT had a high overall quality (no downgrading), with a sample size of 463 (downgraded by one level). The heterogeneity, indirectness, and publication bias were not downgraded, so the overall quality was rated as moderate.	
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☒Possibly important uncertainty or variability ☐ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Although the outcomes used in the published studies differed and were subjectively evaluated by patients, this is reasonable. Currently, there is a lack of a generally accepted and standardized assessment tool for long COVID, and the alleviation of symptoms and improvement of quality of life are essential for these patients. In summary, based on the available evidence, there is likely no significant uncertainty or variability regarding the main outcomes.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the	The available randomized controlled trials (RCTs) suggest that probiotic supplementation in patients with long COVID may lead to improvements in symptoms and overall quality of life. The balance between desirable and	

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comparison ☐ Does not favor either the intervention or the comparison ☒Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know	undesirable effects probably favors probiotics, as they may improve long COVID symptoms with minimal and generally mild adverse effects.	
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☐Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☒ Don't know	Not formally assessed.	
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Very low ☐ Low ☐ Moderate ☐ High ☒ No included studies	The certainty of the evidence on resource requirements is not clear due to limited and indirect cost data.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Not formally assessed.	The available evidence suggests that the cost-effectiveness likely favors the intervention, although data are limited.
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☒ Varies ☐ Don't know	Not formally assessed.	The intervention could improve health equity if made affordable and accessible, especially for populations disproportionately affected by long COVID.
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐ Probably yes ☐ Yes ☐ Varies ☒ Don't know	Not formally assessed.	The intervention is likely acceptable to key stakeholders, as it is safe, non-invasive, and aligned with patients' needs for symptom relief; from a multidisciplinary perspective, it may benefit patients, although the efficacy might vary.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no ☒Probably yes ☐Yes ☐ Varies	Not formally assessed.	The intervention is generally feasible to implement, as probiotics are widely accessible and available in most healthcare and retail settings, though formulation quality and regulatory oversight may vary across regions.

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☐ Don't know		

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☐ Large	☒ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☒ Small	☐ Trivial	☐ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☐ Very low	☐ Low	☒ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☒ Possibly important uncertainty or variability	☐ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☐ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know

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	JUDGEMENT						
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☐ Varies	☒ Don't know	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☒ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation
We suggest using probiotics as a therapy for adults with long COVID to alleviate fatigue, gastrointestinal symptoms, and improve quality of life (conditional recommendation, moderate certainty of the evidence).

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Justification
Evidence from RCTs indicates that specific probiotic or synbiotic formulations (e.g., VSL#3®, multi-strain synbiotics) can reduce fatigue and improve patient-reported well-being. Benefits are clinically meaningful; adverse effects are generally mild. Evidence is limited by heterogeneity in strains, dosing, and outcome measures.
Subgroup considerations
Patients with persistent fatigue, difficulty in concentration, memory loss, or gastrointestinal symptoms may benefit most. Effectiveness may vary across probiotic formulations and populations.
Implementation considerations
Accessibility of high-quality probiotic products may vary by region. Implementation is feasible in outpatient or primary care settings with minimal training. Clinicians should guide patients on strain-specific products and monitor mild gastrointestinal side effects.
Monitoring and evaluation
Track patient-reported outcomes such as fatigue, gastrointestinal symptoms, and quality of life. Monitor adherence and tolerability over the intervention period.
Research priorities
Conduct larger, multicenter RCTs to evaluate strain-specific efficacy and long-term safety. Develop standardized assessment tools for long COVID symptoms. Study cost-effectiveness and accessibility for broader implementation.

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PICO Question 5

PICO Question 5 In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?	
POPULATION:	Adult patients with long COVID
INTERVENTION:	Glucocorticoids
COMPARISON:	No glucocorticoids
MAIN OUTCOMES:	Cardiopulmonary function, measured by 6-minute walk distance (6MWD), peripheral oxygen saturation (SpO_2), FEV_1, FVC, and DLCO
SETTING:	Hospital and Community
PERSPECTIVE:	Population perspective
BACKGROUND:	Glucocorticoids, as a class of steroid hormones, regulate multiple physiological processes and are widely used to treat the acute COVID-19 due to their anti-inflammatory and immunosuppressive effects. However, current long COVID guidelines provide limited recommendations on glucocorticoids use.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes	Glucocorticoids, as a class of steroid hormones, regulate multiple physiological processes and are widely used to treat the acute COVID-19 due to their anti-inflammatory and immunosuppressive effects. However, current long COVID guidelines provide limited recommendations on glucocorticoids use.	Current long COVID guidelines provide limited recommendations on glucocorticoids use.

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☒ Yes ☐ Varies ☐ Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Trivial ☐ Small ☐ Moderate ☐ Large ☒ Varies ☐ Don't know	Five observational studies compared glucocorticoids use versus non-use with different follow-up time. Four studies reported changes in 6MWD: two studies reported statistically significant improvements, while the other two studies found no significant improvement. None of the studies reported SpO_2 outcomes. Regarding FEV_1, one study reported a significant improvement in FEV_1 (L), whereas another study found no significant change in FEV_1 (% predicted). Three studies reported improvement in FVC (% predicted), and two studies showed glucocorticoids could improve FVC (L). Regarding DLCO and DLCO [%], one study reported the use of steroid was associated with a significant improvement in DLCO and DLCO [%]. In the non-steroid group, the changes compared to baseline were not statistically significant. And another study found a significant improvement in DLCO (%) in the corticosteroids group. Therefore, the magnitude of improvement across studies was relatively modest, and findings were inconsistent, which is similar in other respiratory outcomes, including CT score, total lung capacity (TLC), residual volume (RV), transfer factor of the lung for carbon monoxide (TLCO), etc, making it difficult to draw definitive conclusions. One RCT compared effect of different glucocorticoids doses in 130 long COVID adults with persistent dyspnea or hypoxemia. The high-dose group received an initial prednisone dose of 40 mg/day for one week, followed by gradual tapering (30 mg/day for one week, 20 mg/day for two weeks, and 10 mg/day for two weeks). The low-dose group received 10 mg/day of prednisone for six weeks. There were no	Although available evidence suggests that glucocorticoids may improve certain lung imaging and function outcome (e.g., CT sale, radiological response, TLC, TLCO), results are inconsistent regarding the primary outcome, and the magnitude of improvement is generally limited. Thus, the clinical benefit of glucocorticoids in this population remains uncertain.

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	significant differences between the two groups in complete radiological response, complete or good radiological response, FVC, resting SpO_2 and mMRC score at six weeks.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☐ Small ☐Trivial ☐Varies ☒Don't know	Across all included studies, no serious adverse events related to glucocorticoids use were reported during follow-up.	No serious adverse events were reported during follow-up periods, but other non-serious adverse events were not systematically assessed, and the potential long-term risks of glucocorticoids therapy, including infections, osteoporosis, gastrointestinal bleeding, and mental health disorders, remain a concern. Although generally well tolerated in the short term, caution is warranted regarding their long-term safety.
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Very low ☒Low ☐ Moderate ☐ High ☐ No included studies	Six studies (one RCT, three cohort studies and two before-and-after studies) assessed the effect of glucocorticoids on cardiopulmonary function among long COVID adult patients with persistent respiratory symptoms. Significant heterogeneity exists among the glucocorticoids administration methods, dosages, and follow-up periods, preventing conducting the meta-analysis.	Meta-analysis was not performed due to substantial heterogeneity across studies.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		

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JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☐Possibly important uncertainty or variability ☒ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not assessed.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the comparison ☒ Does not favor either the intervention or the comparison ☐Probably favors the intervention ☐ Favors the intervention	Although available evidence suggests that glucocorticoids may improve certain lung imaging and function outcome (e.g., CT sale, radiological response, TLC, TLCO), results are inconsistent regarding the primary outcome, and the magnitude of improvement is generally limited. Thus, the clinical benefit of glucocorticoids in this population remains uncertain. Additionally, while no serious adverse events were reported during follow-up periods, other non-serious adverse events were not systematically assessed, and the potential long-term risks of glucocorticoids therapy, including infections, osteoporosis, gastrointestinal bleeding, and mental health disorders, remain a concern.	

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☐ Varies ☐ Don't know		
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☒Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☐ Don't know	The cost of glucocorticoids is not formally assessed.	Glucocorticoids are generally accessible and the costs are moderate in clinical use.
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate ☐ High ☒No included studies	Not formally assessed.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Favors the comparison ☐ Probably favors the comparison ☐Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Not formally assessed.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☒ Varies ☐ Don't know	Not formally assessed.	Glucocorticoid is generally accessible to patients. However, in areas with limited outpatient resource, monitoring for potential side effects for glucocorticoids is challenging.
Acceptability: Is the intervention acceptable to key stakeholders?		

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JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ No ☐ Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	The glucocorticoids can be taken orally, once daily. This is generally acceptable in patients and for care givers.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ No ☐ Probably no ☐ Probably yes ☒ Yes ☐ Varies ☐ Don't know	Not formally assessed.	The glucocorticoids can be taken orally, once daily. This is generally acceptable in patients and for care givers.

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☐ Large	☒ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☐ Trivial	☐ Varies	☒ Don't know	/

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	JUDGEMENT						
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☒ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☒ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know

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	JUDGEMENT						
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/
FEASIBILITY	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	/	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☒ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☐ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

We suggest against the use of glucocorticoids for improving cardiopulmonary function in adult long COVID patients with persistent respiratory symptoms.

Justification

The panel formulated a conditional negative recommendation for glucocorticoid in treating long COVID patient with persistent respiratory symptoms. Although some studies suggest potential benefit for glucocorticoids, the overall benefit varies in studies with the magnitude of improvement generally limited. The potential risks, especially long-term risks, of glucocorticoids therapy are not carefully evaluated.

Subgroup considerations

Not formally assessed.

Implementation considerations

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Although the benefit for glucocorticoid in treating long COVID is limited, it can be used if the patients meet the diagnosis criteria for other disease, including interstitial lung disease, and this is not within the frame of this document. A duly-informed decision-making process by patients should be followed. The optimal glucocorticoid dosage and regimen should be considered.

Monitoring and evaluation

Long-term safety should be monitored and evaluated, but there is no guidance on how often and for how long the patients are monitored.

Research priorities

There needs to be greater understanding of the effect of glucocorticoids and the pathogenesis for long COVID-related symptoms.
Also, the duration and dosing of treatment should be considered.

PICO Question 5 In patients with long COVID, should glucocorticoids be used versus no glucocorticoids?	
POPULATION:	Adult patients with long COVID
INTERVENTION:	Glucocorticoids
COMPARISON:	No glucocorticoids
MAIN OUTCOMES:	Self-reported olfactory recovery
SETTING:	Hospital and Community
PERSPECTIVE:	Population perspective
BACKGROUND:	Glucocorticoids, as a class of steroid hormones, regulate multiple physiological processes and are widely used to treat the acute COVID-19 due to their anti-inflammatory and immunosuppressive effects. However, current long COVID guidelines provide limited recommendations on glucocorticoids use.
CONFLICT OF INTERESTS:	None

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ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes ☒ Yes ☐Varies ☐Don't know	Glucocorticoids, as a class of steroid hormones, regulate multiple physiological processes and are widely used to treat the acute COVID-19 due to their anti-inflammatory and immunosuppressive effects. However, current long COVID guidelines provide limited recommendations on glucocorticoids use.	Current long COVID guidelines provide limited recommendations on glucocorticoids use.
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☒ Small ☐ Moderate ☐Large ☐Varies ☐Don't know	Results of meta-analysis (involving 138 adult patients with anosmia or olfactory disorders from three RCTs, glucocorticoids was given by inhalation, local injection and systemic administration) showed, when compared to no glucocorticoids, the use of glucocorticoids can improve the olfactory score, but had no effect on patients' self-reported olfactory recovery. One 2024 before-and-after study involving 87 adults with persistent olfactory and gustatory disorders, showed that combination therapy with glucocorticoids, glycosaminoglycans-based antithrombotic drugs, and diuretics can effectively improve the VAS score. Subgroup analysis showed that the glucocorticoids was more effective when initiated within 180 days post SARS-CoV-2 infection. Under the same treatment, olfactory symptoms improved more substantially than gustatory symptoms.	

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Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT ☐Large ☐Moderate ☐ Small ☐Trivial ☐Varies ☒Don't know	RESEARCH EVIDENCE Across all included studies, no serious adverse events related to glucocorticoids use were reported during follow-up.	ADDITIONAL CONSIDERATIONS No serious adverse events were reported during follow-up periods, but other non-serious adverse events were not systematically assessed, and the potential long-term risks of glucocorticoids therapy, including infections, osteoporosis, gastrointestinal bleeding, and mental health disorders, remain a concern. Although generally well tolerated in the short term, caution is warranted regarding their long-term safety.
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Very low ☒Low ☐ Moderate ☐ High ☐ No included studies	Four studies (three RCTs and one before-and-after study) assessed the effect of glucocorticoids on olfactory disorders in adult patients with long COVID.	Two systematic reviews on inhaled glucocorticoids for persistent post-COVID olfactory dysfunction had consistent findings with above.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Important uncertainty or variability ☐Possibly important uncertainty or variability ☒ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not assessed.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the comparison ☒ Does not favor either the intervention or the comparison ☐Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know	Regarding olfactory disorders, both local and systemic glucocorticoids therapy have been shown to improve olfactory scores in long COVID adult patients. However, no significant differences were observed in patients' self-reported olfactory recovery. Furthermore, the magnitude of score improvement does not meet the threshold for clinical significance, indicating limited effectiveness.	
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS

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☐Large costs ☒Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☐ Don't know	The cost of glucocorticoids is not formally assessed.	Glucocorticoids are generally accessible and the costs are moderate in clinical use.
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate ☐ High ☒No included studies	Not formally assessed.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ Favors the comparison ☐ Probably favors the comparison ☐Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Not formally assessed.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased ☒ Varies ☐ Don't know	Not formally assessed.	Glucocorticoid is generally accessible to patients. However, in areas with limited outpatient resource, monitoring for potential side effects for glucocorticoids is challenging.
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐ No ☐ Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	The glucocorticoids can be taken orally, once daily; this is generally acceptable in patients and for care givers. For inhaled glucocorticoids, it is usually once or twice daily and is also generally acceptable.

Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ No ☐ Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	The glucocorticoids can be taken orally, once daily; this is generally acceptable in patients and for care givers. For inhaled glucocorticoids, it is usually once or twice daily and is also generally acceptable.

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☒ Small	☐ Moderate	☐ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☐ Trivial	☐ Varies	☒ Don't know	/

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	JUDGEMENT						
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☒ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☒ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☒ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know

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	JUDGEMENT						
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☒ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☐ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

We suggest against the use of glucocorticoids for the treatment of olfactory disorders in adult patients with long COVID.

Justification

The panel formulated a conditional negative recommendation for glucocorticoid in treating long COVID patient to improve olfactory symptoms. The benefit of glucocorticoid is small and do not meet the threshold for clinical significance. The undesirable effects (especially long-term effects) have not been carefully evaluated.

Subgroup considerations

One study indicates that glucocorticoids may be more effective when initiated within 180 days post infection.

Implementation considerations

Since the overall benefit is limited, whether to use glucocorticoid to improve olfactory symptoms should be based on the judgment of individual care providers. Optimal glucocorticoid dosage and regimen should be considered, and patients should be carefully monitored for potential long-term effects.

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Monitoring and evaluation
Long-term safety should be monitored and evaluated, but there is no guidance on how often and for how long the patients are monitored.
Research priorities
There needs to be greater understanding of the effect of glucocorticoids and the pathogenesis for long COVID-related symptoms. Also, the duration and dosing of treatment should be considered.

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PICO Question 6

PICO Question 6 In patients with long COVID, should immunomodulators be used versus no immunomodulators?	
POPULATION:	Adult patients with long COVID, who have interstitial lung disease
INTERVENTION:	Immunomodulators
COMPARISON:	No immunomodulators
MAIN OUTCOMES:	mMRC score, peripheral oxygen saturation (SpO_2), FEV_1, FVC, and DLCO; time to liberation from oxygen
SETTING:	Hospital
PERSPECTIVE:	Population perspective
BACKGROUND:	Immunomodulators are agents that modulate the immune system by either enhancing or suppressing of immune responses.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒Probably yes ☐ Yes	Immunomodulators are agents that modulate the immune system by either enhancing or suppressing of immune responses. JAK inhibitors, which suppress the JAK-STAT signaling pathway, have demonstrated benefits in reducing inflammation and improving pulmonary involvement in COVID-19 patients.	Currently, clinical guidelines do not provide established recommendations for the use of immunomodulator in long COVID management.

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☐Varies ☐Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☐ Small ☐ Moderate ☐Large ☒Varies ☐Don't know	A cohort study published in 2023 involved 48 long COVID adult patients with interstitial lung disease. Of these, 21 received methylprednisolone, 11 received methylprednisolone combined with mycophenolate mofetil (MMF), and one received MMF alone. After six months of follow-up, all patients demonstrated significant clinical improvements in key outcomes, including mMRC score, SpO_2, FVC, FVC (% of predicted), FEV_1 (% of predicted), and DLCO (% of predicted). However, the study did not include patients who did not receive immunosuppressive therapy. It was unable to compare between the treated group and the untreated group due to the insufficient number of patients. A case-control study published in 2023 included 41 long COVID patients with hypoxia and active inflammation persisting for over four weeks. All patients received corticosteroid upon hospitalization. Among the 21 patients treated with JAK inhibitors (20% with baricitinib and 80% tofacitinib), corticosteroid doses were rapidly tapered within 7-10 days after initiation. Other 20 patients who did not receive JAK inhibitors required continued corticosteroid for 3-4 weeks, with duration adjusted based on clinical improvement. The results showed no significant difference in oxygen support requirements between the two groups. However, the median time to liberation from oxygen support was lower in JAK inhibitor group than steroids group. Two case reports from 2022 and 2024 described tacrolimus treatment for ILD in long COVID adult patients. Both reports showed favorable outcomes, with two patients discontinuing oxygen therapy by day 12 and day 56 post-treatment respectively. Chest X-ray revealed improved pulmonary involvement. Other immunomodulators have been discussed in case series or case reports. A 2023 case series reported the use of methotrexate or leflunomide to treat three adult patients who developed persistent joint pain after recovery	Immunomodulators are commonly used in combination with glucocorticoids, and the effect of immunomodulators alone may not be well assessed. The added-on effect of immunomodulators to steroids have not been well assessed. Also, there are many types of immunomodulators, and there is not enough evidence for each of them.

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	from acute SARS-CoV-2 infection and presented abnormal joint imaging. However, the treatments were discontinued in two patients due to significant adverse events, and one patient did not report the final results. Another 2023 case report described the use of anakinra in adult long COVID patients with persistent steroid-dependent multisystem inflammatory syndrome, showing good tolerability over 12-month period, with no immunosuppression-related symptoms or adverse events. The use of anakinra allowed tapering of the prednisone does with no recurrences of fever or increase in C-reactive protein levels. After removing disease definition restrictions for long COVID, a dynamic systematic review published in 2024 in *The BMJ* included 55 long COVID patients. Among them, 28 patients (mean age 45.5 years) received weekly subcutaneous injections of 700 mg leronlimab for 8 weeks, while the rest received a placebo. Primary outcome was the change in 24 symptom severity at day 56. Compared to baseline, patients receiving leronlimab improved in all symptoms except vomiting, while the placebo group also showed some improvement. There was no compelling evidence in leronlimab in resolving symptoms. No studies have evaluated immunomodulator use in pediatric long COVID populations.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT ☐Large ☐Moderate ☐ Small ☐Trivial ☐Varies ☒Don't know	RESEARCH EVIDENCE Patients in the cohort study tolerated MMF well, with no serious adverse events reported. In another study, patients not receiving JAK inhibitors experienced significantly higher rates of adverse events compared to those treated with JAK inhibitors. No serious adverse events were reported in patients with tacrolimus in the case reports. Significant adverse effects were reported in two patients with arthritis using methotrexate or leflunomide and the treatment was discontinued.	ADDITIONAL CONSIDERATIONS No serious adverse events were reported during follow-up periods, but other non-serious adverse events were not systematically assessed, and the potential long-term risks of immunomodulators, including infections, remain a concern.

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	Another case report described the use of anakinra in adult long COVID patients with persistent steroid-dependent multisystem inflammatory syndrome, showing good tolerability over 12-month period, with no immunosuppression-related symptoms or adverse events.	
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☒Very low ☐Low ☐ Moderate ☐ High ☐ No included studies	Our systematic review included one cohort study, one case-control study, one case series, and three case reports, involving 95 adult patients with long COVID. Data were retrieved from January 1, 2020, to November 30, 2024.	Due to significant heterogeneity across the studies, meta-analysis was not conducted.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☐Possibly important uncertainty or variability ☒ Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not formally assessed.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		

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JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☒Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know	Among the agents studied, MMF has shown potential effectiveness in patient key outcomes, but MMF was used in combination with glucocorticoids, and the study did not include a control group not using glucocorticoids or MMF. Therefore, the effect of MMF alone or in combination with steroids was not definite. The addition of JAKi may render faster tapering of glucocorticoids and shorten median time to liberation from oxygen support. Tacrolimus has also been reported to improve pulmonary symptoms in addition to glucocorticoids in two case reports. However, the sample size is limited, and the effect of tacrolimus alone has not been well assessed. Clinical trial registries indicate that multiple studies are underway investigating immunomodulators such as anakinra, intravenous immunoglobulin (IVIG), and baricitinib for adult long COVID patients.	Current studies are limited by small sample sizes, substantial heterogeneity, and a lack of consensus regarding key aspects such as dosing regimens, duration, combination therapy, and safety precautions.
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☒Large costs ☐Moderate costs ☐ Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☐ Don't know	The cost of immunomodulators is not assessed.	Immunomodulators may not be accessible in certain areas and the costs are generally high in clinical use.

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Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate ☐ High ☒ No included studies	Not formally assessed.	
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the comparison ☐ Probably favors the intervention ☐ Favors the intervention ☐ Varies ☒ No included studies	Not formally assessed.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☒Increased ☐ Varies ☐ Don't know	Not formally assessed.	Immunomodulators may not be accessible in certain areas, and long-term monitoring is necessary.
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒ Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed.	Some immunomodulators can be taken orally, and are generally acceptable.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no ☒Probably yes ☐Yes ☐ Varies	Not formally assessed.	Some immunomodulators can be taken orally, and are generally feasible to implement.

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☐ Don't know		

SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☐ Large	☒ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☐ Trivial	☐ Varies	☒ Don't know	/
CERTAINTY OF EVIDENCE	☒ Very low	☐ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☒ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☒ Large costs	☐ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know

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	JUDGEMENT						
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☒ Increased	☐ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☐ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation
We do not have recommendation favor or against using immunomodulators (mycophenolate mofetil, JAK inhibitors, tacrolimus) for adult patients with long COVID who have interstitial lung disease. (Very low certainty of evidence)

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Justification
The panel formulated a neutral recommendation on use of immunomodulators in treating patients with long COVID who have interstitial lung disease, based on the following: Small-scale studies showed potential benefits for immunomodulators (in combination with glucocorticoids), but the studies lack proper controls and the effect of immunomodulators alone is not clear. The addition of immunomodulator to glucocorticoid steroid was shown to render faster tapering of steroids and shorten time to liberation from oxygen support, but the sample size is limited. There are different types of immunomodulators (MMF, JAKi, etc.), and there is not enough evidence for each of them. The undesirable effects (especially long-term effects) have not been carefully evaluated.
Subgroup considerations
Only patients with interstitial lung disease were studied.
Implementation considerations
The current evidence does not suggest favor or against immunomodulators for patients with long COVID who have interstitial lung disease. The decision whether to use immunomodulators should be individualized and depend on the judgment of care providers. Optimal immunomodulator dosage and regimen should be considered. A duly-informed decision-making process by patients should be followed.
Monitoring and evaluation
Long-term safety should be monitored and evaluated, but there is no guidance on how often and for how long the patients are monitored.
Research priorities
This guideline encourages future clinical research on the use of immunomodulators in patients with long COVID. The use of immunomodulators in pediatric population is also welcome.

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PICO Question 7

PICO Question 7: In patients with long COVID, should cognitive behavioral therapy be used versus no cognitive behavioral therapy?	
POPULATION:	Adult patients with long COVID
INTERVENTION:	Cognitive behavioral therapy (CBT)
COMPARISON:	No CBT
MAIN OUTCOMES:	Fatigue
SETTING:	Hospital
PERSPECTIVE:	Population perspective
BACKGROUND:	Persistent fatigue is a prevalent and incapacitating symptom of long COVID, significantly impairing patients' daily functioning and well-being. CBT has shown potential in alleviating long COVID symptoms. However, current long COVID guidelines provide limited recommendations on CBT.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒Probably yes	CBT has potential in improving emotional regulation, promoting behavioral activation, and returning to normal life, which is reported among patients with long COVID. No dedicated systematic review has evaluated the efficacy of CBT for long COVID patients.	

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☐ Yes ☐ Varies ☐ Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Trivial ☐ Small ☒ Moderate ☐ Large ☐ Varies ☐ Don't know	Fatigue (main outcome): significant improvement Anxiety: no significant improvement Depression: no significant improvement Social functioning: no significant improvement	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Large ☐ Moderate ☐ Small ☒ Trivial ☐ Varies ☐ Don't know	No serious adverse events have been reported.	ADDITIONAL CONSIDERATIONS CBT is a non-pharmacological intervention with a favorable risk profile.
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Very low ☒Low ☐Moderate ☐High ☐No included studies	There was an overall low certainty of evidence across all studies due to 1) the low number of subjects in studies and 2) substantial heterogeneity between the included studies.	The RCTs used different scales.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☐Possibly important uncertainty or variability ☒Probably no important uncertainty or variability ☐No important uncertainty or variability	Not formally assessed	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐Probably favors the comparison ☐Does not favor either the intervention or the	CBT significantly alleviated fatigue symptoms compared to no CBT (or baseline) (SMD: 4.01, 95% CI: 1.17 to 6.85; $I^2 = 99\%$, low certainty of evidence). CBT shows potential for anxiety, depression, and social functioning. No serious adverse events have been reported.	CBT can be utilized as an adjunctive therapy, particularly for patients who do not respond adequately to pharmacological treatments.

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comparison ☒Probably favors the intervention ☐Favors the intervention ☐Varies ☐Don't know		
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☐Moderate costs ☐Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☒Don't know	Not formally assessed.	
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☐ Low ☐ Moderate	Not formally assessed.	

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☐ High ☒No included studies		
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐Probably favors the comparison ☐Does not favor either the intervention or the comparison ☐Probably favors the intervention ☐Favors the intervention ☐Varies ☒No included studies	Not formally assessed.	
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐Probably reduced ☐Probably no impact ☐Probably increased ☐Increased	Not formally assessed.	The implementation of CBT requires trained psychotherapists, and availability of such resources may be limited in certain regions, potentially restricting its practical application.

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☒Varies ☐Don't know		
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes ☒Yes ☐Varies ☐Don't know	Not formally assessed.	Given the non-pharmacological nature and low risk profile, patient preferences likely favor trying CBT, especially for those experiencing prominent fatigue and psychological symptoms.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes ☐Yes ☒Varies ☐Don't know	Not formally assessed.	Limited availability of trained psychotherapists in some regions Need for specialized training in long COVID management

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SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☒ Moderate	☐ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☐ Small	☒ Trivial	☐ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☒ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☐ Moderate costs	☐ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☒ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☐ Low	☐ Moderate	☐ High	☒ No included studies	/	/

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	JUDGEMENT						
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☐ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☒ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
FEASIBILITY	☐ No	☐ Probably no	☐ Probably yes	☐ Yes	☒ Varies	☐ Don't know	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☒ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

We suggest using cognitive behavioral therapy in adult patients with long COVID to alleviate fatigue symptoms. (*Conditional recommendation, low certainty of evidence*).

CBT should primarily be considered for patients experiencing persistent fatigue that significantly impacts daily life.

CBT may be used as an adjunctive treatment, particularly when pharmacological interventions alone are insufficient.

CBT should be delivered by trained psychotherapists, with appropriate monitoring of psychological and physical comorbidities.

CBT should not be interpreted as implying psychological causation of fatigue among long COVID patients.

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Justification
The panel formulated a conditional recommendation for a specific subgroup of long COVID patients, based on the available limited evidence. Evidence suggested that CBT significantly alleviated fatigue symptoms but showed no consistent significant benefit for anxiety, depression, or social functioning. Given the uncertainty regarding the balance of benefits and harms, and the low certainty of evidence, the panel considered that patients with debilitating fatigue may be more willing to accept CBT.
Subgroup considerations
Patients with severe or disabling fatigue are the subgroup most likely to benefit from CBT. Patients with comorbid psychological symptoms (anxiety, depression) may benefit, but the evidence is inconsistent. No sufficient data is available to inform recommendations for pediatric patients. Therefore, this guideline does not provide specific guidance on the use of CBT in children with long COVID.
Implementation considerations
● CBT should be delivered by trained psychotherapists. ● Given its favorable safety profile and lack of reported serious adverse events, CBT poses minimal direct harm. However, systematic reporting of outcomes and adverse events is encouraged.
Monitoring and evaluation
● Baseline psychological assessment is recommended to identify co-existing anxiety, depression, or other psychiatric disorders. ● Monitoring should include follow-up assessments to detect persistence, improvement, or relapse of fatigue and psychological symptoms. ● Implementation may require integration of CBT into existing long COVID rehabilitation services, ensuring multidisciplinary collaboration.
Research priorities
● Large-scale RCTs evaluating the safety and efficacy of CBT in long COVID. ● Development and validation of standardized outcome tools that distinguish between physical and cognitive fatigue, and capture both somatic and psychosocial dimensions. ● Subgroup-specific studies to evaluate pharmacokinetics, safety, and efficacy in vulnerable populations: children, elderly, patients with comorbidities, and those with psychiatric comorbidities.

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PICO Question 8

PICO Question 8: In patients with long COVID, should rehabilitation exercise be used versus no rehabilitation exercise?	
POPULATION:	Adult patients with long COVID
INTERVENTION:	Personalized rehabilitation exercise
COMPARISON:	No personalized rehabilitation exercise
MAIN OUTCOMES:	Physical capacity, quality of life
SETTING:	Multiple healthcare settings (primary care clinics, rehabilitation facilities, telerehabilitation platforms)
PERSPECTIVE:	Population perspective
BACKGROUND:	Rehabilitation exercise is a key component of recovery, aiming to improve physical function, reduce fatigue, and enhance quality of life, and it may be beneficial for patients with long COVID.
CONFLICT OF INTERESTS:	None

ASSESSMENT

Problem: Is the problem a priority?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☐Probably yes	Limited pharmacological treatment options make rehabilitation exercise a crucial intervention.	

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☒Yes ☐Varies ☐Don't know		
Desirable Effects: How substantial are the desirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Trivial ☐ Small ☐ Moderate ☒Large ☐Varies ☐Don't know	SRs included in this guideline indicate that rehabilitation interventions may improve physical capacity (including functional exercise capacity, muscle function, dyspnea, and lung function) and quality of life in long COVID patients.	
Undesirable Effects: How substantial are the undesirable anticipated effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Large ☐Moderate ☒Small ☐Trivial ☐Varies ☐Don't know	There was no compelling evidence for a difference in the odds of adverse events occurring between rehabilitation interventions and control group.	
Certainty of evidence: What is the overall certainty of the evidence of effects?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS

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☐Very low ☒Low ☐ Moderate ☐ High ☐ No included studies	There was an overall low certainty of evidence across all studies due to 1) the low number of subjects in studies and 2) mixed populations with varying COVID-19 severity and hospitalization status.	Wide variety of rehabilitation approaches studied (breathing exercises, aerobic training, resistance training, combinations) making it difficult to determine optimal approach.
Values: Is there important uncertainty about or variability in how much people value the main outcomes?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Important uncertainty or variability ☐Possibly important uncertainty or variability ☒Probably no important uncertainty or variability ☐ No important uncertainty or variability	Not formally accessed.	
Balance of effects: Does the balance between desirable and undesirable effects favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Favors the comparison ☐ Probably favors the comparison ☐ Does not favor either the intervention or the	Meaningful improvements in key outcomes (exercise capacity, dyspnea, quality of life). Low risk of serious adverse events when properly supervised.	Appropriate screening to exclude those with post-exertional malaise (PEM) is suggest.

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comparison ☒Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ Don't know		
Resources required: How large are the resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	VADDITIONAL CONSIDERATIONS
☐Large costs ☐Moderate costs ☒Negligible costs and savings ☐Moderate savings ☐Large savings ☐Varies ☐ Don't know	Not formally assessed.	A study of post-COVID-19 patients after hospitalization demonstrated an incremental cost of £305 per patient (95% CI: -123 to 732) with an ICER of £11,941 per QALY.
Certainty of evidence of required resources: What is the certainty of the evidence of resource requirements (costs)?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Very low ☒Low ☐ Moderate	Not formally assessed.	The evidence is based on a single RCT with economic evaluation.

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☐ High ☐No included studies		
Cost effectiveness: Does the cost-effectiveness of the intervention favor the intervention or the comparison?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐ Favors the comparison ☐ Probably favors the comparison ☐Does not favor either the intervention or the comparison ☒Probably favors the intervention ☐ Favors the intervention ☐ Varies ☐ No included studies	Not formally assessed.	A study demonstrated the cost-effectiveness of rehabilitation programs for long COVID patients previously hospitalized for COVID-19, with data on non-hospitalized patients remain limited.
Equity: What would be the impact on health equity?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐Reduced ☐ Probably reduced ☐ Probably no impact ☐Probably increased ☐Increased	Not formally assessed	Telerehabilitation options improve access for remote/rural populations. Need for healthcare professional supervision can be potential barriers.

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☒Varies ☐ Don't know		
Acceptability: Is the intervention acceptable to key stakeholders?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐Probably no ☒Probably yes ☐ Yes ☐ Varies ☐ Don't know	Not formally assessed	Given the non-pharmacological nature and low risk profile, patient preferences likely favor trying rehabilitation exercise.
Feasibility: Is the intervention feasible to implement?		
JUDGEMENT	RESEARCH EVIDENCE	ADDITIONAL CONSIDERATIONS
☐No ☐ Probably no ☒Probably yes ☐Yes ☐ Varies ☐ Don't know	Not formally assessed.	Multiple intervention types available. Both face-to-face and telerehabilitation delivery options.

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SUMMARY OF JUDGEMENTS

	JUDGEMENT						
PROBLEM	☐ No	☐ Probably no	☐ Probably yes	☒ Yes	☐ Varies	☐ Don't know	/
DESIRABLE EFFECTS	☐ Trivial	☐ Small	☐ Moderate	☒ Large	☐ Varies	☐ Don't know	/
UNDESIRABLE EFFECTS	☐ Large	☐ Moderate	☒ Small	☐ Trivial	☐ Varies	☐ Don't know	/
CERTAINTY OF EVIDENCE	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/
VALUES	☐ Important uncertainty or variability	☐ Possibly important uncertainty or variability	☒ Probably no important uncertainty or variability	☐ No important uncertainty or variability	/	/	/
BALANCE OF EFFECTS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ Don't know
RESOURCES REQUIRED	☐ Large costs	☐ Moderate costs	☒ Negligible costs and savings	☐ Moderate savings	☐ Large savings	☐ Varies	☐ Don't know
CERTAINTY OF EVIDENCE OF REQUIRED RESOURCES	☐ Very low	☒ Low	☐ Moderate	☐ High	☐ No included studies	/	/

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	JUDGEMENT						
COST EFFECTIVENESS	☐ Favors the comparison	☐ Probably favors the comparison	☐ Does not favor either the intervention or the comparison	☒ Probably favors the intervention	☐ Favors the intervention	☐ Varies	☐ No included studies
EQUITY	☐ Reduced	☐ Probably reduced	☐ Probably no impact	☐ Probably increased	☐ Increased	☒ Varies	☐ Don't know
ACCEPTABILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/
FEASIBILITY	☐ No	☐ Probably no	☒ Probably yes	☐ Yes	☐ Varies	/	/

TYPE OF RECOMMENDATION

☐ Strong recommendation against the intervention	☐ Conditional recommendation against the intervention	☐ Conditional recommendation for either the intervention or the comparison	☒ Conditional recommendation for the intervention	☐ Strong recommendation for the intervention

CONCLUSIONS

Recommendation

We suggest personalized rehabilitation training for adult patients with long COVID, following appropriate screening to exclude those with post-exertional malaise, and under the guidance and supervision of healthcare professionals. (*Conditional recommendation, low certainty of evidence*)

Justification

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The panel formulated a conditional recommendation for a specific subpopulation of patients with long COVID who do not present with PEM. The recommendation emphasizes careful patient selection, individualized rehabilitation plans, and professional supervision. Although rehabilitation interventions showed modest improvements in physical function, dyspnea, and quality of life, the overall certainty of evidence was low to very low.

Subgroup considerations

PEM or post-exertional symptom exacerbation (PESE): These patients should not undergo conventional rehabilitation but instead focus on pacing and symptom management.
Children: Although our recommendation didn't include children with long COVID, one RCT suggests that water-based and land-based exercise training programs may offer benefits for this population.

Implementation considerations

- Rehabilitation should be individualized, supervised, and adapted to patient tolerance and symptom severity.
- Programs may be delivered in-person or via telehealth, depending on resources and patient needs.
- Require guidance from trained healthcare professionals.
- Screening for PEM/PESE is mandatory before initiating rehabilitation.

Monitoring and evaluation

- Systematic screening for PEM/PESE prior to enrollment in rehabilitation programs.
- Regular monitoring of patient symptoms, functional outcomes, and potential adverse effects.
- Gradual adjustment of exercise intensity according to individual tolerance and clinical response.

Research priorities

- Comparative trials of different rehabilitation modalities (face-to-face vs telerehabilitation, aerobic vs resistance, multicomponent vs single modality).
- Evaluation of rehabilitation in special populations: children, elderly, patients with PEM, patients with comorbidities.
- Cost-effectiveness analyses in both hospitalized and non-hospitalized long COVID populations.

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